Supercritical exponential metrics in higher dimensions

Tightness, existence, and uniqueness

Research proof manuscript

Abstract

We construct exponential metrics of the whole-space logarithmically correlated Gaussian field in dimension $d \geq 3$ when $Q_d(\psi) > \sqrt{2(d-1)}$. Normalized Liouville first passage percolation is tight in the topology on lower-semicontinuous functions, and its limits are extended length metrics with a complete geodesic nonsingular component. We then prove that the limit is determined by the field and is unique up to a deterministic factor under the original five metric axioms. This gives convergence in probability for compact convolution mollifiers and heat smoothing, with the closed-cube crossing normalization. The construction uses finite intrinsic diameters on separating hypersurfaces. The uniqueness argument uses local discounts and boundary restoration, with surface estimates required only for a constructed reference metric when treating an arbitrary strong assignment.

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1 Introduction

An exponential metric assigns a large cost to paths passing through regions where a logarithmically correlated Gaussian field is large. In the supercritical regime, some points are at infinite distance from every other point. Compactness must therefore be proved in a topology that permits infinite distances. We construct these metrics in dimension $d \geq 3$ when the distance exponent satisfies

$$Q = Q_d(\psi) > \sqrt{2(d-1)}, \qquad \psi > 0\,. \tag{1.1}$$

The part of this range with $Q < \sqrt{2d}$ is strictly supercritical. The lower endpoint in (1.1) is excluded throughout.

The geometric estimate behind the construction is continuity of distances on deterministic hypersurfaces. Such a hypersurface can join two paths which cross an annulus, even when the paths do not intersect. We first prove tightness and construct a local metric determined by the field. We then prove uniqueness in an auxiliary weak class, obtain exact scale covariance, and finally prove uniqueness under the original five axioms of Ding–Gwynne. The main statements below have the forms of [2, Theorems 1.2–1.3] and [1, Theorems 1.3 and 1.8].

1.1 The field and its approximating metrics

Let h be the whole-space logarithmically correlated Gaussian field on $\mathbb{R}^d$, modulo additive constants, with covariance

$$\mathrm{Cov}(h(\mu), h(\nu)) = -\iint \log|x-y|_2\, \mu(dx)\nu(dy) \tag{1.2}$$

for compactly supported zero-mass tests of finite logarithmic energy. Let σ be normalized surface measure on the Euclidean unit sphere. We fix the representative

$$h(\sigma) = 0\,. \tag{1.3}$$

For $r > 0$ and $z \in \mathbb{R}^d$, write $h_r(z) = h(\sigma_{r,z})$ for the Euclidean spherical average. Its jointly continuous version is constructed in Lemma 3.1. The field in (1.2) is not the usual d-dimensional Gaussian free field when $d > 2$.

Geometric balls in the proof are cubes:

$$B_r(z) = z + (-r, r)^d, \qquad S_r(z) = \partial B_r(z), \qquad A_{a,b}(z) = B_b(z) \setminus \overline{B_a(z)}\,. \tag{1.4}$$

The averaging sphere is Euclidean. When an annular field restriction must determine $h_r(z)$, its inner radius is chosen smaller than $r/(8\sqrt{d})$.

Fix a nonnegative radial $\rho \in C_c^\infty(\mathbb{R}^d)$ with integral one. Set $\rho_\varepsilon(x) = \varepsilon^{-d}\rho(x/\varepsilon)$ and $h_\varepsilon = h * \rho_\varepsilon$. For an open connected set U, define

$$D_h^\varepsilon(x, y; U) = \inf_{P \subset U\colon x \to y} \int_0^1 e^{\psi h_\varepsilon(P(t))} |P'(t)|_2\, dt\,. \tag{1.5}$$

The infimum is over piecewise C^1 paths. We omit U when $U = \mathbb{R}^d$. The construction first uses the point-distance normalizer

$$a_\varepsilon = \mathrm{Med}\, D_h^\varepsilon(0, e_1; B_2(0))\,. \tag{1.6}$$

All medians and quantiles are lower quantiles. The all-parameter point-distance exponent of [3, Proposition 1.1 and Lemma 3.6], together with Lemma 2.2, gives

$$a_\varepsilon = \varepsilon^{1-\psi Q + o(1)} \qquad (\varepsilon \downarrow 0)\,. \tag{1.7}$$

An extended metric separates points and may take the value $+\infty$. Its path length is the supremum of the sums of successive endpoint distances over finite ordered partitions. Its internal distance $D(x, y; U)$ is the infimum of these lengths over metric-continuous paths in U, with an empty infimum equal to $+\infty$. A length metric equals its internal metric on the whole space. For a compact set K, write D^K for the compactly constrained distance. If w is positive and continuous, $w \cdot D$ denotes the path metric obtained by integrating w against D-arclength.

We use the lower-semicontinuous topology of [1, Definition 1.2]. Thus $D_n \to D$ means that, for every $(x, y) \in (\mathbb{R}^d)^2$,

$$\begin{aligned} D(x,y) &\le \liminf_{n\to\infty} D_n(x_n, y_n) \quad \text{whenever } (x_n, y_n) \to (x, y), \\ D(x,y) &= \lim_{n\to\infty} D_n(x_n, y_n) \qquad \text{for some } (x_n, y_n) \to (x, y). \end{aligned} \tag{1.8}$$

The second condition includes infinite values. This topology has a metrizable standard Borel realization. The field is viewed in a sufficiently negative local Sobolev space.

1.2 Tightness and the geometry of subsequential limits

The first theorem constructs a metric without assuming that it is determined by the field or that its law is unique.

Theorem 1.1 (Tightness). *Suppose $d \ge 3$ and (1.1) holds. Every deterministic sequence of positive cutoffs tending to zero has a subsequence $\mathcal{E}$ with the following properties.*

(i) The functions $a_\varepsilon^{-1} D_h^\varepsilon$ converge in law, jointly with h, in the topology (1.8) to a lower-semicontinuous function D.

(ii) Every possible subsequential limit D is an extended metric.

(iii) Jointly with the convergence in (i),

$$\left(a_\varepsilon^{-1} D_h^\varepsilon(u, v)\right)_{u,v\in\mathbb{Q}^d} \Longrightarrow \left(D(u, v)\right)_{u,v\in\mathbb{Q}^d}.$$

(iv) For each rational $r > 0$, the deterministic limit

$$c_r = \lim_{\mathcal{E}\ni\varepsilon\to 0} \frac{r a_{\varepsilon/r}}{a_\varepsilon} \in (0, \infty) \tag{1.9}$$

exists, and $c_r = r^{\psi Q + o_r(1)}$ as $r \downarrow 0$ or $r \to \infty$ through rational values.

Call x singular for D if $D(x, y) = \infty$ for every $y \ne x$. Call x lower α-thick for h if

$$\liminf_{r\downarrow 0} \frac{h_r(x)}{\log(1/r)} \ge \alpha. \tag{1.10}$$

Lemma 3.1 shows that smooth averages give the same lower-thickness sets.

Theorem 1.2 (Geometry of the limits). *Let (h, D) be any subsequential coupling in Theorem 1.1. Almost surely, the following hold.*

(i) The distance $D(x, y)$ is finite for Lebesgue-almost every pair $(x, y) \in (\mathbb{R}^d)^2$.

(ii) Every D-bounded set is Euclidean bounded.

(iii) On every Euclidean compact set, the identity from D to Euclidean distance is Hölder continuous with every exponent less than $1/[\psi(Q+\sqrt{2d})]$. If $Q < \sqrt{2d}$, the inverse identity is not continuous.

(iv) The complement of the singular set is one finite-distance component, and its metric is complete.

(v) Every two nonsingular points are joined by a D-geodesic.

(vi) Every lower α-thick point with $\alpha > Q$ is singular. If $Q < \sqrt{2d}$, the singular set is dense, uncountable, and Lebesgue-null.

The inequality $Q > \sqrt{2(d-1)}$ enters through hypersurface estimates. For a fixed coordinate hypersurface, the maximum coarse-field contribution is governed by $\sqrt{2(d-1)}$, so the normalized surface connection costs decay on small scales. The same Hölder argument on full-dimensional sets has threshold $\sqrt{2d}$.

The range covered by the theorems includes a nonempty strictly supercritical interval. Indeed, Q_d is continuous and nonincreasing, and [3, Proposition 1.1 and Lemma 3.7] gives

$$\psi^{-1} - \sqrt{2d} \le Q_d(\psi) \le \psi^{-1} + \sqrt{2}.$$

For small ψ the lower bound exceeds $\sqrt{2d}$, and for large ψ the upper bound is below $\sqrt{2(d-1)}$, since $d \ge 3$. Hence

$$I_d = \{\psi > 0 : \sqrt{2(d-1)} < Q_d(\psi) < \sqrt{2d}\} \tag{1.11}$$

is a nonempty open interval. Every point of I_d is greater than $\psi_{\mathrm{crit}}(d) = \sup\{\psi > 0 : Q_d(\psi) > \sqrt{2d}\}$; a plateau at the critical level is allowed.

1.3 Existence and uniqueness

The final characterization uses the same five axioms as [1, Definition 1.7], with $\mathbb{C}$ replaced by $\mathbb{R}^d$. In particular, the fifth axiom is a finiteness condition for distances between continua.

Definition 1.3 (Strong metric assignment)**.** A strong metric assignment with parameters (ψ, Q) is a Borel assignment $g \mapsto D_g$ to Euclidean lower-semicontinuous extended metrics such that, whenever g is a whole-space LCGF plus a possibly dependent random continuous function, the following hold.

I. Almost surely, $(\mathbb{R}^d, D_g)$ is a length space.

II. For every deterministic open U, the internal metric $D_g(\cdot,\cdot;U)$ is almost surely a measurable function of $g|_U$.

III. Almost surely, simultaneously for all continuous $f : \mathbb{R}^d \to \mathbb{R}$,

$$D_{g+f} = e^{\psi f} \cdot D_g\,. \tag{1.12}$$

IV. For each fixed deterministic $r > 0$ and $z \in \mathbb{R}^d$, almost surely simultaneously for all $u, v \in \mathbb{R}^d$,

$$D_g(ru+z, rv+z) = D_{g(r\cdot+z)+Q\log r}(u,v)\,. \tag{1.13}$$

V. For every deterministic connected open U and every pair of deterministic disjoint compact connected nonsingleton sets $K_1, K_2 \subset U$,

$$D_g(K_1, K_2; U) < \infty \qquad \text{almost surely}\,. \tag{1.14}$$

In (1.14), the distance is an infimum over endpoints. The axiom does not assert a finite diameter for either continuum.

Theorem 1.4 (Existence and uniqueness)**.** *For every $d \geq 3$ and $\psi > 0$ satisfying (1.1), a strong metric assignment with parameters $(\psi, Q_d(\psi))$ exists. If D and $\widetilde{D}$ are two such assignments, there is a deterministic constant $c \in (0, \infty)$ such that, almost surely,*

$$\widetilde{D}_h(x, y) = cD_h(x, y) \qquad \text{for all } x, y \in \mathbb{R}^d . \tag{1.15}$$

On the same full-probability base-field event, this equality holds after every continuous Weyl perturbation, including a dependent random one.

1.4 Convergence with the cube-crossing normalization

To state the limit in the normalization used in the planar papers, let $C_0 = [0, 1]^d$ and let F_0, F_1 be its opposite faces perpendicular to e_1. Define

$$Z_{C_0}(A) = \inf\{\mathrm{len}_A(P) : P \subset C_0,\ P(0) \in F_0,\ P(1) \in F_1\} . \tag{1.16}$$

The constraint is the closed cube. Let

$$k_\varepsilon(x) = (\pi\varepsilon^2)^{-d/2} e^{-|x|_2^2/\varepsilon^2}$$

and write $D_h^{\mathrm{H},\varepsilon}$ for (1.5) with $h * \rho_\varepsilon$ replaced by $h * k_\varepsilon$. For $\kappa \in \{\rho, \mathrm{H}\}$, put $D_h^{\rho,\varepsilon} = D_h^\varepsilon$ and

$$\mathfrak{a}_\varepsilon^\kappa = \mathrm{Med}\, Z_{C_0}(D_h^{\kappa,\varepsilon}) .$$

A coordinate complex is a finite connected union of compact coordinate plaques of dimension $d - 1$ and cubical boundaries.

Theorem 1.5 (Convergence of LFPP)**.** *Suppose $d \geq 3$ and (1.1) holds. There is a strong assignment $\mathcal{D}$ with $\mathrm{Med}\, Z_{C_0}(\mathcal{D}_h) = 1$ such that, for each $\kappa \in \{\rho, \mathrm{H}\}$ and every possibly dependent random bounded continuous f,*

$$(\mathfrak{a}_\varepsilon^\kappa)^{-1} D_{h+f}^{\kappa,\varepsilon} \longrightarrow \mathcal{D}_{h+f} \qquad \text{in probability} \tag{1.17}$$

in the topology (1.8), jointly with h. The limit is the same for every fixed nonnegative smooth radial compactly supported probability mollifier ρ. For each fixed deterministic finite connected coordinate-hypersurface complex S, convergence for $f = 0$ also holds uniformly on $S \times S$.

The normalizers in (1.17) are computed from the pinned base field h, including when a continuous perturbation is added. The heat convolution is used only for the base field and bounded perturbations. The assignment in Definition 1.3 is nevertheless defined for all continuous perturbations.

1.5 Proof outline

The construction begins with a point-distance exponent and a one-sided bound on normalization ratios at selected cutoffs. Surface estimates at those cutoffs produce closed zero-distance clusters. Locality of the clusters and independence in disjoint regions rule out a zero annulus crossing. A comparison of coarse and fine paths then extends the normalization estimates to every cutoff.

Small separating surfaces permit concatenation at arbitrary limiting endpoints, giving the triangle inequality and geodesics.

For locality and field measurability, we first distinguish extracted internal-distance arrays from the actual internal metric of the limit. Compact path limits give one inequality; short connections on rational surfaces give the reverse recovery statement in buffered open sets. Conditional locality then permits resampling in small cells. A deterministic comparison of conditional metric samples bounds the effect of one resampled cell. The arclength measure of a geodesic has no atoms, so the total conditional variance tends to zero.

Uniqueness compares two metrics along geodesics. A local discrepancy supplies a short path for one metric whose endpoints are farther apart for the other. Finitely many such paths are embedded in a tree. Discounting a small neighborhood of the tree gives either a definite shortcut or a small set of boundary contacts in the reference metric. The latter alternative controls restoration of the discounts. A weighted count of the surviving discounts, followed by Gaussian change of measure, rules out distinct optimal comparison constants.

The weak comparison argument uses surface estimates for both metrics. After weak uniqueness yields exact scale covariance and a strong reference assignment, the final argument uses surface estimates only for the reference metric. This distinction allows the competitor to satisfy exactly Definition 1.3. Internal point-distance and closed-cube medians are identified only after their own convergence statements have been proved.

Theorems 1.1 and 1.2 are proved in Sections 10 and 11, after the estimates in Sections 2–9. Sections 12–13 establish locality and field measurability. The weak uniqueness argument occupies Sections 14–19; Section 20 then gives exact covariance and a strong assignment. The proof of Theorem 1.4 concludes in Section 21. Sections 22–23 establish the closed-cube and heat conventions and complete Theorem 1.5.

2 A scale decomposition of the prescribed field

Fix a nonnegative radial $K \in C_c^\infty(\mathbb{R}^d)$ with $\|K\|_2 = 1$, and let W be white noise on $\mathbb{R}^d \times (0,\infty)$. Write

$$K_u(z) = u^{-(d+1)/2} K(z/u), \qquad H_t(x) = \int_{2^{-t}}^1 \int K_u(x-y)\, W(dy\, du), \quad t \geq 0.$$

In particular, $\operatorname{Var} H_t(x) = t \log 2$. The distribution $H_{0,\infty}$ denotes the same integral over $0 < u < 1$. Put

$$g(x) = \int_1^\infty \int (K_u(x-y) - K_u(-y))\, W(dy\, du), \qquad \widehat{h} = H_{0,\infty} + g. \tag{2.1}$$

The field g is smooth on compact sets. Its derivatives have locally bounded variances, by differentiating the kernel and integrating over $u \geq 1$.

Lemma 2.1 (Canonical covariance and pinning)**.** *The field $\widehat{h}$ has covariance (1.2) on zero-mass tests. The Gaussian variable $C = (\widehat{h}, \sigma)$ is well defined, and*

$$h = \widehat{h} - C \tag{2.2}$$

has the law specified by (1.2)–(1.3). On every fixed compact set V, the random variable

$$\|g\|_{\infty,V} + |C|$$

has a Gaussian upper tail, up to a deterministic change of constants.

Proof. Let $f(z) = \int K(w)K(w-z)\,dw$. Radiality makes $f(z)$ a function of $|z|_2$; moreover $f(0) = 1$, f is compactly supported, and $1 - f(z) = O(|z|_2^2)$ near zero. For $r, s > 0$,

$$\lim_{T\to\infty} \int_0^T (f(r/u) - f(s/u))\,\frac{du}{u} = \lim_{T\to\infty} \int_{r/T}^{s/T} f(v)\,\frac{dv}{v} = \log(s/r).$$

Thus integration over all scales gives $-\log|x-y|_2$ up to an additive constant. That constant, and the anchor in (2.1), vanish against zero-mass tests. This proves the covariance assertion.

The logarithmic kernel is integrable on the product of two unit spheres. More directly, the spherical-cap calculation used in Lemma 3.1 gives

$$\|K_u * \sigma\|_2^2 \le Cu^{d-2}, \qquad 0 < u \le 1.$$

Its integral is finite. Thus the short-scale part of C is a well-defined white-noise integral; the large-scale part is the average of the smooth field g. To justify smooth approximation, take $\sigma * \rho_\eta$. Convolution by a probability density contracts the displayed L^2 norm, and for each fixed u the kernel converges in L^2 as $\eta \downarrow 0$. Dominated convergence in u, together with the smooth large-scale part, proves convergence of the pairings. This also justifies the spherical pairing in (1.3). Subtracting C gives the specified pinning. Gaussian concentration on a compact set, together with the finite variance of C, gives the asserted tail bound. □

For real $t \ge 0$, write $\widetilde{H}_t = H_t + g - C$ for the corresponding sharp approximation to the pinned field.

The stationary field to which the probabilistic estimates will first be applied is

$$X_t = H_{0,\infty} * \rho_{2^{-t}}, \qquad t \ge 0. \tag{2.3}$$

Let D_t^{sr} be its exponential path metric with coefficient ψ, and let D_t^{sh} be the corresponding metric of H_t. For a fixed $q \in (0,1)$, sufficiently close to one, set

$$b_t = \inf\{a : \mathbb{P}[D_t^{\mathrm{sr}}(0, e_1; B_2) \le a] \ge q\}. \tag{2.4}$$

Here b_0 is its actual positive value; it is not set equal to one. We use

$$\chi = 1 - \psi Q_d(\psi), \qquad s = d - 1.$$

The exact decomposition, for real $0 \le m \le t$, is

$$X_t = H_m * \rho_{2^{-t}} + X_{m,t}^{\mathrm{f}}, \qquad X_{m,t}^{\mathrm{f}} = H_{m,\infty} * \rho_{2^{-t}}. \tag{2.5}$$

The fine field is independent of the white noise at scales $u \ge 2^{-m}$, and

$$\big(X_{m,t}^{\mathrm{f}}(2^{-m}x)\big)_x \overset{d}{=} \big(X_{t-m}(x)\big)_x. \tag{2.6}$$

If R_K, R_ρ are support radii of K, ρ, its restriction to a set V uses white noise only where

$$u \le 2^{-m}, \qquad \operatorname{dist}(y, V) \le R_K 2^{-m} + R_\rho 2^{-t} \le (R_K + R_\rho)2^{-m}. \tag{2.7}$$

Consequently the good-site processes used below have a fixed finite dependence range in lattice units, uniformly in both cutoffs. Nonnegativity of K and ρ also implies nonnegative covariance of X_t, so Gaussian positive association applies to its increasing functionals [9].

Lemma 2.2 (Gaussian estimates and the point exponent)**.** *For $t \geq 0$, the error $R_t = X_t - H_t$ satisfies*

$$\sup_x \operatorname{Var} R_t(x) \leq C, \qquad \mathbb{E}|R_t(x) - R_t(y)|^2 \leq C \min\{1, 2^{2t}|x-y|_2^2\}. \tag{2.8}$$

Also,

$$\operatorname{Var} X_t(x) = t \log 2 + O(1), \qquad \mathbb{E} \sup_{x \in B_{2^{-t}}(z)} (X_t(x) - X_t(z)) \leq C, \tag{2.9}$$

uniformly in t, z. On each fixed bounded set V, $\sup_V |R_n| = O_{\mathbb{P}}(\sqrt{n})$. For fixed distinct endpoints in the interior of a fixed cube U,

$$\frac{1}{n} \log_2 D_n^{\mathrm{sh}}(x, y; U) \longrightarrow -\chi, \qquad \frac{1}{n} \log_2 D_n^{\mathrm{sr}}(x, y; U) \longrightarrow -\chi \tag{2.10}$$

in probability. The convergence is uniform in marginal probability when the endpoints range over a compact family with positive separation and positive distance from ∂U. In particular,

$$b_n = 2^{-\chi n + o(n)}. \tag{2.11}$$

The exponent statement (1.7) holds for all positive cutoffs tending to zero.

Proof. For $u \geq \varepsilon$, translation of the smooth kernel and Minkowski's inequality give

$$\|K_u * \rho_\varepsilon - K_u\|_2^2 \leq C\varepsilon^2 u^{-3}.$$

For $u < \varepsilon$, Young's inequality gives

$$\|K_u * \rho_\varepsilon\|_2^2 \leq \|K_u\|_1^2 \|\rho_\varepsilon\|_2^2 \leq C u^{d-1} \varepsilon^{-d}.$$

Their integrals over $u \geq \varepsilon$ and $u < \varepsilon$ are bounded. Differentiating once gives an integrated squared norm at most $C\varepsilon^{-2}$, which proves (2.8). The mixed inner product of K_u with its convolution error is at most $C\varepsilon u^{-2}$ for $u \geq \varepsilon$; its integral is bounded. This proves the variance assertion in (2.9). The increment estimate for H_t on a cutoff-size cube, and (2.8), give a uniformly bounded canonical metric and expected supremum there. Gaussian entropy and concentration estimates now give the second assertion and the stated $O_{\mathbb{P}}(\sqrt{n})$ bound after a cover by $O(2^{dn})$ cubes.

The first assertion in (2.10) follows from [3, Proposition 1.1, Lemmas 2.7, 3.2 and 3.6]. These results precede the subcritical restriction in that paper. On U, adding R_n multiplies all path lengths by factors between $e^{-\psi\|R_n\|_\infty}$ and $e^{\psi\|R_n\|_\infty}$. Their logarithms are $o_{\mathbb{P}}(n)$, proving the second assertion and (2.11).

The pinned mollified field equals $X_t + g * \rho_{2^{-t}} - C$. The last two terms have a uniformly bounded-in-probability supremum on a fixed bounded domain. Thus the same logarithmic exponent holds for the pinned metric and each fixed quantile. For $t \in [n, n+1]$, the supremum on a fixed compact set of $H_t - H_n$ is $O_{\mathbb{P}}(\sqrt{n})$, uniformly in t: on a cutoff-size space-time box its variance and entropy are bounded, and there are $O(2^{dn})$ such spatial boxes. The corresponding convolution-error estimate is uniform on the same unit time interval. This proves the assertion for arbitrary cutoffs. In particular, different auxiliary kernels yield the same exponent for the one prescribed field law. □

Lemma 2.3 (Coarse oscillations)**.** *Fix $R, A < \infty$ and write $L_m = (1+m)^2$. Then*

$$O_m = \sup_{\substack{x,y \in B_R \\ |x-y|_\infty \leq A L_m 2^{-m}}} |H_m(x) - H_m(y)| = o_{\mathbb{P}}(m). \tag{2.12}$$

The same bound controls $|(H_m * \rho_{2^{-t}})(x) - H_m(z)|$ uniformly for $t \geq m$ and $|x - z|_\infty \leq A L_m 2^{-m}$, after enlarging A, R.

Proof. The kernel difference bound gives

$$\operatorname{Var}(H_m(x) - H_m(y)) \le C \log(2 + L_m)$$

for the pairs in (2.12). A grid of mesh 2^{-m} has at most $C2^{dm}L_m^d$ relevant pairs. Their maximal difference is $O_{\mathbb{P}}(\sqrt{m \log(2+m)})$ by Gaussian tails. The oscillations within individual grid cells have uniformly bounded means and variances, so their maximum is $O_{\mathbb{P}}(\sqrt{m})$. Both quantities are $o_{\mathbb{P}}(m)$. Finally, nonnegativity of ρ bounds the convolution difference by the oscillation on an enlarged ball, since $2^{-t} \le 2^{-m}$. □

3 Sphere averages and thickness

This section supplies the regularization comparisons used in the geometric theorem. They concern the field, not convergence or universality of the path metrics.

Lemma 3.1 (Comparison of thickness conventions)**.** *The spherical pairings $h_r(x) = (h, \sigma_{r,x})$ admit a jointly continuous version for $x \in \mathbb{R}^d$, $r > 0$. For every $R < \infty$, almost surely,*

$$\frac{1}{m} \sup_{\substack{x \in B_R \\ 2^{-m-1} \le r \le 2^{-m}}} |h_r(x) - \widetilde{H}_{-\log_2 r}(x)| \longrightarrow 0, \tag{3.1}$$

$$\frac{1}{m} \sup_{\substack{x \in B_R \\ 2^{-m-1} \le r \le 2^{-m}}} |(h * \rho_r)(x) - \widetilde{H}_{-\log_2 r}(x)| \longrightarrow 0, \tag{3.2}$$

$$\frac{1}{m} \sup_{\substack{x \in B_R \\ m \le t \le m+1}} |\widetilde{H}_t(x) - \widetilde{H}_m(x)| \longrightarrow 0. \tag{3.3}$$

Consequently the exact and lower thickness sets defined by spherical averages, the fixed smooth mollifier, and the dyadic sharp approximation coincide, simultaneously at all points.

Proof. The additive anchors cancel in every difference under consideration. For the spherical error, the white-noise kernel is $K_u * \sigma_r$ for $u < r$, and $K_u * \sigma_r - K_u$ for $u \ge r$.

For $u \le r$, a spherical-cap estimate gives

$$\|K_u * \sigma_r\|_2^2 \le C u^{d-2} r^{-(d-1)}. \tag{3.4}$$

Indeed, the relevant cap has normalized area at most $C(u/r)^{d-1}$, the kernel amplitude is at most $Cu^{-(d+1)/2}$ times that area, and the supporting spherical shell has volume at most $Cr^{d-1}u$. Multiplication of these bounds proves (3.4); comparable u, r are included by increasing C. For $u \ge r$, the translation estimate gives

$$\|K_u * \sigma_r - K_u\|_2^2 \le C r^2 u^{-3}.$$

Both integrals in u are finite and bounded uniformly in r. In particular, the spherical pairing is well defined in L^2 and its difference from the sharp approximation has uniformly bounded variance.

A spatial derivative, or a derivative with respect to the sphere radius, in the small-scale estimate costs a factor u^{-1} in the L^2 norm. Thus an increment of size $\delta \le r$ is bounded in squared norm by

$$C \min\{1, \delta^2/u^2\}\, u^{d-2} r^{-(d-1)}.$$

For $d = 3$, its integral over $u \le r$ is at most

$$C(\delta/r)^2(1 + \log(r/\delta)) \le C\delta/r;$$

for $d > 3$ it is bounded by $C(\delta/r)^2$. At larger scales the increment estimate follows by differentiating the smooth convolution difference and integrating Cr^2u^{-5}. Varying the sharp cutoff from r to a comparable radius s adds a white-noise band of variance at most $C|r - s|/r$. Therefore the spherical error has, on comparable radius ranges, canonical squared increments bounded by

$$C\left(\frac{|x - y|_2}{r} + \frac{|r - s|}{r}\right).$$

The same estimates for the spherical field itself on compact positive-radius intervals give a jointly continuous version, using Gaussian moments and Kolmogorov's criterion.

For the smooth-mollifier error, the bounds in Lemma 2.2 apply also to scales $u > 1$ in the full representation: their integrals only become smaller at large u. Differentiating the mollifier gives the corresponding spatial and radius increment estimates. The change of sharp cutoff again contributes a band variance bounded by $C|r - s|/r$. A unit-time increment of H_t has bounded variance and the same rescaled spatial and time entropy bounds; $g - C$ cancels from (3.3).

On one cutoff-size cube and one unit interval in $-\log_2 r$, each of these error processes has bounded expected absolute supremum and bounded maximal variance. This follows from the displayed canonical metric estimates by the Gaussian entropy bound. Cover B_R by $O_R(2^{dm})$ such cubes. Gaussian concentration and a union bound imply, for every $\eta > 0$ and all sufficiently large m,

$$\mathbb{P}[\text{one of the three suprema above exceeds } \eta m] \le C_R 2^{dm} e^{-c\eta^2 m^2}.$$

The probabilities are summable. Borel–Cantelli, followed by countable choices of R and $\eta \downarrow 0$, proves (3.1)–(3.3). Dividing by $\log(1/r)$, which differs from $m \log 2$ by a bounded amount on the indicated band, proves the equality of thickness sets. □

Lemma 3.2 (Thick points in every open ball)**.** *For each fixed $\alpha \in (0, \sqrt{2d})$, almost surely every nonempty open ball contains an exact-α thick set of Hausdorff dimension at least $d - \alpha^2/2$. In particular, this set is uncountable.*

Proof. Apply [4, Theorem 2.2] to the independent-scale field in natural logarithmic time, $\mathcal{H}_j = H_{j/\log 2}$, on a fixed bounded ball. We verify its hypotheses. If $f = K * K$ is the radial covariance function, then

$$q_j(x, y) = \mathbb{E}[\mathcal{H}_j(x)\mathcal{H}_j(y)] = \int_{e^{-j}}^{1} f(|x - y|_2/u)\,\frac{du}{u}.$$

Here $0 \le f \le 1$, $f(0) = 1$, $1 - f(v) = O(v^2)$ as $v \to 0$, and f has compact support. These properties give the uniform logarithmic covariance bound on each bounded ball. They also give, for $|x - y|_2 \le e^{-N}$ and $j \ge N$,

$$q_j(x, y) - q_N(x, y) \le \log\frac{1}{|x - y|_2} - N + C.$$

Moreover, $q_j(x, x) = j$. The increment kernels $p_j = q_j - q_{j-1}$ are positive definite, since they are covariances of independent white-noise scale bands, and satisfy $p_j(x, x) = 1$. In condition (D) of the cited theorem one may therefore take $c_j = 1$, for which $\sum_{j\ge1} c_j/j^2 < \infty$. Together with the two displayed covariance bounds and $q_j(x, x) = j$, this verifies its conditions (C) and (D).

That theorem gives the stated lower dimension for points with $\mathcal{H}_j(x)/j \to \alpha$. The bounded-time increment estimate in the proof of Lemma 3.1 applies equally to unit natural-logarithmic time bands, so the conclusion is unchanged on passage to dyadic time. Adding the locally bounded field $g - C$ does not change thickness. Lemma 3.1 then gives the intrinsic spherical and mollifier conventions. Apply the result on a countable basis of balls to obtain the simultaneous assertion for all nonempty open balls. The lower bound is positive for the stated range of α, so each such thick set is uncountable. □

4 Gaussian localization and annular decoupling

This section gives three Gaussian estimates used below: changes of measure under finitely many smooth shifts, decoupling of well-separated annular restrictions, and smoothness of conditional means away from the conditioning set. We work modulo constants for events invariant under a common constant Weyl factor.

The Fourier form of (1.2) is

$$\operatorname{Var} h(\phi) = c_d \int_{\mathbb{R}^d} |\xi|^{-d} |\widehat{\phi}(\xi)|^2 \, d\xi, \qquad \int \phi = 0, \tag{4.1}$$

for the positive constant determined by the Fourier convention. Thus the Cameron–Martin norm of a compactly supported smooth perturbation, modulo constants, is

$$\|f\|_{\mathcal{H}}^2 = c_d' \int_{\mathbb{R}^d} |\xi|^d |\widehat{f}(\xi)|^2 \, d\xi. \tag{4.2}$$

Only its translation and dilation invariance, not the numerical Fourier constant, enters the proof.

Lemma 4.1 (Fixed-cardinality changes of measure)**.** *Let F be a deterministic Cameron–Martin shift and let $0 < \alpha < 1$. For every measurable event A,*

$$\mathbb{P}[A(h)] \le \exp\left\{ \frac{\alpha \|F\|_{\mathcal{H}}^2}{2(1-\alpha)} \right\} \mathbb{P}[A(h+F)]^{\alpha}. \tag{4.3}$$

If F is a sum of at most k translated and dilated members of a fixed finite family in C_c^∞, the exponential factor is bounded in terms of that family, k, and α, independently of positions and scales. For the sphere-pinned representative the actual shift is $F - F(\sigma)$; (4.3) has the displayed interpretation for constant-invariant events.

Proof. Let μ_F be the shifted Gaussian law and write $R = d\mu_F/d\mu = \exp(Z - \|F\|_{\mathcal{H}}^2/2)$, where Z is centered Gaussian with variance $\|F\|_{\mathcal{H}}^2$. Apply Hölder to $\mu(A) = \int_A R^{-1} \, d\mu_F$ with exponents $1/(1-\alpha)$ and $1/\alpha$. The Gaussian moment of R^{-1} gives (4.3).

A Fourier change of variables in (4.2) shows that $f(x) \mapsto f((x-z)/r)$ preserves the norm. If every template has squared norm at most B, the Hilbert-space triangle inequality gives

$$\left\| \sum_{i=1}^{k} f_i \right\|_{\mathcal{H}}^2 \le B k^2. \tag{4.4}$$

No orthogonality of spatially disjoint bumps is used. Subtracting $F(\sigma)$ changes only the representative of its class modulo constants and makes the shift preserve the pin. □

Lemma 4.2 (Separated restriction blocks)**.** *Fix $0 < a < b < \infty$ and a sufficiently small $\kappa > 0$. Let $r_1 > \cdots > r_n$ have $r_{j+1}/r_j \le 2^{-L}$, where L is sufficiently large, and let $|z_j - x|_\infty \le \kappa r_j$. Let $\mathcal{F}_j$ be the Gaussian sigma field generated by zero-mass finite-energy tests supported in*

$$z_j + \{ar_j < |y|_\infty < br_j\}.$$

There is $P_0 < \infty$, depending only on the geometry and separation, such that, for nonnegative $\mathcal{F}_j$-measurable F_j,

$$\mathbb{E}\prod_{j=1}^{n} F_j \le \prod_{j=1}^{n} (\mathbb{E}F_j^{P_0})^{1/P_0}. \tag{4.5}$$

If $G_j \in \mathcal{F}_j$ and $\mathbb{P}(G_j) \ge p$, then for $0 < \vartheta < p$,

$$\mathbb{P}\left[\sum_{j=1}^{n} \mathbf{1}_{G_j} \le \vartheta n\right] \le \exp\{-nI_p(\vartheta)/P_0\}, \tag{4.6}$$

$$I_p(\vartheta) = \vartheta \log\frac{\vartheta}{p} + (1-\vartheta)\log\frac{1-\vartheta}{1-p}.$$

Completed sigma fields are permitted. For fixed $\vartheta < 1$, the decay rate tends to infinity as $p \uparrow 1$.

Proof. Rescale an outer annulus to unit size and an inner annulus to size $t = r_j/r_i$. Their supports have positive separation in these coordinates. Subtract from the logarithmic kernel a term constant in the inner variable. The resulting kernel and all its derivatives on the fixed supports are $O(t)$. A term constant in the outer variable can be subtracted as well. Multiplying by fixed smooth cutoffs in both variables gives a kernel of $H^m_x H^m_y$ norm at most Ct, for any fixed integer $m > d/2$.

Sobolev duality and $(1 + |\xi|^2)^{-m} \le C|\xi|^{-d}$ imply

$$|\operatorname{Cov}(h(\phi_i), h(\phi_j))| \le C(r_j/r_i)\|\phi_i\|_{\dot{H}^{-d/2}}\|\phi_j\|_{\dot{H}^{-d/2}}, \qquad j > i. \tag{4.7}$$

The small moving-center errors change only C. The covariance norm of a zero-mass test is invariant under these rescalings.

For finitely many Gaussian test coordinates in each block, (4.7) and the geometric row-sum bound show that their full covariance matrix is at most P_0 times its block-diagonal part in quadratic-form order. The Gaussian Hölder inequality of Chen–Dafnis–Paouris [5, Theorem 1(i)] gives (4.5). Approximate each restriction sigma field by increasing finite collections of tests, apply conditional expectation, and then truncate and pass to the limit. Smooth approximation includes the finite-energy spherical averages. Null-set completion does not change the inequality.

Apply (4.5) to $F_j = \exp(-t\mathbf{1}_{G_j})$. Exponential Markov and optimization, with scalar variable tP_0, give (4.6). The formula for I_p proves the last assertion. □

Lemma 4.3 (Additional conditionally local randomness)**.** *Suppose events G_j are conditionally independent given the entire field, and $q_j = \mathbb{P}(G_j \mid h)$ is measurable with respect to the corresponding restriction block in Lemma 4.2. If $\mathbb{E}q_j \ge p$, then (4.6) remains valid.*

Proof. Conditional independence gives

$$\mathbb{E}e^{-t\sum_j \mathbf{1}_{G_j}} = \mathbb{E}\prod_j (1 - (1 - e^{-t})q_j).$$

Apply (4.5). The convex function $q \mapsto (1 - (1 - e^{-t})q)^{P_0}$ lies below the chord joining its endpoint values on $[0, 1]$. Its expectation is therefore at most $1 - p + pe^{-tP_0}$. The same optimization as before proves the bound. □

Lemma 4.4 (Simultaneous covers)**.** *Fix a bounded macroscopic region, an inner-ball fraction, and finitely many annular templates. At radius r use a lattice of mesh κr. Suppose each designated event has probability at least p, uniformly in center and radius, and is determined modulo constants by a fixed annular field block at that scale. The conditionally local version of Lemma 4.3 is also allowed.*

For every $M < \infty$ and every fixed $\delta > 0$, one can choose $p < 1$ so that, from any deterministic subset containing at least a δ fraction of the geometrically spaced levels in $[\varepsilon^2 R, \varepsilon R]$, the good inner balls cover every point of the macroscopic region with probability $1 - O(\varepsilon^M)$. One can require a positive fraction of successful levels at every point. Constants are uniform in the chosen subset and in a fixed multiplicative shift of the radius block.

Proof. Use a finest grid of mesh at most $\kappa\varepsilon^2 R/10$ in a fixed enlargement of the region. It has $O(\varepsilon^{-2d})$ points. For each grid point w and each available radius choose a nearest point $z_r(w)$ of the radius-r lattice. Its distance from w is $O(\kappa r)$.

Split the radius levels into a fixed number of residue classes so their annular blocks have the separation required by Lemma 4.2. One class contains a positive fraction of the designated levels, hence at least a fixed multiple of $\log \varepsilon^{-1}$ levels. Choose p so that the exponent in (4.6) exceeds the finest-grid entropy and M. A union bound over the grid and finitely many classes proves the assertion at each w. Any point within one finest-grid spacing of w lies in the prescribed inner part of every successful ball for w, after decreasing κ once. The same environment consequently works for every path, without a union bound over paths. □

Recall the white-noise representation from Section 2, with scale kernel

$$K_u(x) = u^{-(d+1)/2} K(x/u). \tag{4.8}$$

Write F_{Lr} for the field from scales below Lr and G_{Lr} for its smooth complement. A pinning constant can be included in either part; it cancels from normalized local events.

Lemma 4.5 (Coarse oscillations and small units)**.** *On a box of radius Cr, the oscillation of G_{Lr} tends to zero in probability as $L \to \infty$, uniformly in r and the center. Fine restrictions whose $O(Lr)$ white-noise footprints are disjoint are independent.*

Suppose deterministic positive constants c_r satisfy, for each $\eta > 0$,

$$C_\eta^{-1}(r/R)^{\psi Q+\eta} \le c_r/c_R \le C_\eta (r/R)^{\psi Q-\eta}, \qquad 0 < r \le R, \tag{4.9}$$

with $Q > 0$. For any fixed finite collection of centers $|z| \le CR$,

$$\frac{c_{\rho R} e^{\psi h_{\rho R}(z)}}{c_R e^{\psi h_R(0)}} \longrightarrow 0 \tag{4.10}$$

in probability as $\rho \downarrow 0$, uniformly in R.

Proof. The squared L^2 norm of a coarse increment over distance $O(r)$ is at most

$$C \int_{Lr}^{\infty} (r/u)^2 \, \frac{du}{u} \le CL^{-2}.$$

Rescaled derivatives satisfy the corresponding bounds. Gaussian Sobolev estimates on a fixed rescaled box and Gaussian concentration give the oscillation assertion. Compact support of K gives the footprint independence.

The covariance (1.2), applied to the zero-mass difference of the two spherical measures, gives

$$\operatorname{Var}(h_{\rho R}(z) - h_R(0)) \le \log \rho^{-1} + C \tag{4.11}$$

for the stated centers. Choose $\eta < \psi Q$ in (4.9). The logarithm of (4.10) is at most $-(\psi Q - \eta) \log \rho^{-1} + O(1)$ plus a centered Gaussian variable of variance $O(\log \rho^{-1})$. A Gaussian tail bound proves the result. The finite union needs no independence. □

4.1 Conditional means outside an arbitrary open set

For the locality argument one needs a smooth conditional mean away from the conditioning region. The proof below uses the covariance kernel, not a local Markov operator, and therefore applies in odd dimensions as well.

Lemma 4.6 (Smooth conditional mean)**.** *Let V be a nonempty open set and pin the field at a smooth probability density supported in V. Put*

$$m = \mathbb{E}[h \mid h|_V], \qquad R = h - m.$$

Then R is independent of $h|_V$ and vanishes on V. The random distribution m has an almost surely smooth version on $\mathbb{R}^d \setminus \overline{V}$.

Proof. Gaussian orthogonal projection gives independence and vanishing. Fix $W \Subset \mathbb{R}^d \setminus \overline{V}$, a smooth probability density η_0 supported near W, and a smooth cutoff χ which is zero near $W \cup \operatorname{supp} \eta_0$ and one on V. For x near W set

$$g_x(y) = \chi(y) \left[-\log |x - y|_2 + \int \eta_0(w) \log |w - y|_2 \, dw \right]. \tag{4.12}$$

There is no singularity on the support of χ. Cancellation of the logarithmic leading term at infinity gives, for all multi-indices,

$$|\partial_x^\alpha \partial_y^\beta g_x(y)| \le C_{\alpha,\beta} (1 + |y|)^{-1-|\beta|}, \tag{4.13}$$

uniformly for x in a compact neighborhood of W.

The map $x \mapsto g_x$ is smooth with values in $\dot{H}^{d/2}$. Indeed, if $d = 2k$, its derivatives of order k in y are square integrable, with integrable radial bound r^{-3} at infinity. If $d = 2k + 1$, its derivatives of orders k and $k + 1$ have radial bounds r^{-2} and r^{-4}; Fourier interpolation gives the required homogeneous half derivative. The same statements hold after every x-derivative, and dominated difference quotients give smooth dependence in the Hilbert norm.

For a zero-mass test ϕ supported in V, the covariance with the formal increment $h(\delta_x - \eta_0)$ equals $\langle \phi, g_x \rangle$. By Sobolev duality this is a bounded linear functional on the closed Gaussian Hilbert space generated by $h|_V$. It has a Riesz representative v_x, smooth in x. This does not evaluate h at a point: testing in x identifies the conditional mean of $h - h(\eta_0)$ on W with the Gaussian process represented by v_x.

Every derivative of this process has bounded second moment on compact subsets. Gaussian Sobolev estimates, followed by Sobolev embedding at arbitrarily high orders, give a smooth version. Add the finite scalar $\mathbb{E}[h(\eta_0) \mid h|_V]$. A countable exhaustion proves the assertion. □

Remark 4.7 (Returning to the sphere pin). The sphere pin need not lie in V. Apply Lemma 4.6 to $h^V = h - h(\eta_V)$, where η_V is a smooth probability density in V. The original representative is $h = h^V + T$, with $T = -h^V(\sigma)$. Its local restriction determines $(h^V|_V, T)$, since $T = h(\eta_V)$. If a

local metric kernel for h^V depends only on $h^V|_V$ and is conditionally independent of the entire exterior field and metric, this remains true after conditioning additionally on T, which is a function of the full h^V. Constant Weyl scaling then restores the original representative. No independence of this scalar from the field is asserted.

5 Surface bounds from a one-sided ratio

The fine-field paths in this section remain in the full ambient space. Only the region on which the coarse field is estimated is restricted to a thin neighborhood of a surface.

We use the following deterministic lattice fact from the proof of [3, Lemma 4.1]. Suppose all sites outside a finite box are good, and every bad $*$-component has lattice diameter at most L. There is a unique infinite nearest-neighbor good component $\mathcal{O}$. It comes within $C_d(L+1)$ of every site. If $u, v \in \mathcal{O}$ lie in a connected open set V, they can be joined by a good nearest-neighbor path in the $C_d(L+1)$-neighborhood of V. In a ball of radius $C(L+1)$, such a path can be chosen with at most $C'(L+1)^d$ edges, by erasing loops. These assertions concern only the site labels. The proof in the cited lemma surrounds a finite good component, or a component cut off by the boundary of the neighborhood, by a bad $*$-connected external boundary; a component that violates the assertion would force that boundary to have diameter greater than L. It does not use a restriction on Q.

The finite-dependence estimate needed with this fact is elementary. A self-avoiding $*$-path of length k contains at least ck sites whose determining white-noise regions are disjoint. If each site is bad with probability at most p, the probability that the path is bad is at most p^{ck}. The number of paths from a given site is at most $(3^d)^k$. Taking p sufficiently small gives an exponential bound in k, uniformly in the cutoffs. We also use the high-density conclusion that the probability for a specified site not to belong to the infinite good component tends to zero with p; this is [3, Lemma 2.10] (the same external-boundary argument).

A *surface* is a nondegenerate compact coordinate plaque, a cube boundary, or a finite connected union of these sets. Put

$$H_n(S;U) = \sup_{x,y\in S} D_n^{\mathrm{sr}}(x,y;U), \qquad T_\ell(S) = \{x : \mathrm{dist}_\infty(x,S) \le A(1+\ell)^2 2^{-\ell}\},$$

where the fixed constant A is chosen large enough for the constructions below.

Lemma 5.1 (Fine-field chaining)**.** *There is $q_0 < 1$, depending on d, K, ρ, such that the following holds for $q > q_0$. On each fixed bounded region there are events $G_{M,n}$, increasing in M, with*

$$\sup_{n\ge M} \mathbb{P}[G_{M,n}^c] \le C \sum_{\ell \ge M-1} 2^{d\ell} e^{-c\ell^2} \longrightarrow 0. \tag{5.1}$$

On $G_{M,n}$, for $n \ge r \ge M$ and a compact set S whose relevant tubes are contained in U and the fixed region,

$$\sup_{\substack{x,y\in S\\ |x-y|_\infty \le 2^{-r}}} D_n^{\mathrm{sr}}(x,y;U) \le C \sum_{\ell=r}^{n-1} (1+\ell)^{2d} 2^{-\ell} b_{n-\ell} e^{\psi \sup_{T_\ell(S)} H_\ell} + C(1+n)^{2d} 2^{-n} e^{\psi \sup_{T_n(S)} X_n}. \tag{5.2}$$

If S is a cube boundary of diameter at most $C_0 2^{-r}$, the same bound, with a changed constant, bounds $H_n(S;U)$.

Proof. For $0 \le \ell \le n$, call $z \in 2^{-\ell}\mathbb{Z}^d$ good if each signed coordinate edge from z has fine-field internal distance at most $2^{-\ell}b_{n-\ell}$, measured in $B_{2^{1-\ell}}(z)$ using $X^{\mathrm{f}}_{\ell,n}$. The case $\ell = n$ uses the positive reference quantile b_0. Declare vertices outside a fixed enlargement of the region good. By (2.6), stationarity, and radiality, the failure probability is at most $2d(1-q)$. The dependence range is bounded by (2.7).

At level $0 \le \ell < n$, call a vertex nice if it is good and all level-$\ell+1$ vertices within distance $2^{1-\ell}$ are good. This process has a bounded dependence range and failure probability at most $C_d(1-q)$. Its infinite nearest-neighbor component is contained in both successive infinite good components: every coarse edge subdivides into two fine good edges. Apply the path-counting estimate to both processes. Except on an event of probability at most $C2^{d\ell}e^{-c\ell^2}$, their bad $*$-components have diameter at most a sufficiently small constant times $(1+\ell)^2$. The lattice fact then gives nearby good vertices, nearby vertices common to successive good components, and local routes with at most $C(1+\ell)^{2d}$ edges. Intersect these events from level $M-1$ to $n-1$ to obtain (5.1).

Fix $x \in S$. Join it by a straight segment to a level-$n-1$ good vertex at distance at most $Cn^2 2^{-n}$. The cost of this segment is bounded by the last term of (5.2). Successively join to vertices in the good components at adjacent levels. At level ℓ, the displacement from x is at most

$$C\sum_{k\ge\ell}(1+k)^2 2^{-k} \le C'(1+\ell)^2 2^{-\ell}.$$

Every fine edge domain used at this level is therefore in the stated tube. The coarse part $H_\ell * \rho_{2^{-n}}$ is bounded above there by $\sup_{T_\ell(S)} H_\ell$, after enlarging A. Concatenating the fine paths gives the ℓth term of the sum. Fine internal distances can be approximated arbitrarily closely; prelimit geodesics are not needed.

Do the same for y. Their level-r vertices are at distance at most $Cr^2 2^{-r}$, and the lattice fact joins them with at most $C(1+r)^{2d}$ good edges in the same tube. Sum the costs. When $n = r$, use a straight segment directly. Finally, any two points of a cube boundary of diameter $C_0 2^{-r}$ can be joined on that boundary by a bounded number, depending on C_0, d, of links of size at most 2^{-r}. This proves the diameter assertion. □

Fix a set $I \subset \mathbb{N}_0$ of integer cutoffs and suppose, for some $A_0 < \infty$ and $\delta > 0$, that

$$\frac{b_{n-m}}{b_n} \le A_0 2^{(\chi+\delta)m}, \qquad n \in I, \quad 0 \le m \le n, \qquad \theta := Q - \delta/\psi > \sqrt{2s}. \tag{5.3}$$

The first application of the next results will be to record cutoffs, where (5.3) is immediate. No all-cutoff ratio estimate is assumed here.

Lemma 5.2 (Surface compactness and a field bound)**.** *Under (5.3), the restrictions $b_n^{-1}D_n^{\mathrm{sr}}(\cdot,\cdot;U)$ to $S \times S$, for a fixed surface $S \Subset U$ and $n \in I$, are tight in the uniform topology. Their Hölder seminorms are tight for every*

$$0 < \beta < \min\{1, \psi(\theta - \sqrt{2s})\}. \tag{5.4}$$

The same assertion holds for a fixed connected compact set whose cutoff-size covering number is at most $C2^{s\ell}$ at level ℓ.

In a joint subsequential limit on all rational surfaces and rational internal domains, retain the coarse fields $(H_\ell)_{\ell\ge0}$. On each fixed bounded region there is an almost surely finite random M such that every rational cube boundary S of diameter at most $C_0 2^{-m}$, with $m \ge M$, satisfies

$$H(S;V) \le CA_0 F_m(S), \qquad F_m(S) = \sum_{\ell\ge m}(1+\ell)^{2d}2^{-\psi\theta\ell}e^{\psi\sup_{T_\ell(S)}H_\ell}. \tag{5.5}$$

Here V is a fixed larger domain containing the tubes and $H(S;V)$ is the diameter in the limiting internal restriction. With $p=\theta/\psi$,

$$\mathbb{E}[F_m(S)^p] \le C(1+m)^C 2^{-\theta^2 m/2}, \tag{5.6}$$

uniformly over deterministic cube boundaries of the indicated size and location.

Proof. Fix β as in (5.4) and choose $\sqrt{2s} < a < \theta - \beta/\psi$. A tube around a fixed surface can be covered by at most $C(1+\ell)^C 2^{s\ell}$ cutoff-size cubes. On one cube, the expected supremum of H_ℓ is bounded: subtract its value at the center and use the rescaled increment estimate. Gaussian concentration and a union bound give

$$\mathbb{P}[\sup_{T_\ell(S)} H_\ell > a\ell\log 2] \le Ce^{-c\ell}. \tag{5.7}$$

The same bound holds with X_ℓ in place of H_ℓ, by (2.9).

On the intersection of these events and $G_{r_*,n}$, division of (5.2) by b_n gives

$$b_n^{-1} \sup_{\substack{x,y\in S\\ |x-y|_\infty\le 2^{-r}}} D_n^{\mathrm{sr}}(x,y;U) \le CA_0 \sum_{\ell\ge r}(1+\ell)^C 2^{-\psi(\theta-a)\ell} \le C_\beta 2^{-\beta r}, \qquad r_* \le r \le n.$$

The terminal term uses $b_n \ge b_0 A_0^{-1} 2^{-(\chi+\delta)n}$; b_0 is a fixed positive constant. The failure probability tends to zero with r_*. For $|x-y| \le 2^{-n}$, a straight segment and the same lower bound for b_n yield

$$b_n^{-1} D_n^{\mathrm{sr}}(x,y;U) \le C|x-y|_2\, 2^{(1-\psi(\theta-a))n} \le C|x-y|_2^{\beta}.$$

For distances above 2^{-r_*}, a deterministic finite chain in the connected compact set gives a diameter bound. Such a chain exists by compactness and connectedness; equivalently, use a finite cover by small balls and its connected intersection graph. The finitely many cutoffs below r_* have smooth densities on a neighborhood of the set. This proves tight Hölder bounds. The triangle inequality supplies joint equicontinuity of the restricted distance functions, so Arzelà–Ascoli gives the claimed tightness. The argument used only the specified covering-number bound.

For a cube boundary of diameter $O(2^{-m})$, the sharper tube cover has at most

$$C(1+\ell)^C 2^{s(\ell-m)}, \qquad \ell \ge m,$$

cutoff-size cubes. For every fixed $u>0$, Gaussian concentration on a single cube and summation over the cover give

$$\mathbb{E}e^{u\sup_{T_\ell(S)} H_\ell} \le C_u(1+\ell)^C 2^{s(\ell-m)} 2^{u^2\ell/2}.$$

Take $u = p\psi = \theta$. If $p \le 1$, use subadditivity of the pth power; if $p > 1$, use Minkowski's inequality. The resulting series converges because $\theta^2/2 > s$, and its first-scale factor is $2^{-\theta^2 m/2}$. This proves (5.6).

We specify the passage to (5.5). Use strictly increasing integer cutoffs and extract jointly the countable surface arrays, the fields, the finite random variables $F_m(S)$, and the binary variables $G_{M,n}$. The joint law of the fields and all $F_m(S)$ is fixed, so their functional relation is preserved in the limit, without a continuity claim for the infinite series. In a Skorokhod representation the binary limits G_M are increasing in M and satisfy $\mathbb{P}[G_M = 0] \to 0$. Thus some G_M equals one almost surely.

For each fixed surface, the normalized terminal term in (5.2) tends to zero almost surely in this representation. Indeed, (5.7) for X_n has summable failure probabilities along the increasing cutoffs,

and $a < \theta$. This statement is simultaneous for the countable family of rational surfaces. The nonterminal sum, divided by b_n, is bounded by $CA_0F_m(S)$. Passing the inequality to the limiting restriction proves (5.5), simultaneously for all the indicated surfaces. Countable exhaustions handle the ambient region and domain choices. □

Corollary 5.3 (Separators at an independent center)**.** *Let*

$$Q_m(z) = B_{3\cdot 2^{-m}}\big(2^{-m}\lfloor 2^m z\rfloor\big),$$

with coordinatewise rounding. These rational cubes are nested, contain $B_{2^{-m}}(z)$, and shrink to z. For every deterministic z,

$$F_m(\partial Q_m(z)) \longrightarrow 0 \qquad \textit{almost surely.}$$

The same conclusion holds for a random z independent of the entire sequence $(H_\ell)_{\ell\ge 0}$. Whenever (5.5) applies, the limiting internal diameters of these separators tend to zero.

Proof. The uniform estimate (5.6), Markov's inequality, and Borel–Cantelli give the deterministic-center assertion. For an independent random center, condition on its value and use the same uniform bound. The last assertion follows pathwise from (5.5) after its finite random starting scale. No independence from that starting scale is needed. □

Remark 5.4 (An accumulating set). For every fixed z, the connected compact set

$$S_z^* = [z, z+e_1] \;\cup \bigcup_{m\ge 0} \partial B_{2^{-m}}(z)$$

has cutoff-size covering number at most $C2^{s\ell}$. The boundaries with $m \le \ell$ require at most $C\sum_{m\le\ell} 2^{s(\ell-m)}$ cubes; all the smaller ones lie in a bounded number of cutoff-size cubes. The segment requires $O(2^\ell)$ cubes, and $s \ge 1$. Hence Lemma 5.2 applies to S_z^* as well. This supplies continuity at a deterministic point together with its shrinking boundaries, not just on each boundary separately.

6 Noncollapse without volume continuity

Throughout this section assume (5.3). The proof uses continuous restrictions to deterministic surfaces, not continuity of a limiting distance throughout the volume.

Lemma 6.1 (Positive probability of a crossing)**.** *For every fixed cubical annulus $A = B_R(z)\setminus\overline{B_r(z)}$, $0 < r < R$, there are $c, p > 0$ such that*

$$\inf_{n\in I} \mathbb{P}[D_n^{\mathrm{sr}}(\partial B_r(z), \partial B_R(z)) > cb_n] \ge p. \tag{6.1}$$

Proof. First take $z = 0$, $R = 2r$, and $r < 1/(100\sqrt{d})$. For $u \in \partial B_r$, $v \in \partial B_{2r}$, let $U = B_{3r}$ and $\ell = |u - v|_2$. A polygonal chain from 0 to e_1 can be made with at most $J_0 = J_0(r, d)$ segments, each of length ℓ. Take steps of length ℓ along e_1 and use two equal-length segments in the (e_1, e_2)-plane for the remaining displacement. The chain remains in a fixed small neighborhood of $[0, e_1]$. Isometric images of the triple (u, v, U) along its segments all lie in B_2 by the choice of r.

Let $Y_1, \dots, Y_J$ be the internal distances in these images. Their marginal laws equal that of $D_n^{\mathrm{sr}}(u, v; U)$, by stationarity and radiality, and

$$D_n^{\mathrm{sr}}(0, e_1; B_2) \le \sum_{j=1}^{J} Y_j, \qquad J \le J_0.$$

Since $\mathbb{P}[D_n^{\mathrm{sr}}(0, e_1; B_2) \ge b_n] \ge 1 - q$, a union bound gives

$$\mathbb{P}[D_n^{\mathrm{sr}}(u, v; U) > b_n/(2J_0)] \ge (1 - q)/J_0 =: p_0.$$

No independence of the Y_j is used.

The tight Hölder constants of Lemma 5.2 allow finite deterministic plaque covers of the two boundaries so fine that, with probability at least $1 - p_0/2$, every patch has internal diameter at most $b_n/(8J_0)$, uniformly in $n \in I$. Choose a center in each patch. The triangle inequality shows that each patch-pair distance exceeds $b_n/(4J_0)$ with probability at least $p_0/2$. These lower-bound events are increasing functions of X_n. Positive Gaussian association bounds the probability of their finite intersection below by a positive product. On this intersection the distance between the two full boundaries has the same lower bound.

For a general annulus, cover its inner boundary by finitely many cubes $B_\rho(z_i)$ with ρ sufficiently small for the preceding argument and with $B_{2\rho}(z_i) \subset B_R(z)$. Every crossing traverses one of these smaller shells. Positive association applied to all their lower-bound events gives (6.1). Any finitely many small cutoffs can be included by reducing c. □

We next describe the limiting profiles precisely. It is enough initially to use rational z, r, R. Set

$$A_1 = \overline{B_r(z)}, \qquad A_2 = \mathbb{R}^d \setminus B_R(z).$$

Give $A_1 \cup A_2$ zero length density and its complement density $e^{\psi X_n}$. Let d_n be the resulting pseudometric. A last-exit/first-entry truncation shows that

$$C_n := b_n^{-1} d_n(A_1, A_2) = b_n^{-1} D_n^{\mathrm{sr}}(\partial B_r(z), \partial B_R(z)).$$

Define $f_{i,n}(x) = b_n^{-1} d_n(A_i, x)$. All nonzero costs occur in the fixed compact cube $\overline{B_R(z)}$.

Let Σ be the union of all compact rational coordinate plaques. A rational polycubical domain will mean a connected open set with boundary a finite union of such plaques, which is either bounded or has compact complement. For each such U containing the closed source A_i with positive distance from its compact boundary, retain the internal profile

$$f_{i,n}^U(x) = b_n^{-1} d_n(A_i, x; U), \qquad x \in U.$$

Also retain the corresponding profiles after replacing X_n by $X_{a,n}^{\mathrm{f}}$, for each fixed integer $a \ge 0$ and $n \ge a$.

For an internal profile, use only rational plaques compactly contained in U; their union covers $\Sigma \cap U$. These are countably many arrays of continuous functions on compact plaques. They are tight: local moduli follow from Lemma 5.2 and the triangle inequality, while a fixed polygonal route inside U from the source to a plaque, covered by finitely many coordinate plaques, bounds a reference value. A connected open domain admits such a route with positive boundary clearance. For coarse-deleted profiles use

$$e^{-\psi \|H_a\|_{\infty, V}} f_{i,n}^U \le f_{i,n}^{U,a} \le e^{\psi \|H_a\|_{\infty, V}} f_{i,n}^U, \tag{6.2}$$

where V is a fixed enlargement of $B_R(z)$ accounting for the support of ρ; the same comparison bounds their local moduli. Extract all these arrays jointly with C_n, the fields, and the surface bounds of Lemma 5.2, and use a Skorokhod representation. Denote the limits by $C, f_i, f_i^U, f_i^{U,a}$. Limits agree on intersections of plaques. Set

$$E_i = \mathrm{cl}\{x \in \Sigma : f_i(x) = 0\}, \qquad E_i^U = \mathrm{cl}\{x \in \Sigma \cap U : f_i^U(x) = 0\}. \tag{6.3}$$

All closures are Euclidean. They are measurable random closed sets: on each compact plaque the zero-set map is Borel, and then one takes a countable union and closure.

Lemma 6.2 (Locality of closed zero clusters)**.** *For every fixed eligible domain U,*

$$E_i^U \subset E_i, \qquad \{E_i^U \subset U\} \subset \{\min_{\partial U} f_i > 0\} \subset \{E_i^U = E_i\}. \tag{6.4}$$

Also, $E_i \subset U$ implies $E_i^U = E_i$.

Proof. Internal distances dominate global distances, so $f_i^U \geq f_i$ on $\Sigma \cap U$ and $E_i^U \subset E_i$.

Suppose $\min_{\partial U} f_i = 0$. Uniform convergence on the finitely many boundary plaques gives source-to-boundary paths of normalized cost tending to zero. Choose rational polycubical inner approximations U_η containing A_i, whose closures lie in U and whose boundaries are within $C\eta$ of ∂U. They can be constructed from grid cubes whose η-enlargements lie in U. For cofinite U, keep the domain unchanged outside a fixed large cube. The source has positive clearance from the compact boundary, so these approximations contain it for sufficiently small η.

Stop each cheap path at its first hit of ∂U_η. Its preceding segment lies in U. Compactness of the hit points and uniform convergence of $f_{i,n}^U$ on the boundary plaques give a point of E_i^U within $C\eta$ of ∂U. Letting $\eta \downarrow 0$, and using closedness of E_i^U and compactness of ∂U, shows that E_i^U meets ∂U. This proves the first event inclusion in (6.4).

For the second inclusion, every path from the source to $x \in U$ either stays in U or first meets ∂U. Therefore, before taking limits,

$$f_{i,n}(x) \geq \min\{f_{i,n}^U(x), \min_{\partial U} f_{i,n}\}.$$

For $x \notin U$ the lower bound is $\min_{\partial U} f_{i,n}$. Pass to the limits on the relevant plaques. When the boundary minimum is positive, the global and internal zero sets on Σ coincide and there are no global zero points outside U. Their closures coincide as well. Finally, if $E_i \subset U$, no boundary point is a zero of f_i. Continuity on the finite compact union ∂U makes its minimum positive, and the preceding implication applies. □

The second implication of (6.4) asserts equality, not containment in U. A sequence of zero points on different plaques may approach the boundary even if its trace is positive there. The proof never infers volume continuity from separate surface continuity.

Lemma 6.3 (Independence and separation)**.** *The pair (E_1, E_2) is independent of $(H_\ell)_{\ell \geq 0}$. Both sets are connected and contain their respective sources. Almost surely,*

$$\{C > 0\} = \{E_1 \cap E_2 = \varnothing\} = \{E_1 \cap A_2 = \varnothing\} = \{E_2 \cap A_1 = \varnothing\}. \tag{6.5}$$

For deterministic domains U_1, U_2 with disjoint closures, the two local clusters $E_1^{U_1}$ and $E_2^{U_2}$ are independent.

Proof. For every fixed a, comparison (6.2) passes to the limiting functions. The zero sets of the original and coarse-deleted profiles, and hence their closures, agree. The coarse-deleted arrays are independent of the white noise at scales $u \geq 2^{-a}$, both before and after taking limits. Thus (E_1, E_2) is independent of $(H_0, \dots, H_a)$ for every a. A monotone-class argument gives independence of the entire sequence. This is not an assertion that the clusters are measurable tail variables.

To see connectedness, fix a base point in the wired source. For each zero point $x \in \Sigma$, take a Hausdorff subsequential limit K_x of paths from that base point to x of normalized cost tending to zero. The paths can be confined to a fixed large cube: excursions outside it can be replaced by free paths in A_2 along its boundary. That boundary is connected since $d \geq 2$. The compact limit K_x is connected and contains both endpoints. If $w \in K_x$ is different from the base point, a sufficiently small rational cube containing w but not the base point is met by the approximating paths. Their boundary-hit points have a subsequential limit at which the limiting profile is zero, by uniform convergence on that boundary. Shrink the cube to conclude $w \in E_i$. Thus $K_x \subset E_i$. The closure of the union of these continua is exactly E_i, proving connectedness. The sources belong to E_i since the profiles vanish there and Σ is dense.

If $C = 0$, the prelimit inequality $|f_{1,n} - f_{2,n}| \leq C_n$ gives $f_1 = f_2$ and $E_1 = E_2$. If $C > 0$, the profiles are equal to C on the opposite sources, so $E_1 \subset \overline{B_R(z)}$ and E_2 avoids the interior of A_1. Suppose the clusters meet. Select a point Z from their intersection in a fixed large closed cube, using a Borel selector for nonempty compact sets and a fixed default point on the empty-set event. The selection depends only on (E_1, E_2), not on C separately; it is independent of all H_ℓ.

For each sufficiently large m, $\partial Q_m(Z)$ has a zero of each profile. Indeed, choose a zero point of that profile inside $Q_m(Z)$, which is possible by the closure definition. Start a cheap approximating path at a source point outside the cube, using free movement in the source to change the starting point. Its first hit of the fixed rational boundary has a subsequential limit where the profile vanishes. Uniform convergence on that boundary justifies the assertion even when Z is only a closure point.

The prelimit triangle inequality, and the two zero points on the boundary, now imply

$$C \leq H(\partial Q_m(Z); V),$$

for a fixed larger cube V. Corollary 5.3 makes the right-hand side tend to zero. Hence intersection is impossible when $C > 0$. Together with the $C = 0$ case and source containment, this proves (6.5).

Finally, fix eligible U_1, U_2 whose closures have positive separation. Choose a sufficiently large that the determining regions (2.7) for the two domains are disjoint for every $n \geq a$. Their coarse-deleted internal profile arrays are independent. Independence passes to their joint weak limit. Their closed zero sets equal the original local clusters by (6.2), giving the last assertion. Although U_2 may be cofinite, all nonzero costs occur in $B_R(z)$; the same finite-range conclusion applies. □

Proposition 6.4 (Noncollapse under the one-sided bound)**.** *Under* (5.3)*, every fixed cubical annulus satisfies*

$$\lim_{M \to \infty} \sup_{n \in I} \mathbb{P}[D_n^{\mathrm{sr}}(\partial B_r(z), \partial B_R(z)) < M^{-1} b_n] = 0. \tag{6.6}$$

Proof. Consider any subsequential limit C of the normalized crossing of a rational annulus, with the joint construction above. Write $Z = \{C > 0\}$ and $p = \mathbb{P}[Z]$. For a closed set E and $\eta = 2^{-j}$, define

$$N_j(E) = \bigcup_{\substack{v \in \eta \mathbb{Z}^d \\ \mathrm{dist}_\infty(v, E) < \eta}} B_{2\eta}(v).$$

This neighborhood contains E with a positive buffer and is contained in its 3η-neighborhood. Every selected cube intersects E, so $N_j(E)$ is connected when E is connected. If E is bounded or cofinite, the neighborhood is rational polycubical and of the same type.

On Z, E_1 is compact, E_2 is closed and cofinite, and their distance is positive. For all sufficiently large j, their neighborhoods consequently have disjoint closures, contain their closed sources with positive clearance, and avoid the opposite sources. On Z^c, the neighborhoods are identical. Hence the probability of the separated-neighborhood event tends to p.

Fix possible deterministic neighborhoods U_1, U_2 with these properties. Locality gives the exact identity

$$\{N_j(E_i) = U_i\} = \{N_j(E_i^{U_i}) = U_i\}. \tag{6.7}$$

For the forward direction, $E_i \subset U_i$ and Lemma 6.2 identifies the clusters. For the reverse direction, $E_i^{U_i} \subset N_j(E_i^{U_i}) = U_i$ and the same lemma applies. Lemma 6.3 therefore yields

$$\mathbb{P}[N_j(E_1) = U_1,\ N_j(E_2) = U_2] = \mathbb{P}[N_j(E_1) = U_1]\mathbb{P}[N_j(E_2) = U_2].$$

There are countably many possible neighborhoods. Sum over the separated pairs. Each marginal sum over eligible neighborhoods is at most p, because its neighborhood avoids the opposite source and (6.5) then forces $C > 0$. The separated-neighborhood probability is thus at most p^2. Let $j \to \infty$ to obtain $p \le p^2$.

Lemma 6.1 and the closed-set part of Portmanteau give $\mathbb{P}[C \ge c] \ge p_0 > 0$ for some c, p_0, so $p = 1$. Every subsequential crossing limit is therefore positive. Since these crossings are tight from above by a fixed surface connection, this proves (6.6) for rational annuli. For an arbitrary fixed annulus, place a rational cubical annulus between its two boundaries; every crossing of the original annulus crosses that one. The finitely many bounded cutoffs are positive random variables and do not affect the limit in (6.6). □

7 Cutoff comparisons and the record envelope

The preceding noncollapse result needs only a one-sided ratio bound. We first use it at record cutoffs. The next deterministic estimate is what transfers its conclusion to the remaining cutoffs.

Lemma 7.1 (All-path shortcut estimate)**.** *Let $\varepsilon \in (0, 1/100)$ and let $L \ge 1$ satisfy $(L+4)\varepsilon < 1/4$. Partition space into the closed mesh cells*

$$Q_z = z + [-\varepsilon/2, \varepsilon/2]^d, \qquad z \in \varepsilon\mathbb{Z}^d.$$

Declare sites good or bad, with every site outside B_4 good, and suppose every bad $*$-component has ℓ^∞-diameter at most L in lattice units. Let f, c, v be continuous fields, let D_f, D_v, D_{c+v} be their exponential path metrics with parameter ψ, and let $a, M > 0$. Suppose that every good site z whose cell meets B_2 satisfies

$$D_v(\partial B_\varepsilon(z), \partial B_{2\varepsilon}(z)) \ge M^{-1}\varepsilon a. \tag{7.1}$$

Assume also that $O \ge 0$ bounds $|f(x) - f(y)|$ and $|c(x) - f(y)|$ whenever the points in question lie in B_3 and are at ℓ^∞-distance at most $C_d(L+4)\varepsilon$. Then

$$D_f(0, e_1; B_3) \le C_d M (L+4) e^{2\psi O} a^{-1} D_{c+v}(0, e_1; B_2). \tag{7.2}$$

Here the lattice is assumed to contain $0, e_1$, as it does when $\varepsilon = 2^{-m}$.

Proof. Fix a piecewise C^1 path $P \subset B_2$ from 0 to e_1. Its visited mesh cells form a finite $*$-connected graph. This remains true when the path touches cell boundaries or visits a cell repeatedly; no generic-crossing assumption is required.

For each visited good center z, the path enters $B_\varepsilon(z)$. At least one endpoint is outside $B_{2\varepsilon}(z)$. Choose a subpath P_z crossing from $\partial B_\varepsilon(z)$ to $\partial B_{2\varepsilon}(z)$ and contained in the closed intervening shell. Such a subpath is obtained by taking a first exit from the larger cube and the last visit to the smaller boundary before that exit, reversing the path if needed. By (7.1) and the coarse comparison,

$$\operatorname{len}(P_z; D_{c+v}) \geq M^{-1}\varepsilon a\, e^{\psi f(z) - \psi O}.$$

Each parameter point of P belongs to at most C_d of the selected subpaths, since their supporting cubes have radius 2ε. Therefore

$$\sum_{\substack{z:\, Q_z \cap P \neq \varnothing \\ z \text{ good}}} \varepsilon e^{\psi f(z)} \leq C_d M e^{\psi O} a^{-1} \operatorname{len}(P; D_{c+v}). \tag{7.3}$$

Repeated visits do not increase this multiplicity: only one subpath is selected for each center.

Contract each bad $*$-component in the visited-cell graph to a single vertex, then take a simple graph path between the endpoint cells. Distinct contracted bad vertices are not adjacent. Consecutive retained good vertices are either neighboring cells or are joined through one bad component. Replace a traversal of a bad component by the straight segment between the corresponding good centers. This segment has Euclidean length at most $C_d(L+4)\varepsilon$. If an endpoint belongs to a bad component, join it directly to the adjacent retained good center. The endpoint cells cannot belong to the same bad component, by the diameter assumption; hence a retained good center always exists.

The resulting polygonal path joins 0 to e_1 and stays in B_3: its vertices are visited good centers, together with the two endpoints, and B_3 is convex. Assign each segment to one of its adjacent retained good centers. Each good center receives at most two assignments. The coarse oscillation bound makes the D_f-length of the constructed path at most

$$C_d(L+4)e^{\psi O} \sum_{\substack{z:\, Q_z \cap P \neq \varnothing \\ z \text{ good}}} \varepsilon e^{\psi f(z)}.$$

Combine this with (7.3), and take the infimum over P. This proves (7.2), including the endpoint cases without an additive error. □

Lemma 7.2 (One-scale lower comparison)**.** *Let $J \subset \mathbb{N}_0$ be such that, for every fixed cubical annulus A, the family*

$$\left\{ \frac{b_k}{D_k^{\mathrm{sr}}(\textit{across } A)} : k \in J \right\}$$

is tight. For every $\eta > 0$ there is $c_\eta > 0$ such that

$$b_{m+k} \geq c_\eta b_k 2^{-(\chi+\eta)m}, \qquad m \in \mathbb{N}_0, \quad k \in J. \tag{7.4}$$

Proof. Set $\varepsilon = 2^{-m}$ and use the exact decomposition

$$X_{m+k} = H_m * \rho_{2^{-(m+k)}} + X^f_{m,m+k}.$$

A site $z \in \varepsilon\mathbb{Z}^d \cap B_4$ is good when the fine-field crossing of the shell with radii $\varepsilon, 2\varepsilon$ centered at z is at least $M^{-1}\varepsilon b_k$. The hypothesis, stationarity, and fine-field scaling let us choose M so that the bad probability is arbitrarily small, uniformly in m and $k \in J$. These events have a bounded dependence range in lattice units. Declare sites outside B_4 good. The path-counting estimate then gives

$$\mathbb{P}[\text{some bad component has diameter greater than } L_m] \le C2^{dm}e^{-cL_m}, \qquad L_m = (1+m)^2,$$

uniformly in $k \in J$.

For all sufficiently large m, apply Lemma 7.1 with $f = H_m$, $c = H_m * \rho_{2^{-(m+k)}}$, $v = X^f_{m,m+k}$, and $a = b_k$. Lemma 2.3, with an enlarged constant to account for the support of ρ, supplies an $O_m = o_{\mathbb{P}}(m)$ independent of k in the bound. Thus, on an event of probability tending to one uniformly in k,

$$D^{\mathrm{sh}}_m(0, e_1; B_3) \le CM(L_m + 4)e^{2\psi O_m} b_k^{-1} D^{\mathrm{sr}}_{m+k}(0, e_1; B_2). \tag{7.5}$$

The point exponent in Lemma 2.2 makes the left side at least $2^{-(\chi+\eta/2)m}$ with probability tending to one. The prefactor on the right is at most $2^{\eta m/2}$ with probability tending to one. Hence

$$\mathbb{P}\big[D^{\mathrm{sr}}_{m+k}(0, e_1; B_2) \ge b_k 2^{-(\chi+\eta)m}\big] \longrightarrow 1$$

uniformly in $k \in J$. In particular, the lower-tail probability is eventually less than q, which gives the claimed lower bound for the q-quantile.

For each of the finitely many remaining m, fix a small annulus separating 0 and e_1. After scaling the fine field, its crossing is 2^{-m} times a crossing for X_k in one fixed annulus. Its reciprocal in units of b_k is tight by the stated hypothesis. The minimum of the fixed coarse field on a sufficiently large bounded set is bounded below with arbitrarily high probability. Combining these two events gives a positive lower quantile uniformly in $k \in J$; no independence is needed. Decrease c_η to include these m. The case $m = 0$ is immediate. The hypothesis for all fixed annuli, rather than only one unit annulus, is used precisely in this finite-scale step. □

Proposition 7.3 (Uniform one-sided ratios and noncollapse)**.** *Suppose $Q > \sqrt{2(d-1)}$. For every $\delta > 0$ there is $C_\delta < \infty$ such that*

$$\frac{b_{n-m}}{b_n} \le C_\delta 2^{(\chi+\delta)m}, \qquad 0 \le m \le n. \tag{7.6}$$

For every fixed surface $S \Subset U$, the restrictions $b_n^{-1} D^{\mathrm{sr}}_n|_{S\times S}$ are tight in the uniform topology. Every fixed annulus crossing, and every fixed point distance in a connected domain, is tight together with its reciprocal after division by b_n.

Proof. Choose first

$$0 < \delta < \psi(Q - \sqrt{2(d-1)}).$$

By the exponent for b_n, the sequence $s_n = 2^{(\chi+\delta)n} b_n$ tends to infinity. Let J be its successive record indices, including 0. For every $n \in J$ and $m \le n$,

$$\frac{b_{n-m}}{b_n} \le 2^{(\chi+\delta)m}.$$

Thus Proposition 6.4 applies along J, for every fixed annulus. Lemma 7.2, with loss $\delta/2$, gives

$$b_{m+k} \ge c b_k 2^{-(\chi+\delta/2)m}, \qquad k \in J. \tag{7.7}$$

Define the deterministic envelope

$$A_n = \max_{0\le k\le n} b_k 2^{-(\chi+\delta)(n-k)} = 2^{-(\chi+\delta)n} \max_{0\le k\le n} s_k.$$

A maximizing index k may be chosen to be a record index. Applying (7.7) at that k yields $b_n \ge cA_n$, whereas $b_n \le A_n$ by definition. Since the running maximum of s_k is nondecreasing,

$$\frac{A_{n-m}}{A_n} \le 2^{(\chi+\delta)m}.$$

These inequalities prove (7.6). Any larger value of δ follows by choosing a smaller positive one in the preceding argument.

Apply Lemma 5.2 and Proposition 6.4 again, now with $I = \mathbb{N}_0$. They give the surface upper bounds and crossing lower tails at every cutoff. A fixed pair of boundary points connected through finitely many coordinate plaques supplies the crossing upper bound. The same construction bounds any fixed point distance from above: an open connected domain contains a polygonal coordinate path between the points, with a compact neighborhood still inside the domain. A small separating annulus supplies its lower bound. The same bounds are uniform in marginal probability over compact families of the stated geometries with uniformly positive separations and boundary clearances; no simultaneous estimate over a continuum of centers is asserted here. □

Lemma 7.4 (One-scale upper comparison)**.** *For every $\eta > 0$ there is $C_\eta < \infty$ such that*

$$b_{m+k} \le C_\eta b_k 2^{-(\chi-\eta)m}, \qquad m, k \in \mathbb{N}_0. \tag{7.8}$$

Proof. At mesh $\varepsilon = 2^{-m}$, call a vertex good when all its fine-field distances to signed coordinate neighbors, measured inside $B_{2\varepsilon}$ about the vertex, are at most $M\varepsilon b_k$. Proposition 7.3 makes the bad probability arbitrarily small by choosing M large, uniformly in k. The dependence range is bounded. Choose M so that both endpoints lie in the original infinite good component with probability at least $1 - (1-q)/4$, using the high-density percolation bound in [3, Lemma 2.10]. With probability tending to one as $m \to \infty$, uniformly in k, all bad components meeting B_4 have diameter at most $L_m = (1+m)^2$.

Let $P \subset B_{3/2}$ be a near-minimizer for $D^{\mathrm{sh}}_m(0, e_1; B_{3/2})$. For the deterministic routing argument, declare all sites outside B_4 good. Endpoint membership in the original infinite good component implies membership in the modified one. The deterministic routing fact used in Lemma 5.1, applied to an arbitrarily thin connected open neighborhood of P, gives a good nearest-neighbor path from 0 to e_1 within distance $CL_m\varepsilon$ of P. Erase loops. Assign every remaining vertex to a mesh cell visited by P at distance at most $CL_m\varepsilon$. Each visited cell receives at most CL_m^d assignments, since the route has no repeated vertices.

For a visited cell centered at z, choose a subpath of P crossing its shell of radii $\varepsilon, 2\varepsilon$. Its Euclidean length is at least ε, and its coarse density is at least $e^{\psi H_m(z) - \psi O_m}$. The same bounded-overlap argument as in (7.3) gives

$$\sum_{z:\, Q_z \cap P \ne \varnothing} \varepsilon e^{\psi H_m(z)} \le C e^{\psi O_m} \operatorname{len}(P; D^{\mathrm{sh}}_m).$$

Concatenate fine-metric paths along the good route. Their defining edge domains lie in B_2 for all sufficiently large m. Comparing the convolved coarse field on each edge path to the assigned visited cell, using Lemma 2.3, gives

$$D^{\mathrm{sr}}_{m+k}(0, e_1; B_2) \le CML_m^d e^{C\psi O_m} b_k D^{\mathrm{sh}}_m(0, e_1; B_{3/2}). \tag{7.9}$$

An arbitrarily small near-minimization error can be removed before taking this infimum. The point exponent and $O_m = o_{\mathbb{P}}(m)$ imply that the two coarse factors on the right are at most $2^{-(\chi-\eta)m}$ with probability tending to one. The total success probability exceeds q for all sufficiently large m, uniformly in k. This proves the upper bound for the q-quantile.

For the finitely many smaller m, use the exact field split and bound the coarse factor on B_2. The fine point distance becomes, after scaling, a distance for X_k between 0 and $2^m e_1$ inside $2^m B_2$. Proposition 7.3 gives a uniform upper tail in units of b_k for each of these finitely many geometries. Increasing C_η proves (7.8) for every m. □

Combining the two comparisons, for every $\eta > 0$,

$$c_\eta 2^{(\chi-\eta)m} \le \frac{b_{n-m}}{b_n} \le C_\eta 2^{(\chi+\eta)m}, \qquad 0 \le m \le n. \tag{7.10}$$

The median and every fixed quantile of the reference point distance are uniformly comparable to b_n, because that distance divided by b_n is tight together with its reciprocal.

8 Quantitative annulus bounds and the pinned field

Lemma 8.1 (Uniform lower tail)**.** *For every fixed cubical annulus A, there are $c, C > 0$ such that*

$$\sup_{n\ge 0} \mathbb{P}[D_n^{\mathrm{sr}}(\textit{across } A) < T^{-1} b_n] \le C \exp\left(-c \frac{(\log T)^2}{\log\log T}\right), \qquad T \ge e^e. \tag{8.1}$$

In particular, normalized annulus crossings have uniformly bounded negative moments of every positive order.

Proof. We give the proof for $A = B_2 \setminus \overline{B_1}$; replacing it by a fixed annulus changes only constants. Choose M so that the fine good-crossing process in Lemma 7.2 has a sufficiently small bad probability uniformly over all fine cutoffs, by Proposition 7.3. At mesh 2^{-m} the probability of a bad $*$-component of diameter at least $c_0 2^m$ meeting B_3 is at most

$$C 2^{dm} e^{-c2^m}.$$

On its complement, every path crossing the middle third of A visits a good cell whose doubled shell lies in A. Otherwise the visited cells in that middle region would form a bad $*$-connected set of diameter at least $c_0 2^m$. For $n \ge m$ this good cell gives

$$\begin{aligned} \frac{D_n^{\mathrm{sr}}(\text{across } A)}{b_n} &\ge M^{-1} 2^{-m} \frac{b_{n-m}}{b_n} \exp(\psi \inf_{B_{3+R_\rho}} H_m) \\ &\ge c_\eta 2^{-(\psi Q+\eta)m} \exp(\psi \inf_{B_{3+R_\rho}} H_m). \end{aligned}$$

The enlargement accounts for the coarse convolution. The last inequality is (7.10).

Take $m = \lfloor 2 \log_2 \log T \rfloor$. For large T, the last expression exceeds T^{-1} whenever

$$\inf_{B_{3+R_\rho}} H_m \ge -\frac{\log T}{2\psi}.$$

The expected supremum of $-H_m$ on this fixed cube is $O(m)$ and its maximal point variance is $m \log 2$. Gaussian concentration bounds the complementary probability by $C \exp(-c(\log T)^2/m)$. The bad-component probability is smaller than the claimed bound.

If $n < m$, use the direct lower bound by the Euclidean shell width times $\exp(\psi \inf_A X_n)$. The upper comparison with fine cutoff zero bounds b_n by $C_\eta 2^{-(\chi-\eta)n}$. Since $n \le m = O(\log\log T)$, and $\operatorname{Var} X_n \le C(n+1)$, the same Gaussian argument applies. Increase C for bounded values of T. Integrating the lower-tail estimate gives the negative moments. □

Lemma 8.2 (Real cutoffs and deterministic normalizations)**.** *For real $t \ge 0$, $b_t \asymp b_{\lfloor t \rfloor}$ uniformly in t. For every $\eta > 0$,*

$$c_\eta 2^{(\chi-\eta)(v-u)} \le \frac{b_u}{b_v} \le C_\eta 2^{(\chi+\eta)(v-u)}, \qquad 0 \le u \le v. \tag{8.2}$$

The surface bounds, point-distance bounds, and (8.1) hold uniformly for real cutoffs. For the prescribed pinned field and its median,

$$a_{2^{-t}} \asymp b_t, \tag{8.3}$$

and the same bounded-domain estimates hold for $D_h^{2^{-t}}$ with normalizer $a_{2^{-t}}$. Ratios of these medians satisfy (8.2) with b_t replaced by $a_{2^{-t}}$.

Proof. Write $t = N + s$, where $N \in \mathbb{N}_0$ and $s \in [0,1)$. Scaling the white noise gives

$$X_t(x) = (H_s * \rho_{2^{-t}})(x) + X'_N(2^s x), \tag{8.4}$$

where X'_N has the law of X_N and is independent of the first term. The coarse term on each fixed bounded set has a Gaussian supremum tail uniformly in s, t. The scaled reference endpoints are $0, 2^s e_1$ and the domain is $2^s B_2$.

Their normalized fine point distances have uniform upper and lower tails. For the upper tail, split $[0,1]$ into two intervals of length $1/2$. On each interval choose a fixed coordinate plaque containing all endpoints and a fixed cube containing that plaque compactly and lying inside every corresponding $2^s B_2$. Surface tightness applies to this finite collection. For the lower tail, one fixed small annulus about zero separates all these endpoint pairs. Multiplying by the bounded coarse factor shows that the reference distance for X_t, divided by b_N, is tight with tight reciprocal. Its fixed q-quantile is therefore bounded above and below by positive constants times b_N. This proves $b_t \asymp b_N$. Rounding u, v in (7.10) gives (8.2).

The same argument applies to surfaces and annuli: the dilated geometries range over a compact family with uniform covering constants and positive boundary clearances. A finite family of fixed smaller annuli supplies lower bounds for the dilated annulus crossings. The Gaussian tail of the coarse factor, combined with (8.1), preserves that estimate. This proves the real-cutoff assertions.

For the pinned field, the exact identity is

$$h * \rho_{2^{-t}} = X_t + g * \rho_{2^{-t}} - C. \tag{8.5}$$

On a fixed bounded domain V, the absolute value of the correction is at most

$$G_V = \sup_{V + B_{R_\rho}} |g| + |C|,$$

uniformly for $t \ge 0$. This variable has a Gaussian tail by Lemma 2.1. Consequently internal distances are bounded between $e^{-\psi G_V} D_t^{\mathrm{sr}}$ and $e^{\psi G_V} D_t^{\mathrm{sr}}$. Annulus crossings have the same comparison after trimming paths to the annulus. The upper and reciprocal bounds for the reference point distance transfer without any independence assumption.

Choose a deterministic M so that the probability that this pinned reference distance lies in $[M^{-1} b_t, M b_t]$ is greater than $3/4$, uniformly in t. Every median lies in that interval, which

proves (8.3). The same product comparisons give the surface bounds. For the lower tail, split according to $G_V \le (\log T)/(2\psi)$ and use (8.1) with $T^{1/2}$ on the other factor; the Gaussian failure probability is no larger than the claimed bound after adjusting constants. The deterministic comparison of medians transfers (8.2) as well. □

The argument proves estimates for the actual convolution metrics of h. It does not identify a metric limit for one regularization with a limit for another, and does not move a random factor through a deterministic median.

9 Separating annuli at every point

For $r = 2^{-m}$ and $z \in \mathbb{R}^d$, let

$$A_m(z) = B_{4r}(z) \setminus \overline{B_r(z)}, \qquad S_m(z) = \partial B_{2r}(z).$$

For $t \ge 0$, write

$$C_{t,m}(z) = D_h^{2^{-t}}(\partial B_r(z), \partial B_{4r}(z)), \qquad H_{t,m}(z) = \sup_{x,y \in S_m(z)} D_h^{2^{-t}}(x, y; A_m(z)).$$

Any two paths crossing $A_m(z)$ can be joined at cost at most $H_{t,m}(z)$, because both meet $S_m(z)$. This statement does not use planar intersection of paths.

Lemma 9.1 (Many good annuli)**.** *There are integers $L, \ell_0 \ge 1$ such that the following holds. For every $B > 0$ there is $M_B < \infty$ such that, for any center z, any increasing list of nonnegative integers $m_1, \dots, m_k$ with spacings at least L, and any $t \ge m_k + \ell_0$,*

$$\mathbb{P}\left[\#\{j : H_{t,m_j}(z) \le M_B C_{t,m_j}(z)\} < k/2\right] \le 2e^{-Bk}. \tag{9.1}$$

Proof. Choose ℓ_0 so that $R_\rho 2^{-\ell_0} < 1/4$, where R_ρ is a support radius of ρ. For $r = 2^{-m}$ and $t \ge m + \ell_0$, every mollifier neighborhood of the closed shell $\overline{B}_{4r}(z) \setminus B_r(z)$ lies in

$$U_m(z) = \{x : r/2 < |x - z|_\infty < 5r\}.$$

Trim a crossing path between its last inner-boundary contact before its first outer-boundary contact. This shows that $C_{t,m}(z)$ is the crossing distance constrained to the closed shell. Thus the event

$$G_{t,m}(z) = \{H_{t,m}(z) \le M C_{t,m}(z)\}$$

depends only on $h|_{U_m(z)}$ modulo constants. Indeed, after subtracting the average against any smooth probability density supported in $U_m(z)$, all mollified field values in the shell are zero-mass tests supported there. The ratio is unchanged by that subtraction.

Set $h^{r,z} = h(z + r\cdot) - h_r(z)$. Its zero-mass covariance is (1.2), and its unit-sphere average is zero, so it has the same law as h. Exact prelimit scaling, with $\delta = 2^{-t}/r$, gives

$$\frac{H_{t,m}(z)}{C_{t,m}(z)} \overset{d}{=} \frac{H_{-\log_2 \delta,0}(0)}{C_{-\log_2 \delta,0}(0)}.$$

The factor $re^{\psi h_r(z)}$ cancels. Lemma 8.2 supplies tight normalized separator diameters and tight reciprocals of normalized crossings. Hence for every $p < 1$ one can choose $M = M(p)$ with $\mathbb{P}[G_{t,m}(z)] \ge p$, uniformly in the stated centers and cutoffs.

Choose L for Lemma 4.2 with annular radii $1/2$ and 5. Its Gaussian Hölder bound and the associated Chernoff estimate yield

$$\mathbb{P}\left[\sum_{j=1}^{k} \mathbf{1}_{G_{t,m_j}(z)} < k/2\right] \le \exp\{-kI_p(1/2)/P_0\}\,.$$

Since $I_p(1/2) \to \infty$ as $p \uparrow 1$, choose p so that $I_p(1/2)/P_0 \ge B$ and put $M_B = M(p)$. □

Corollary 9.2 (Simultaneous separating connectors)**.** *In any joint subsequential limit of the normalized rational annulus crossings and internal separator metrics, almost surely the following holds for every $z \in \mathbb{R}^d$. There are infinitely many pairwise disjoint, nested rational cubical annuli surrounding z, with diameters tending to zero and with positive inner buffer around z, whose limiting crossings C_j and middle-separator diameters H_j satisfy*

$$H_j \le MC_j \tag{9.2}$$

for one deterministic $M < \infty$.

Proof. Fix a bounded cube of possible centers. For an integer a, use centers on $2^{-2a-4}\mathbb{Z}^d$ and scales in $[a, 2a]$ separated by L. There are at most $C_R 2^{2da}$ centers and at least $c_L a$ scales per center for large a. Choose B in Lemma 9.1 sufficiently large that the union-bound failure probability is summable in a.

For each fixed a only finitely many scalar crossings and diameters are involved. The event that at least half the inequalities $H \le M_B C$ hold is closed in this finite array. Portmanteau therefore transfers the probability estimate to the joint limit. Borel–Cantelli implies that for all sufficiently large a, every grid center has a good annulus among the chosen scales.

Given z, take a nearest grid center. Its distance from z is less than 2^{-2a-4}, so each annulus under consideration surrounds z with a definite inner buffer. Take $a = 3^j$. Consecutive scale ranges are widely separated, and the errors in the chosen centers are much smaller than the relevant inner radii. Thus the selected annuli are nested and have disjoint closures for all sufficiently large j. A countable exhaustion of space proves the simultaneous assertion. Fixing the preceding B fixes $M = M_B$. □

10 Construction of the extended metric

We now work with the prescribed metrics D_h^ε and their medians a_ε. All restrictions and crossings in this section are divided by a_ε before taking limits.

Proof of Theorem 1.1. Let $\mathcal{Q}$ be the countable collection of cubes with rational centers and positive rational radii. From any sequence $\varepsilon \to 0$, extract a subsequence along which the following arrays converge jointly: distances between all pairs of boundaries in $\mathcal{Q}$; unrestricted surface metrics and internal surface metrics in rational polycubical domains; rational point distances; unrestricted metric restrictions to S_z^* for rational z; and all the deterministic normalization ratios in part (iv). Include the fields h, $(H_m)_{m\ge 0}$, g, C, and their convolution processes. These field coordinates have a fixed joint law. They can be realized in their usual Polish spaces of local continuous functions and distributions, using, for example, $H_{\mathrm{loc}}^{-(d+1)}$ for h.

Lemma 8.2 gives tightness of each metric coordinate and positive finite bounds for each fixed normalization ratio. Countable-product tightness gives a joint extraction. Use a Skorokhod

representation. The prelimit metric arrays retain their defining relations with the prelimit field copies, so the full prelimit path metrics and paths remain available in this representation. Write

$$\mathring{D}(S,T) = \lim_{\varepsilon\to 0} a_\varepsilon^{-1} D_h^\varepsilon(S,T)$$

for the extracted boundary-set distances. All limits in this proof are along the extracted subsequence.

Every fixed rational annulus crossing is positive almost surely by Lemma 8.2. This holds simultaneously for the countable collection. More generally, $\mathring{D}(S,T) > 0$ for disjoint rational cube boundaries: cover S by finitely many small rational inner cubes whose doubled cubes avoid T. Every path from S to T crosses one of the corresponding annuli. The minimum of their positive limiting crossings is a lower bound. All these boundary distances are finite, since fixed point connections give uniform upper bounds.

For $x \neq y$, define

$$D(x,y) = \lim_{m\to\infty} \mathring{D}(\partial Q_m(x), \partial Q_m(y)), \qquad D(x,x) = 0, \tag{10.1}$$

where Q_m is the nested cube family in Corollary 5.3. Once the two cubes are disjoint, the terms are nondecreasing: every path between smaller nested boundaries crosses the larger ones. The limit exists in $[0,\infty]$. Equivalently, $D(x,y)$ is the supremum of $\mathring{D}(\partial U, \partial V)$ over all disjoint rational cube neighborhoods $U \ni x$, $V \ni y$. This equivalence follows by nesting sufficiently small Q_m inside any such pair.

The supremum description proves that D is a measurable lower semicontinuous random function: for each fixed pair U, V, the corresponding nonnegative constant on the open set $U \times V$, extended by zero elsewhere, is lower semicontinuous, and the supremum is countable. Symmetry is immediate. A fixed rational annulus separating any two distinct points proves $D(x,y) > 0$, simultaneously for all $x \neq y$.

Lower-semicontinuous convergence. Suppose $x_\varepsilon \to x$, $y_\varepsilon \to y$ with $x \neq y$. For any fixed pair of disjoint rational cube neighborhoods of x, y, the endpoints eventually lie inside them. The approximating point distance is at least their boundary distance. Passing to the limit and taking the supremum gives

$$\liminf_{\varepsilon\to 0} a_\varepsilon^{-1} D_h^\varepsilon(x_\varepsilon, y_\varepsilon) \geq D(x,y). \tag{10.2}$$

For $x = y$ the inequality follows from nonnegativity.

Conversely, for each fixed m choose endpoints on $\partial Q_m(x), \partial Q_m(y)$ realizing, or arbitrarily closely approximating, their set distance. Let m tend to infinity sufficiently slowly along the cutoff sequence that the fixed-boundary limits have been approximated. These endpoints converge to x, y and their normalized distances converge to (10.1), also when that value is infinite. On the diagonal, take identical endpoints. These are exactly the two defining properties of the lower-semicontinuous topology. They also justify measurability in that function space: the prelimit continuous distance functions are measurable (for fixed endpoints, their distances are infima over polygonal paths with rational intermediate vertices), and the pathwise limit in this metrizable topology is measurable. The countable supremum description above separately gives joint measurability of $(\omega, x, y) \mapsto D(x,y)$.

The all-point triangle inequality. It remains to treat distinct points with $D(x,z) + D(z,y) < \infty$. Apply Corollary 9.2 at z, omitting finitely many annuli so that none contains x or y. Every

path from a sufficiently small neighborhood of z to a sufficiently small neighborhood of x crosses any fixed finite collection of these disjoint annuli. Recovery pairs for (x, z) and the prelimit length inequality imply

$$\sum_j C_j \le D(x, z) < \infty. \tag{10.3}$$

Thus $H_j \le MC_j \to 0$.

Fix one such annulus. Choose recovery pairs and near-minimizing prelimit paths for (x, z) and (z, y). Their two endpoints near z need not agree. Both paths cross the middle separator. Truncate them there and connect their hit points inside the annulus, at cost at most its internal separator diameter plus an arbitrarily small error. Inequality (10.2) at the two outer endpoints gives

$$D(x, y) \le D(x, z) + D(z, y) + H_j.$$

Let $j \to \infty$. This proves the triangle inequality without any continuity assumption at z. Hence D is a separating extended metric.

Rational-point consistency. Let $\overline{D}(u, v)$ denote the extracted limit of the approximating distance at rational u, v. Inequality (10.2) gives $D(u, v) \le \overline{D}(u, v)$. On S_u^* the normalized unrestricted metrics converge uniformly to a continuous pseudometric. Since $u \in S_u^*$ and the cube boundaries shrink to u,

$$\lim_{m\to\infty} \lim_{\varepsilon\to 0} \sup_{x\in\partial B_{2^{-m}}(u)} a_\varepsilon^{-1} D_h^\varepsilon(u, x) = 0.$$

The same holds at v. The prelimit triangle inequality then gives

$$\overline{D}(u, v) \le \mathring{D}(\partial B_{2^{-m}}(u), \partial B_{2^{-m}}(v)) + o_m(1).$$

Shrinking these boundaries proves the reverse inequality. Thus all rational-point evaluations agree with their extracted limits, jointly with the lower-semicontinuous limit.

Normalizer ratios and arbitrary subsequential limits. For each fixed rational $r > 0$, Lemma 8.2 bounds the ratio $ra_{\varepsilon/r}/a_\varepsilon$ above and below by positive constants for all sufficiently small ε. After the joint extraction, its limit c_r therefore lies in $(0, \infty)$. More precisely, for every $\eta > 0$,

$$c_\eta r^{\psi Q+\eta} \le c_r \le C_\eta r^{\psi Q-\eta} \quad (0 < r \le 1),$$

and the signs of η reverse when $r \ge 1$. Dividing logarithms by $\log r$, then letting r tend to zero or infinity and $\eta \downarrow 0$, proves part (iv).

Finally, start from any subsequence already converging in law in the lower-semicontinuous topology. Further extract the additional tight arrays just used. The same construction proves convergence to an extended metric in that topology. Uniqueness of the limit in its metrizable Hausdorff realization identifies the original limiting law with this one. Consequently the metric assertion concerns every possible subsequential limit, not only the particular extraction constructed above. The retained field coordinates give the claimed joint coupling with h. □

11 Geometry and singular points

Work in the limiting coupling of Theorem 1.1. We use the sharp approximations $\widetilde{H}_m$ from Section 2; Lemma 3.1 identifies their thickness sets with those defined by spherical or smooth averages.

Lemma 11.1 (Uniform field-dependent barriers)**.** *Almost surely, for every $R < \infty$ and $\eta > 0$, for all sufficiently large integers m, all $z \in 2^{-m}\mathbb{Z}^d \cap B_R$, and all $x \in B_{2^{-m}}(z)$,*

$$\mathring{D}(\partial B_{2^{-m}}(z), \partial B_{4\cdot 2^{-m}}(z)) \geq 2^{-(\psi Q+\eta)m} e^{\psi \widetilde{H}_m(x)}. \tag{11.1}$$

Also, almost surely for all sufficiently large m,

$$\inf_{B_R} \widetilde{H}_m \geq -(\sqrt{2d}+\eta)m \log 2. \tag{11.2}$$

Proof. Fix a small $\delta > 0$. For $t \geq m$, the scale decomposition, the lower ratio bound, and (8.3) give

$$\begin{aligned} &a_{2^{-t}}^{-1} D_h^{2^{-t}}(\partial B_{2^{-m}}(z), \partial B_{4\cdot 2^{-m}}(z)) \\ &\qquad \geq c_\delta 2^{-(\psi Q+\delta)m} \exp\left(\psi \inf_{B_{4\cdot 2^{-m}}(z)} [(H_m + g) * \rho_{2^{-t}} - C]\right) Y_{t,m,z}, \end{aligned}$$

where $Y_{t,m,z}$ has the law of a normalized fixed-annulus crossing for the short-range convolution field at cutoff $t-m$. No independence between $Y_{t,m,z}$ and the coarse expression is asserted. Lemmas 8.1 and 8.2 imply

$$\mathbb{P}[Y_{t,m,z} < 2^{-\delta m}] \leq C \exp(-c_\delta m^2 / \log(m+2)).$$

The union bound over $O_R(2^{dm})$ centers is summable in m, uniformly in $t \geq m$.

The oscillation of H_m on a cube of radius $C2^{-m}$ has bounded variance and bounded expected supremum, uniformly in its center and m. Gaussian concentration and a union bound over the same grid show that, except on an event of probability at most $C_R e^{-c_\delta m^2}$,

$$\sup_{x \in B_{2^{-m}}(z)} \widetilde{H}_m(x) - \inf_{y \in B_{4\cdot 2^{-m}}(z)} [(H_m + g) * \rho_{2^{-t}}(y) - C] \leq \delta m \log 2$$

for all centers, uniformly in $t \geq m$. Here the support of ρ only enlarges the cutoff-size cube by a fixed factor. The compact-set Gaussian bound on g handles its contribution, and the constant C cancels.

For each fixed m the resulting inequalities concern finitely many crossing variables and continuous coarse fields. Their success event is closed, since it can be expressed using the supremum of $\widetilde{H}_m$ on each inner cube. Pass the estimates to the joint limit. Their failure bounds are summable, so Borel–Cantelli applies. Choose δ small relative to η and ψ, and absorb the fixed factor c_δ into the arbitrarily small exponent loss. This proves (11.1) simultaneously for all x in the indicated cubes.

For (11.2), cover B_R by $O_R(2^{dm})$ cutoff-size cubes. At their centers, use Gaussian tails with variance $m \log 2$ and threshold $(\sqrt{2d} + \eta/2)m \log 2$. The resulting failure probabilities are summable. The maximal within-cell oscillation is $o(m)$ almost surely by the same bounded-variance concentration estimate, and $g - C$ is bounded on B_R almost surely. This proves (11.2). Countable choices of R and of arbitrarily small positive losses give the asserted quantifiers. □

Lemma 11.2 (Pinned-field scaling and large annuli)**.** *Let σ_r be normalized surface measure on the Euclidean sphere of radius r. The field*

$$h^{(r)}(x) = h(rx) - (h, \sigma_r)$$

has the law of h, and

$$D_h^\varepsilon(rx, ry) = r e^{\psi(h,\sigma_r)} D_{h^{(r)}}^{\varepsilon/r}(x, y). \tag{11.3}$$

The scalar and the field on the right need not be independent. In every limiting coupling under consideration,

$$\mathring{D}(\partial B_{2^j}, \partial B_{2^{j+1}}) \longrightarrow \infty \qquad (j \to \infty) \tag{11.4}$$

almost surely.

Proof. The logarithmic covariance on zero-mass tests is invariant under dilation, since the extra constant $-\log r$ vanishes on those tests. Subtracting (h, σ_r) imposes unit-sphere average zero after the dilation. This proves the equality in law. The convolution identity

$$(h * \rho_\varepsilon)(r\,\cdot) = h^{(r)} * \rho_{\varepsilon/r} + (h, \sigma_r)$$

and a change of variables along paths prove (11.3).

For completeness, put

$$L_d(r) = \iint \log|ru - v|_2 \, \sigma(du)\sigma(dv).$$

The pinned covariance gives

$$\mathrm{Var}(h, \sigma_r) = -\log r - 2L_d(1) + 2L_d(r). \tag{11.5}$$

Since $L_d(r) = \log r + o(1)$ as $r \to \infty$ and $L_d(r) = o(1)$ as $r \downarrow 0$, this variance is $|\log r| + O_d(1)$ at either end. In dimensions above two it is not asserted to equal $|\log r|$ exactly.

Take $r = 2^j \geq 1$. By (11.3), the normalized crossing of $B_{2r} \setminus \overline{B_r}$ has the law of

$$\frac{r a_{\varepsilon/r}}{a_\varepsilon} e^{\psi(h,\sigma_r)} Y_{\varepsilon/r},$$

where $Y_{\varepsilon/r}$ has the law of a normalized unit-annulus crossing for the pinned field. The real-cutoff ratio bound makes the deterministic factor at least $c_\delta r^{\psi Q - \delta}$. Gaussian tails from (11.5) and the uniform lower tail for Y show that the whole expression is at least $r^{\psi Q/2}$, except with probability at most Ce^{-cj}, uniformly in sufficiently small ε. For example, take $\delta = \psi Q/8$ and intersect the events

$$(h, \sigma_r) \geq -\tfrac{1}{8} Q \log r, \qquad Y_{\varepsilon/r} \geq r^{-\psi Q/8}.$$

The fixed constant is absorbed for large r. A union bound suffices; no independence is required.

For each fixed j, pass this closed lower-bound event to the subsequential crossing limit. Borel–Cantelli then gives (11.4), since $Q > 0$ in the stated regime. □

Proof of Theorem 1.2. Work on the probability-one event on which the countable limits, separating connectors, barriers, and large-annulus bounds above hold.

Finiteness and the rational surfaces. For every deterministic pair x, y, the normalized approximating point distances are tight by Lemma 8.2. Inequality (10.2) therefore gives $D(x, y) < \infty$ almost surely. Explicitly, for each T, the event $D(x, y) > T$ forces those approximating distances eventually to exceed T, so its probability is at most their uniform upper-tail bound. Let $T \to \infty$. Fubini proves finiteness for Lebesgue-almost every pair. Applying the same argument with one endpoint fixed at 0 gives

$$\mathrm{Leb}\{x : D(0, x) = \infty\} = 0. \tag{11.6}$$

Every point of a rational coordinate plaque has finite distance to a rational point on that plaque. Indeed, the lower-liminf property bounds its distance above by the finite continuous limit of the

restricted metrics. Rational points have finite mutual distances by Theorem 1.1. Thus every point of the deterministic union Σ lies in one finite-distance component. One can also identify D with the continuous unrestricted limit on each plaque: approximate its two points by rational points and use rational-point consistency, the triangle inequality, and the continuous upper bound. We do not identify an internal-domain limit with the intrinsic internal metric of D.

Euclidean boundedness and the inverse Hölder bound. Given any x, paths from a small neighborhood of x to points tending to Euclidean infinity must cross each sufficiently large annulus centered at zero. Lemma 11.2 and the boundary-distance construction imply that their D-distances tend to infinity. Every D-bounded set is therefore Euclidean-bounded.

Let $a > \psi(Q + \sqrt{2d})$. Combining (11.1) and (11.2), with sufficiently small losses, shows that all sufficiently small grid-annulus crossings in a fixed bounded region are at least 2^{-am}. For distinct nearby x, y, choose m such that

$$8 \cdot 2^{-m} \le |x - y|_\infty < 16 \cdot 2^{-m},$$

and take a nearest grid center to x. The annulus with inner radius 2^{-m} and outer radius $4 \cdot 2^{-m}$ separates the two points. Hence

$$D(x, y) \ge c|x - y|_2^a.$$

For pairs at a fixed positive separation in a compact region, a finite cover by separating annuli gives a positive uniform lower bound. This proves the inverse Hölder assertion with exponent $1/a$. Countably many choices of a and a compact exhaustion give every exponent below the one in the theorem, simultaneously.

One nonsingular component. Suppose $D(x, y) < \infty$ for some $y \neq x$. Choose a rational cube boundary separating them. Recovery endpoints and near-minimizing prelimit paths have first-hit points on this boundary. A subsequence of those points converges to z on the boundary, and (10.2) gives

$$D(x, z) \le D(x, y) < \infty.$$

Since $z \in \Sigma$, it belongs to the common rational component. The triangle inequality puts x in that component as well. All nonsingular points consequently have finite mutual distances. Their set is exactly $\{x : D(x, 0) < \infty\}$, because 0 is nonsingular and the rational component is nontrivial. This set and its complement are Borel. Equation (11.6) shows that the singular set is Lebesgue-null.

Completeness. A D-Cauchy sequence of nonsingular points is D-bounded and hence Euclidean-bounded. The inverse Hölder estimate makes it Euclidean-Cauchy. Let x be its Euclidean limit. Lower semicontinuity gives

$$D(x, x_n) \le \liminf_{k \to \infty} D(x_k, x_n) \longrightarrow 0.$$

The limit is nonsingular: unless the sequence is eventually equal to x, it has finite distance to a distinct nonsingular point. In the eventual-equality case this is already known. Thus the nonsingular component is complete.

Geodesics. For distinct nonsingular x, y, choose recovery endpoints and prelimit paths whose normalized lengths tend to $D(x, y)$. Their images lie in a fixed Euclidean compact set for all sufficiently small cutoffs. Otherwise one such path would cross a fixed sufficiently large annulus whose limiting crossing exceeds $D(x, y) + 1$, contradicting its length bound. Parameterize these paths on $[0, 1]$ with constant normalized length speed.

They are Euclidean-equicontinuous. To verify this without assuming a prelimit estimate below the cutoff, fix a Euclidean separation $\rho > 0$. A finite family of small rational annuli covers the compact region in the following sense: any two points in it at separation at least ρ are separated by one of these annuli. Their limiting crossings have a positive minimum. For all sufficiently small cutoffs, the normalized distance between any such pair is therefore bounded below by a positive constant. The common speed bound gives the required modulus of continuity.

Arzelà–Ascoli gives a uniformly Euclidean-convergent subsequence with limit P. For every $s, t \in [0, 1]$, the lower-liminf property yields

$$D(P(s), P(t)) \le D(x, y)|t - s|.$$

The triangle inequality between the endpoints forces equality for every s, t. Thus P is a D-geodesic. Its points have finite distance to x, so it stays in the nonsingular component. The case $x = y$ is the constant path.

Thick points and the discontinuous inverse. By Lemma 3.1, the spherical thickness condition is equivalent to

$$\liminf_{m\to\infty} \frac{\widetilde{H}_m(x)}{m \log 2} \ge \alpha.$$

If $\alpha > Q$, choose $\eta > 0$ and $\nu > 0$ with $\eta < \psi(\alpha - Q - \nu)$. The lower-thickness condition gives $\widetilde{H}_m(x) \ge (\alpha - \nu)m \log 2$ eventually. The barrier (11.1), with loss η at a nearest dyadic grid center to x, makes its surrounding small-annulus crossings tend to infinity. Every fixed $y \ne x$ is outside these annuli for large m. Hence $D(x, y) = \infty$. The barrier estimates are simultaneous at all centers and points, so this conclusion holds for all thick points, not merely deterministic ones.

If $Q < \sqrt{2d}$, choose $\alpha \in (Q, \sqrt{2d})$. Lemma 3.2 gives an exact-α thick set of positive Hausdorff dimension in every nonempty open ball. These points are singular. A countable basis of balls makes the singular set dense and uncountable almost surely. Distinct singular points approach every Euclidean point, while their distance from that point is infinite. Thus the identity from Euclidean space to the extended metric is not continuous. This completes all six assertions. □

11.1 Estimates retained in the construction

A coordinate complex is a finite connected union of compact coordinate plaques of dimension $d - 1$ and cubical boundaries. Let Σ be the union of all compact rational coordinate plaques. There are only countably many rational complexes and rational polycubical buffers.

Proposition 11.3 (Joint construction estimates)**.** *Under* (1.1)*, the construction has the following additional properties.*

(T1) Every cutoff sequence has a further joint extraction of the field, global lower-semicontinuous distances, global restrictions and buffered internal arrays on the countable rational complexes, and rational annulus crossings. The global limit is a separating extended length metric. Its nontrivial finite-distance component is complete and geodesic, its metric-bounded sets are Euclidean bounded, and each fixed deterministic point is nonsingular almost surely.

(T2) If $S \Subset U$ is a fixed coordinate complex, the normalized internal restrictions on S^2 have tight Hölder seminorms for every

$$0 < \beta < \min\{1, \psi(Q - \sqrt{2(d-1)})\}. \tag{11.7}$$

The estimates are uniform in marginal probability over families contained in a fixed bounded region, with a bounded number of plaques, bounded diameters, positive lower bounds on nonzero plaque side lengths and buffer clearances, and connecting coordinate chains of uniformly bounded length and positive clearance.

(T3) Fixed normalized annulus crossings and their reciprocals are tight. Their lower tails satisfy

$$\mathbb{P}[X < T^{-1}] \le C \exp\left\{-c\frac{(\log T)^2}{\log\log T}\right\}, \qquad T \ge e^e, \tag{11.8}$$

uniformly in the cutoff, with constants depending on the annular template.

(T4) For every $\eta > 0$ there is $C_\eta < \infty$ such that, for $0 < \varepsilon \le r \le R$,

$$C_\eta^{-1}(r/R)^{\psi Q+\eta} \le \frac{r a_{\varepsilon/r}}{R a_{\varepsilon/R}} \le C_\eta (r/R)^{\psi Q-\eta}. \tag{11.9}$$

(T5) In every joint extraction retaining the arrays in (T1), simultaneously at every point there are shrinking nested rational separating annuli with disjoint closures for which the extracted internal middle-separator diameter is at most a deterministic multiple of the extracted crossing. Finite limiting paths, parametrized continuously for the limiting metric, are Euclidean continuous.

The internal arrays in (T2) and (T5) are extracted limits. Their comparison with actual internal metrics is proved in Section 12.

Proof. Lemma 5.2, including its field-dependent estimate (5.5), and Proposition 7.3 give the surface estimates. The crossing lower tail is Lemma 8.1; the ratio bound follows from Lemma 8.2. Corollary 9.2 supplies the simultaneous separators. Theorems 1.1 and 1.2, together with the fixed-point access argument in their proofs, give the metric and path conclusions. The proof in Section 11 identifies the retained global surface restrictions with D, so each retained global annulus crossing is its actual set-to-set distance. Finite pairs admit geodesics; constant paths handle the diagonal, and a path between points at infinite distance has infinite length. Thus D is an extended length metric. All arrays form a countable family and can be retained in the same extraction. □

12 Compact paths and intrinsic metrics

The arguments below distinguish three objects: a global metric, its actual intrinsic metric in an open set, and a subsequential array extracted from prelimit intrinsic distances. Only the first two are metrics defined by the limiting length structure. The arrays are used through a buffered inequality and an exhaustion identity.

Lemma 12.1 (Compact separation)**.** *Let D be a separating Euclidean lower-semicontinuous extended metric. If K_1, K_2 are disjoint Euclidean compact sets, then $D(K_1, K_2) > 0$. Consequently, for compact K and $a > 0$,*

$$\inf\{D(x,y) : x, y \in K,\ |x-y|_2 \ge a\} > 0. \tag{12.1}$$

Proof. If the first infimum were zero, compactness would give a convergent sequence of endpoint pairs with distances tending to zero. Lower semicontinuity would give two distinct limiting points at distance zero. Apply the same argument to the compact set of pairs in (12.1). □

Lemma 12.2 (Compact constraints and a stable selector)**.** *For a compact set K and $x, y \in K$, the compactly constrained metric satisfies*

$$D^K(x,y) = \sup_{m\geq 1} \inf \left\{ \sum_{j=0}^{N-1} D(x_j, x_{j+1}) : \begin{array}{l} x_0 = x,\ x_N = y,\ x_j \in K,\ N < \infty, \\ \max_j D(x_j, x_{j+1}) \leq m^{-1} \end{array} \right\}. \tag{12.2}$$

It is lower semicontinuous in its endpoints. Every finite point-to-point or compact-set-to-compact-set constrained distance is attained. The distance and a selected minimizing path, parametrized on $[0,1]$ at constant speed, are Borel functions of the lower-semicontinuous metric and the endpoint data.

The selection can be made so that multiplying the metric by a positive constant does not change the geometric path. It can also be made stable in this sense: if $D' \geq D$ and the selected D^K-minimizer retains its old length in D', then it is the selected $(D')^K$-minimizer.

Proof. Subdividing a finite rectifiable path into links of length at most m^{-1} proves that the right side of (12.2) is at most the left side. Conversely, suppose the right side is $L < \infty$. Choose, for each m, a chain with cost at most $L + m^{-1}$. Parametrize its vertices by accumulated cost, use a step path between consecutive vertex times, and extend constantly to $[0, L+1]$. Its increments satisfy

$$D(P_m(s), P_m(t)) \leq |s-t| + 2m^{-1}. \tag{12.3}$$

The compact inverse modulus (12.1) gives asymptotic Euclidean equicontinuity. A diagonal extraction at a dense set of times gives a uniformly Euclidean-convergent subsequence in K. Lower semicontinuity in (12.3) makes the limit D-1-Lipschitz. Its duration is at most L, giving the reverse inequality and attainment. Applying this construction to varying convergent endpoints proves lower semicontinuity. Minimizing over compact endpoint sets then proves the set-to-set assertion.

Here and below a constrained minimizing path is geodesic for D^K. Its D-length on a subinterval equals the corresponding D^K-distance, but its ambient D-distance need not equal that length. This distinction is needed when a path has a cheaper competitor outside K.

For each fixed finite chain length, the sublevel sections of the optimization problem in its intermediate vertices are compact. Joint evaluation of a lower-semicontinuous function is Borel. The measurable projection theorem for Borel sets with sigma-compact sections [7, Theorem 18.18] therefore makes each finite-chain value Borel. Countable infima and suprema give the Borel assertion for (12.2).

For fixed endpoint sets, let L be the finite constrained minimum. A minimizing path is specified in $C([0,1], K)$ by its endpoint membership and

$$D(P(s), P(t)) \leq L|s-t|, \qquad s, t \in \mathbb{Q} \cap [0,1]. \tag{12.4}$$

Lower semicontinuity extends these inequalities to all times. They imply length at most L, and the endpoint minimum forces total length L. Thus the path is a constant-speed constrained minimizer. The sections of this Borel family are compact, by (12.1) and Arzelà–Ascoli. Successively minimize one fixed countable family of continuous functions separating paths. Each minimum is Borel by the same measurable projection theorem; the decreasing compact fibers have a unique intersection. This gives a Borel lexicographic selector.

If $D' \geq D$ and the old selected path retains length L, both constrained endpoint minima are L. Every new minimizer is an old minimizer, and the old selected path belongs to the smaller family. The lexicographic selector therefore keeps it. Domination of the two length measures and equality of their total masses give equality on every subpath, so their normalized constant-speed parametrizations agree. Positive scalar multiplication preserves the same parametrization and the same lexicographic choice. □

If $E \leq CD$, partition sums give $\mathrm{len}_E(P) \leq C\,\mathrm{len}_D(P)$ on every finite D-path. Taking path infima gives the same comparison for all actual intrinsic metrics and compact constraints. This observation never requires a finite D-length for a path known only to be E-rectifiable.

Lemma 12.3 (Locality of compact constraints)**.** *Let D be a separating Euclidean lower-semicontinuous length metric whose finite rectifiable paths are Euclidean continuous, and suppose D has the locality axiom in Definition 1.3. For deterministic $K \Subset U$, with K compact and U open, the metric D^K and its selected finite minimizing paths are measurable in the completed local field sigma field of U. No lower semicontinuity of $D(\cdot,\cdot;U)$ is assumed.*

Proof. Choose bounded open V with $K \Subset V \Subset U$. The global barrier $a = D(K, \partial V)$ is positive. If $x, y \in K$ and $D(x,y) < a$, sufficiently close almost-minimizers cannot leave V, so $D(x,y;U) = D(x,y)$. If instead $D(x,y;U) < a$, its global distance is below a and the same equality holds. Hence, for all sufficiently large m, the short-link chains in (12.2) are unchanged when D is replaced by $D(\cdot,\cdot;U)$. The chain formula reconstructs D^K from that local intrinsic metric. The random barrier is used to prove equality, not as an input to the reconstruction.

For a precise measurability argument, condition on the field in U and take two independent copies of the remaining field and assigned metric. Their actual intrinsic metrics in U agree almost surely by locality; the reconstruction makes their compactly constrained metrics agree. Lemma 12.2 already makes D^K a Borel random element in a standard Borel space. Two independent conditional samples can agree almost surely only when their conditional law is a point mass. Thus D^K is local. Its fixed Borel selector is local as well. □

Lemma 12.4 (Routing through separating boundaries)**.** *Let A, B be extended length metrics with Euclidean-continuous finite paths, and let P be a finite A-path. Suppose its image is covered by the inner halves of good cubes, of radii in $[r_-, r_+]$ with $r_- > 0$, such that*

$$\mathrm{diam}_{B(\cdot,\cdot;U_{z,r})}\, S_r(z) \leq M\, A(S_{r/2}(z), S_r(z)). \tag{12.5}$$

Here $U_{z,r}$ is an annular buffer contained in a fixed multiple of $B_r(z)$. Then a B-path joins points at Euclidean distance $O(r_+)$ from the endpoints of P, with length at most $M\,\mathrm{len}_A(P)$ plus arbitrary excess. The new pieces stay in the indicated buffers.

Proof. Choose a good cube whose inner half contains the initial point. If it contains the terminal point, stop. Otherwise follow P to its first exit from the cube, charge that time interval, and repeat from the exit point. Each charged interval pays at least the crossing on the right of (12.5). The intervals are disjoint in time. There are finitely many steps: every charged step has Euclidean displacement at least $r_-/2$, and P is uniformly Euclidean continuous.

The chosen open cubes have a connected intersection graph from the initial cube to the uncharged terminal cube. Remove duplicates and cubes contained in others. The remaining maximal cubes cover the same union. Two intersecting nonnested cubes have intersecting boundaries. Take a simple graph path from a maximal cube containing the initial point to one containing the terminal point. Stop if the uncharged terminal cube is reached, so it is not an intermediate cube. Every

intermediate cube is charged. Join successive boundary-intersection points by its B-separator connection. Each such separator is used at most once; its cost is charged to one of the original disjoint A-intervals.

The first and last intersection points are within $O(r_+)$ of the original endpoints. If one maximal cube contains both endpoints, use a constant path instead. Approximate the finitely many intrinsic infima with errors of arbitrarily small total. This proves the claim without requiring intersections between old and new paths. □

12.1 Buffered domination comes before intrinsic identification

An enhanced convergence will mean global convergence in (1.8), uniform convergence of restrictions on the retained rational coordinate complexes, and convergence of the retained rational crossings and internal arrays. We also require Euclidean equicontinuity for prelimit paths in a fixed compact set, parametrized on a common compact interval at uniformly bounded metric speed. Positive limiting annular crossings imply this last property: a finite family of annuli separates any two points of that compact set at a prescribed positive Euclidean distance, and the speed bound gives a positive lower bound on the time needed to cross one.

Lemma 12.5 (Confinement from the limiting metric)**.** *Suppose $D_n \to D$ in (1.8), every D-bounded set is Euclidean bounded, and the prelimit paths under consideration are Euclidean continuous. Fix a Euclidean compact set K and $L < \infty$. For all sufficiently large n, every such path starting in K with D_n-length at most L is contained in one common Euclidean compact set.*

Proof. Otherwise choose an increasing sequence n_j and paths of length at most L, starting at $x_j \in K$, whose images reach outside $B_j(0)$. Pass to a subsequence with $x_j \to x \in K$. For every sufficiently large integer R, Euclidean continuity gives a point $z_{j,R} \in S_R(0)$ on each sufficiently late path. A diagonal extraction makes $z_{j,R} \to z_R \in S_R(0)$ for every such R. Since $D_{n_j}(x_j, z_{j,R}) \le L$, the lower-limit inequality gives $D(x, z_R) \le L$. The points z_R form an unbounded subset of one D-ball, a contradiction. □

Lemma 12.6 (Buffered surface domination)**.** *Suppose $D_n \to D$ in an enhanced convergence, and an internal array L^V is extracted on S^2, where $S \Subset V$ and $\overline{V} \Subset U$. Then*

$$D(x, y; U) \le L^V(x, y), \qquad x, y \in S. \tag{12.6}$$

In particular, extracted internal diameter bounds become actual intrinsic diameter bounds in a slightly larger open buffer.

Proof. The assertion is vacuous when the right side is infinite. Otherwise choose prelimit paths in V from x to y with lengths tending to $L^V(x, y)$. Their images lie in the fixed compact set $\overline{V}$. Parametrize at bounded constant speed. Positive limiting annular barriers give Euclidean equicontinuity, so a subsequence converges uniformly to a path in $\overline{V} \subset U$. For every finite partition, the lower-limit inequality bounds the sum of limiting endpoint distances by the liminf prelimit length. Taking the supremum over partitions proves (12.6). □

For the input (T5), enlarge each annular buffer by a fixed factor before applying Lemma 12.6. A subsequence of the shrinking shells has pairwise disjoint enlarged closures. Their bounded geometric eccentricities and positive inner buffers permit this thinning at every point. Thus (T5) gives actual intrinsic separator diameters in the enlarged buffers. This use of compact path limits precedes, and does not invoke, the intrinsic reconstruction below.

Lemma 12.7 (Small separators and a rational core)**.** *Let E be a separating lower-semicontinuous length metric with Euclidean-continuous finite paths, finite continuous global restrictions on rational coordinate complexes, and finite actual intrinsic restrictions in their open buffers. Suppose every point has nested disjoint shrinking rational shells, with middle-separator intrinsic diameters H_j and crossing costs X_j, satisfying $H_j \le \Lambda X_j$.*

Every point lying on a nonconstant finite path has arbitrarily small rational separators with vanishing intrinsic diameter and vanishing access from that point. These connections can be confined to any prescribed open neighborhood. Rational points are metrically dense in the unique nontrivial finite-distance component. Its intersection with every connected open set is rectifiably connected.

Proof. Start a finite nonconstant path at x, removing an initial pause. It crosses every sufficiently small shell about x. Disjoint shells can be charged to disjoint path intervals, so $\sum_j X_j < \infty$ and $H_j \to 0$. The path length up to its first middle-separator contact tends to zero. Indeed, any sufficiently early nonconstant time has image different from x and eventually outside the shrinking separator. The first contact occurs before that time.

Follow this prefix and then a separator connection. The resulting access cost from x to every point of the separator tends to zero, and the entire connection lies in a shrinking neighborhood. A rational point on the rational separator gives local metric approximation by rational points.

Any nontrivial finite component contains a finite path crossing a rational cube boundary. That boundary has finite diameter and contains rational points. All rational points have finite mutual distance through a fixed finite coordinate complex, so the nontrivial component is unique. Inside a connected open U, first join each nonsingular endpoint to a nearby rational point by a small separator. An axis-parallel polygonal chain between the two rational points has a compact neighborhood in U. A finite connected family of coordinate plaques containing the chain has finite actual intrinsic diameter in that neighborhood. Concatenation proves the local connectivity assertion. □

Lemma 12.8 (Buffered intrinsic reconstruction)**.** *In an enhanced convergence, suppose D has the properties of Lemma 12.7. For every open U and $x, y \in \Sigma \cap U$,*

$$D(x, y; U) = \inf_{V \Subset U} L^V(x, y) = \inf_{V \Subset U} \limsup_n D_n(x, y; V), \tag{12.7}$$

where V ranges through a fixed countable family of bounded connected rational polycubical domains containing both endpoints. An empty infimum is infinite. The actual intrinsic metric is lower semicontinuous on U^2, and, for distinct $x, y \in U$,

$$D(x, y; U) = \lim_{\delta \downarrow 0} \inf_{\substack{p,q \in \mathbb{Q}^d \cap U \\ |p-x|_2 < \delta,\ |q-y|_2 < \delta}} D(p, q; U). \tag{12.8}$$

The diagonal value is zero.

Proof. Lemma 12.6 gives the lower bound in (12.7). To prove the upper bound, take a finite D-path P in U between the specified endpoints. Its Euclidean image is compactly contained in U. Cover it by finitely many small rational balls with larger buffers still contained in one rational $V \Subset U$. A finite family of annuli separating the small balls from the complements of their buffers has a positive minimum limiting crossing cost.

Partition P into D-short pieces lying in the small balls and shorter than half that minimum. Interior partition vertices may be chosen in Σ. To see this, rational coordinate hyperplanes are met

densely in the parameter of every nonconstant Euclidean-continuous path segment: two distinct image points have different values in some coordinate, and the intermediate value theorem gives a rational coordinate value between them. Metric continuity permits the partition times to be adjusted by arbitrarily small length. Constant intervals have zero length and may be omitted.

All finitely many chosen vertices lie on one of the retained finite rational complexes, after adjoining connecting plaques. Uniform convergence on that complex gives convergence of the global D_n-distances between consecutive vertices. A sufficiently close prelimit minimizing link cannot leave its buffer, because its cost is below the separating annulus barrier. Concatenating these links in V gives

$$\limsup_n D_n(x, y; V) \le \mathrm{len}_D(P). \tag{12.9}$$

Take the infimum over P. This proves both equalities in (12.7), including the version that uses only global convergence and global surface restrictions rather than an identified internal array.

We next prove lower semicontinuity of the actual internal metric. Suppose $x_n \to x$, $y_n \to y$, $x \neq y$, and internal paths with bounded lengths join x_n to y_n. Stop the initial pieces on leaving a fixed small rational cube about x whose closure lies in U and avoids y. Compactness of these stopped paths, using (12.1), gives a finite nonconstant D-path from x to that boundary. Thus x is nonsingular; the same applies to y.

Choose shrinking separators at x and y with arbitrarily small internal access, all contained in U. For large n, the original paths meet both separators. Join x and y to those contacts and retain the intervening original path. The extra length tends to zero as the separators shrink. This proves

$$D(x, y; U) \le \liminf_n D(x_n, y_n; U). \tag{12.10}$$

No compact confinement of the middle parts away from ∂U was needed. On the diagonal, nonnegativity suffices.

The local separator access also approximates each finite-distance endpoint by rational points in its component of U. The triangle inequality gives the upper bound in (12.8); (12.10) gives the lower bound, also at an infinite-distance pair. □

The formulas above also give joint measurability for translated open cubes. For rational endpoints, write their internal distance as the infimum of D^K over rational compact polycubical K contained in the open cube and containing the endpoints. The condition $K \subset U + z$ is Borel in z, and Lemma 12.2 makes each term Borel in D. The rational envelope gives all other endpoints. This observation justifies the independent random grid used in Section 13; it avoids an uncountable intersection of separate locality events.

Lemma 12.9 (Fixed complexes with arbitrary coordinates)**.** *Suppose $D_n \to D$ enhancedly on the rational complexes, with D a separating extended metric. Let S be one fixed coordinate complex, whose coordinates need not be rational, and suppose the restrictions $D_n|_{S^2}$ are tight in the uniform topology. In any further joint extraction their uniform limit is the actual restriction $D|_{S^2}$. Buffered internal upper bounds on S pass by Lemma 12.6.*

Proof. The intersection $S \cap \Sigma$ is dense in S. Indeed, on each nondegenerate coordinate plaque at least one coordinate varies over an interval; requiring that varying coordinate to be rational gives a dense subset contained in rational coordinate hyperplanes. Each such point belongs to a compact rational plaque. Choose a countable dense subset $T \subset S \cap \Sigma$.

Write L for a further uniform limit on S^2. For every $p, q \in T$, a fixed connected rational complex contains both points, so $L(p, q) = D(p, q)$ by the already retained uniform convergence.

The lower-limit inequality gives $D(x,y) \le L(x,y)$ for every $x, y \in S$. If $p_j, q_j \in T$ tend to x, y, respectively, continuity of L gives

$$D(x,p_j) \le L(x,p_j) \longrightarrow 0, \qquad D(y,q_j) \le L(y,q_j) \longrightarrow 0.$$

The triangle inequality therefore yields $D(x,y) = L(x,y)$. This argument concerns a fixed complex, or a countable prescribed family of them; it makes no simultaneous samplewise assertion over all real translates. □

Lemma 12.10 (Weighted recovery)**.** *In the preceding setting, assume in addition that finite D_n-rectifiable paths are Euclidean continuous and that D-bounded sets are Euclidean bounded. Let $f_n \to f$ locally uniformly with a common global bound. Then*

$$e^{\psi f_n} \cdot D_n \longrightarrow e^{\psi f} \cdot D \tag{12.11}$$

in the lower-semicontinuous topology and uniformly on each retained fixed coordinate complex. Buffered recovery in an open domain needs only bounds on a compact neighborhood of the paths in question, and does not require the extra global confinement hypothesis. On an event of enhanced convergence, these conclusions are deterministic and simultaneous for every bounded continuous f and any approximations having these properties.

Proof. Put $w_n = e^{\psi f_n}$ and $w = e^{\psi f}$. The global bound makes the weighted and unweighted metrics uniformly bi-Lipschitz. A bounded-cost weighted prelimit path has bounded unweighted cost. Lemma 12.5 confines it to one Euclidean compact set whenever its initial points stay in a compact set. The enhanced annular barriers then give Euclidean equicontinuity. On each interval of a finite path partition, its weighted cost is at least the minimum of w_n on that interval times the corresponding unweighted endpoint distance. Pass to the limit using (1.8) and uniform convergence of the weights on the common compact image. Refine the partition to obtain the weighted length of the limit path. This proves the lower-limit inequality.

For recovery, choose a finite limiting path with weighted cost arbitrarily close to the infimum. Trim its endpoints to nearby Σ-points along the path when necessary. Cover the compact image by finitely many buffered balls on which f oscillates by at most η. Use the short-link construction in the proof of Lemma 12.8, with Σ-vertices and separating barriers. The weighted prelimit cost of each recovered link is at most its unweighted cost times the maximum of w_n on its ball. The sum has limsup at most $e^{O(\eta)}$ times the limiting weighted cost. Let $\eta \downarrow 0$ and remove the endpoint trimming. Infinite values are handled by the lower-limit inequality, and the diagonal by constant paths.

For fixed endpoints on a retained complex the same construction gives exact-endpoint recovery. Its unweighted intrinsic modulus, multiplied by an upper bound on the weights on the buffer, gives equicontinuity. Thus every possible uniform limit on the complex is the asserted weighted restriction. All choices use only deterministic path arguments on the already convergent arrays, so no exceptional set is indexed by f. □

Lemma 12.11 (Stability of short geodesics and shortcuts)**.** *Suppose (A_n, B_n) and (A, B) are pairs of Euclidean lower-semicontinuous extended length metrics, with $B_n \le M A_n$ and enhanced convergence of (A_n, B_n) to (A, B). Assume that finite A_n- and B_n-rectifiable paths are Euclidean continuous, and that every A-bounded set and every B-bounded set is Euclidean bounded. Fix a smooth weight w bounded above and below by positive constants. Let S be a fixed coordinate complex retained in the enhanced convergence and contained in a compact set K_0. Let K be compact with $K_0 \subseteq K \Subset U$. Suppose every path from S leaving K_0 has $w \cdot A_n$ cost at least $a > 0$ for all large n.*

A sequence of $(w \cdot A_n)^K$-geodesics with endpoints in S and lengths below $a/2$, parametrized on $[0,1]$ at constant speed, has a Euclidean-uniformly convergent subsequence, with convergence of lengths, to a $(w \cdot A)$-geodesic. If the limiting path has a strict $w \cdot B$ shortcut compactly contained in U, on an interval of length strictly above a positive threshold, the corresponding assertion with any fixed relaxation of these margins holds for the approximating paths.

Proof. Each constrained geodesic stays in K_0, since its length is below the exit barrier. Every ambient competitor of cost below a also stays in $K_0 \subseteq K$, so the constrained endpoint distance equals the global weighted distance. Lemma 12.10 applies by the Euclidean boundedness assumptions and gives weighted uniform convergence on S^2. Hence the lengths converge after extracting the endpoints. The constant-speed parametrization, compact barriers, and a positive lower bound on w give Euclidean equicontinuity. The limiting path has length at most the limiting length, while its endpoints have exactly that distance. It is therefore a geodesic. If the limiting length is zero, the path is constant; no positive-threshold shortcut assertion is made in that case.

For a strict shortcut on a nonconstant limiting path, choose small fixed rational cubes about its two interval endpoints, compactly contained in U. Their boundary contacts along the limiting geodesic can be chosen within an arbitrarily small time error δ of the prescribed times. Injectivity and compactness of a geodesic ensure that all contacts with sufficiently small cubes have this time property. Uniform Euclidean convergence gives corresponding contacts of the prelimit paths. Extract their times and boundary points. The new endpoints lie on fixed rational boundaries, where the weighted B_n restrictions converge uniformly.

Append the small limiting A-path pieces to the limiting shortcut and use buffered recovery to join the corresponding prelimit boundary contacts. The extra B_n cost is at most M times the extra A_n length, hence $O(\delta)$ times a fixed length bound. If a shortcut endpoint is a path endpoint, use the fixed complex S directly. Choose δ smaller than the strict cost and length reserves. This gives the relaxed shortcut on a prelimit interval. The same exact-endpoint recovery proves persistence of a strict intrinsic B upper bound between the two path endpoints. □

Lemma 12.12 (Borel reconstruction convention)**.** *The separating extended metrics form a Borel subset of the space of nonnegative lower-semicontinuous functions on $(\mathbb{R}^d)^2$. The lower-semicontinuous envelope of a countable rational-pair array is a Borel function of that array. It can be replaced by the infinite discrete metric on the Borel set where it is not a metric.*

Proof. Symmetry is equality of a function and its transpose, a Borel condition in the epigraph realization. Vanishing on rational diagonal points implies vanishing on the whole diagonal by lower semicontinuity. Separation is equivalent to positivity of the minimum on each compact set

$$\{(x,y) : |x|_\infty, |y|_\infty \le m,\ |x-y|_2 \ge m^{-1}\}.$$

These minima are Borel by compact-section projection.

A triangle violation is witnessed in a compact cube by positive rational a, b, δ and a triple with $D(x,y) \le a$, $D(y,z) \le b$, and $D(x,z) > a + b + \delta$. For fixed D, the first two inequalities define a compact set and the last a relatively open subset. Its section is sigma-compact. The measurable projection theorem and a countable union detect all violations.

For a rational array, take the supremum over rational open product rectangles of the infimum of its values at rational pairs in that rectangle, assigning this infimum on the rectangle and zero elsewhere. Each such function is lower semicontinuous and depends Borelly on the array; the countable supremum is the required envelope. The infinite discrete metric is separating and lower semicontinuous. The convention affects no input on which the reconstruction has been proved valid. □

Lemma 12.13 (Local Weyl length calculus)**.** *Let D be a separating Euclidean lower-semicontinuous length metric whose finite rectifiable paths are Euclidean continuous. For every positive continuous w, the construction $D^w = w \cdot D$ is a separating extended length metric. Its topology controls the Euclidean topology. Every finite D^w-rectifiable path has compact Euclidean image, is D-rectifiable, and satisfies*

$$\operatorname{len}_{D^w}(P) = \int_P w \, ds_D. \tag{12.12}$$

For positive continuous v, w,

$$v \cdot (w \cdot D) = (vw) \cdot D. \tag{12.13}$$

No surface upper bound is used in these assertions. Lower semicontinuity of D^w is not asserted by this lemma.

Proof. Any defining path from x to the complement of a fixed Euclidean ball about x pays at least the minimum of w on that ball times the positive D-distance from x to its boundary. Thus D^w separates points and its topology controls the Euclidean topology at each fixed point. A D^w-continuous path on a compact parameter interval has compact Euclidean image K.

Choose a compact buffer K' with $K \subset \operatorname{int} K'$. Compact separation gives a positive D-barrier from K to $\partial K'$. Multiplying by $\min_{K'} w$ gives a positive weighted exit barrier; take a finite positive lower bound a for it. If $x, y \in K$ and $D^w(x, y) < a/2$, near-minimizing weighted paths cannot leave K', and

$$D^w(x, y) \geq (\min_{K'} w) D(x, y). \tag{12.14}$$

This makes a D^w-continuous path D-continuous and bounds its original length by a fixed multiple of its weighted length.

For the length identity, cover the compact path image by neighborhoods on which w oscillates by at most a prescribed positive amount. In smaller compact neighborhoods, the same exit-barrier argument confines sufficiently short near-minimizing connections for both metrics. Their distances are bounded above and below by the local supremum and infimum of w times the original distance. Subdivide the path into sufficiently short pieces and take the supremum over refinements of the corresponding partition sums. Upper and lower Riemann sums against D-arclength give (12.12) as the oscillation tends to zero. The defining infimum is therefore a length metric in its own topology. Applying the identity twice gives (12.13). All these arguments are local on compact path images, so neither weight needs a global bound. □

Lemma 12.14 (Lower semicontinuity after arbitrary continuous weights)**.** *Suppose in addition that E has small local separators at its nonsingular points and the compact path geometry above. Then $E^w = w \cdot E$ is Euclidean lower semicontinuous for every positive continuous w. Together with Lemma 12.13, it is a separating extended length metric with the length and composition identities above. The assertions hold simultaneously, pathwise, for all such weights.*

Proof. Take converging endpoint sequences with bounded weighted path costs. For distinct limiting endpoints, stop the initial and terminal pieces at fixed small boundaries in neighborhoods of those endpoints. There w is bounded below, so their original E-lengths are bounded. Compactness gives finite nonconstant limiting E-paths from the limiting endpoints to those boundaries. Both endpoints are therefore nonsingular.

Shrinking local separators join them to the original weighted paths at costs tending to zero, since w is locally bounded above. Retain the original middle portions, as in (12.10). This proves the lower-limit inequality without claiming common compact confinement of the middle paths.

Nonnegativity handles the diagonal. The proof uses only deterministic local bounds on w, so no exceptional set is indexed by the weight. □

Only the last lemma uses canonical small separators. The local length calculus applies also to the arbitrary competitor under the original axioms. In particular, removing an unbounded smooth coarse field in a local trial does not assume a surface estimate or global coercivity for that modified competitor.

13 Locality and field measurability

Fix a subsequential coupling (h, E) retaining the joint arrays of Proposition 11.3. At this stage E is not assumed measurable from h. Lemma 12.6, applied before any intrinsic identification, gives finite actual intrinsic surface bounds and the enlarged-buffer version of the small separators. Lemmas 12.7 and 12.8 therefore apply. In particular, the actual intrinsic metrics are lower semicontinuous, have a rational reconstruction, and inherit the local Hölder upper bounds.

Proposition 13.1 (Raw deterministic scale constants)**.** *There are positive Borel constants c_r, $r > 0$, satisfying (4.9) for every $\eta > 0$, such that the following laws are uniformly tight in deterministic $r > 0$ and $z \in \mathbb{R}^d$:*

(a) the Hölder upper constants, for each exponent in (11.7), of

$$(x, y) \longmapsto c_r^{-1} e^{-\psi h_r(z)} E(z + rx, z + ry; z + rU), \qquad x, y \in S, \tag{13.1}$$

for each fixed coordinate complex $S \Subset U$;

(b) each fixed-template annulus crossing and its reciprocal, normalized in the same units.

One may take

$$c_r = \mathrm{Med}\big[e^{-\psi h_r(0)} E(S_r(0), S_{2r}(0))\big]. \tag{13.2}$$

No equality $c_r = r^{\psi Q}$ is asserted here.

Proof. Set $b_\varepsilon(r) = r a_{\varepsilon/r}/a_\varepsilon$. The exact prelimit change of variables is

$$a_\varepsilon^{-1} D_h^\varepsilon(z + r\cdot, z + r\cdot) = b_\varepsilon(r) e^{\psi h_r(z)} a_{\varepsilon/r}^{-1} D_{T_{z,r}h}^{\varepsilon/r}, \quad T_{z,r}h = h(z + r\cdot) - h_r(z). \tag{13.3}$$

The field $T_{z,r}h$ has the pinned base law. For each fixed r, the ratios $b_\varepsilon(r)$ are bounded above and below. Pass to a further extraction on which they tend to $b(r) > 0$. The uniform cutoff estimates of Proposition 11.3 on the right of (13.3) pass to the scaled limiting arrays. Buffered domination gives the required actual intrinsic upper bounds, in a slightly larger buffer. For a desired open buffer use a smaller fixed buffer first; its closure is contained in the desired one.

For rational complexes, rational-point consistency in Theorem 1.1, the continuous surface upper bounds, and the triangle inequality identify the global limiting restrictions with the actual metric. For a fixed translated or dilated complex, add its tight arrays to the extraction and apply Lemma 12.9. This also covers nonrational centers and radii; it does not assume that a nonrational plaque contains points of $\mathbb{Q}^d$. The crossing in (13.2) is consequently the limit of its prelimit surface infimum. Uniform upper and reciprocal quantile bounds show that $c_r/b(r)$ lies in one fixed compact subinterval of $(0, \infty)$. The same constants work at every center, since the transformed field in (13.3) has the same marginal law there. This proves the normalized estimates.

For any fixed r, R, extract the two ratios together. Their quotient satisfies (11.9). Comparison of each b with its median c proves (4.9), with constants enlarged by a fixed factor. This is a separate argument for each deterministic pair r, R, not an uncountable simultaneous extraction. Finally, the crossing is a Borel function of r and the metric: it is a compact endpoint infimum after rescaling the two boundaries to fixed sets. Joint measurability of the spherical averages and the lower-quantile definition make (13.2) Borel in r. □

Definition 13.2 (Conditional locality)**.** A coupled metric E is conditionally local for h if, for every deterministic open V, its actual intrinsic metric in V is conditionally independent of

$$(h, E(\cdot, \cdot; \mathbb{R}^d \setminus \overline{V}))$$

given $h|_V$. The same definition for a finite tuple of metrics is imposed jointly on the tuple of intrinsic metrics. Restrictions and exhaustions are represented by countable coordinates.

Proposition 13.3 (Conditional locality of the coupled limit)**.** *Every LFPP subsequential coupling is conditionally local. This holds jointly for any finite number of subsequential metric coordinates extracted from the same field. Constant shifts act by their exact Weyl factor, and continuous perturbations obey the simultaneous local weighted-recovery rule.*

Proof. Fix $W \Subset V$ and $W_0 \Subset \mathbb{R}^d \setminus \overline{V}$. Re-pin the field inside V, as in Remark 4.7, and write $h = m + R$ as in Lemma 4.6. Choose a smooth cutoff ϕ supported outside $\overline{V}$, equal to one near $\overline{W_0}$, and put $f = \phi m$. It is a random smooth compactly supported function, and $h - f = R$ near $\overline{W_0}$.

For sufficiently small mollification scales, every internal prelimit metric coordinate in W is a function of $h|_V$. Every internal coordinate for $h - f$ in W_0 is a function of R. The two blocks

$$(h|_V, \text{inner coordinates}), \qquad (R, \text{outer coordinates for } h - f) \tag{13.4}$$

are independent. Retain m, R, f, their relevant smooth restrictions, and these coordinates in the joint extraction. The Gaussian coordinates have a fixed joint law; their projection identities and functional relations are preserved in the limit. Independence of the two entire blocks is preserved as well.

Weighted recovery identifies the outer shifted metric as $e^{-\psi f} \cdot E$. Buffered exhaustion identifies its actual intrinsic restrictions, and the same exhaustion identifies the inside metric. Given $h|_V$, the full field is $m + R$, with m known. The independent outer block retained the entire residual distribution, not only an exterior restriction. The inside kernel is therefore independent of the entire field beyond its local restriction and of the shifted exterior metric. Adding the known smooth f back to the exterior metric preserves this conditional independence.

Exhaust V by inner buffers and the open exterior by compact buffers. The countable intrinsic reconstruction proves Definition 13.2. The same argument retains tuples of coordinates and gives joint locality. Remark 4.7 returns to the prescribed sphere representative.

For a constant shift the prelimit identity is exact. For a continuous perturbation on a compact buffer, mollification converges uniformly and Lemma 12.10 is pathwise. These give additivity and the simultaneous local Weyl rule without yet claiming that E is a function of h. □

Lemma 13.4 (Conditional products on disjoint open cells)**.** *For pairwise disjoint deterministic open sets U_i, the intrinsic metric coordinates $E(\cdot, \cdot; U_i)$ are conditionally independent given the entire field. Their conditional kernels depend only on the respective local field restrictions. The same factorization holds for tuples of independent samples of the conditional law of E given h.*

Proof. For a finite family, conditional locality for one member makes its metric independent of the full field and all the other intrinsic coordinates given its local field. Conditioning further on the full field leaves that same local kernel. Iterate over the members to factor the joint kernel. If boundaries touch, inner exhaustions have positive separations and recover the actual intrinsic metrics. Countable families follow from their finite-dimensional distributions. Taking independent copies of the whole conditional law preserves the product factorization across cells. □

Lemma 13.5 (Supportwise comparison)**.** *There is a deterministic $M < \infty$ such that two conditionally independent samples $E, \widetilde{E}$ given h satisfy*

$$M^{-1}E \le \widetilde{E} \le ME \tag{13.5}$$

almost surely simultaneously on all pairs. For almost every h, the same comparison holds under every coupling with these conditional marginals.

Proof. For each annular template, the normalized intrinsic separator diameter of one copy is tight and the normalized crossing of the other has tight reciprocal, with the same constants c_r. Their ratio is bounded by a sufficiently large deterministic M with probability as close to one as desired, uniformly in center and scale.

To apply Lemma 4.3, re-pin in each annular block at a smooth probability density supported there. Conditional locality and constant additivity imply that the conditional probability of the normalized, constant-invariant event depends only on this locally pinned field, hence on the original block modulo constants. Lemma 13.4 gives conditional independence across separated annuli, also for the two-copy tuple. Lemmas 4.3 and 4.4 give simultaneous good-inner-ball covers. Apply Lemma 12.4 to every finite path and pass the approximating endpoints to their limits by lower semicontinuity. A summable mesh sequence and a countable exhaustion of space give both inequalities in (13.5) on one event.

Fix a rational pair and condition on h. Its distance is positive and finite almost surely. If two independent values with this law have ratio at most M almost surely, the essential supremum of the law is at most M times its essential infimum. Indeed, otherwise two separated intervals of positive conditional mass violate that ratio with positive product probability. This is a support statement, so it holds under every coupling of the same marginal law. Apply it to the countably many rational pairs. Formula (12.8), with $U = \mathbb{R}^d$, extends the comparisons to all pairs under any such coupling. □

Proposition 13.6 (No residual metric randomness)**.** *Every subsequential metric E in the construction is almost surely measurable with respect to h.*

Proof. Fix distinct rational p, q and let $X = E(p, q)$. Select an E-geodesic P using Lemma 12.2 in a deterministic exhaustion of confinement boxes. Its geometric selection is independent of any grid. Write μ_P for its E-arclength occupation measure on $\mathbb{R}^d$. The path is simple, so this finite measure is nonatomic.

1: reconstruction from cell metrics. Choose a grid of mesh δ with an independent uniform shift θ. Its open cells are U_i. For each fixed Euclidean point, the probability of lying on a shifted grid boundary is zero. Fubini therefore gives

$$\mu_P(\text{grid boundaries}) = 0 \tag{13.6}$$

almost surely. Joint measurability in the shift follows from the countable compact-constraint reconstruction following Lemma 12.8.

Condition on h, θ and all the cell intrinsic metrics, and sample two copies of the remaining metric randomness. Lemma 13.5 compares them with deterministic constant M, despite this nonindependent coupling. The first sample's geodesic is rectifiable for the second. The two length measures agree on each open-cell interval of that path, because the intrinsic cell metrics agree. They give zero mass to the remaining boundary times by (13.6) and length domination. The path has the same total length for the two metrics. Reverse the samples to conclude equality of their endpoint distances. Repeat for rational pairs. Thus the global metric is a measurable function of h, θ and the cell coordinates.

2: conditional resampling. Given h, θ, the cell coordinates have product law by Lemma 13.4. Resample cell i alone and reconstruct a metric $E^{(i)}$ with the same conditional marginal law. Put $X_i = E^{(i)}(p, q)$. On the original selected path, lengths outside U_i agree, and lengths inside it increase by at most a factor M. Hence

$$(X_i - X)_+ \le (M-1)\mu_P(U_i). \tag{13.7}$$

For a finite product, conditional variance tensorization follows by writing the Doob martingale of a square-integrable function of the coordinates and applying conditional Jensen to each martingale difference. It gives $\mathrm{Var}(X) \le \frac{1}{2}\sum_i \mathbb{E}(X - X_i)^2$. Martingale approximation gives the countable-product version. Exchangeability of the original and resampled coordinate replaces each half square by its positive-part square. Thus

$$\mathrm{Var}(X \mid h, \theta) \le (M-1)^2 \mathbb{E}\left[X \max_i \mu_P(U_i) \,\middle|\, h, \theta \right]. \tag{13.8}$$

Here $\sum_i \mu_P(U_i) \le X$ was used.

3: vanishing influence. The conditional law of X has finite essential supremum almost surely. It is positive and finite, and its support has bounded ratio by Lemma 13.5; comparison to a positive finite conditional median proves the assertion. Conditional second moments in (13.8) are therefore finite.

For every realized compact geodesic image, atomlessness implies

$$\sup_{x \in \mathbb{R}^d} \mu_P(B^{\mathrm{Eucl}}_{\sqrt{d}\delta}(x)) \longrightarrow 0.$$

Otherwise centers of balls with a fixed positive mass would have a compactly convergent subsequence and produce an atom. This supremum bounds the cell maximum for every shift. Since θ is independent of (h, E), the left side of (13.8) is $\mathrm{Var}(X \mid h)$. Integrate in the shift and use conditional dominated convergence, with bound X^2. The variance is zero. Countably many rational distances and (12.8) determine the entire metric. □

Proposition 13.7 (One assignment on all continuous perturbations)**.** *A subsequential limit has a Borel assignment $g \mapsto E_g$ satisfying length, actual intrinsic locality, and simultaneous continuous Weyl scaling for a base field plus any possibly dependent random continuous function. It is translation covariant for each fixed deterministic translation. On the base field it retains Proposition 13.1.*

Proof. Joint convergence to a field-measurable metric implies convergence in probability on the original field space. For completeness, adjoin the fixed Borel random variable E_h to any subsequential joint limit of $(h, a_\varepsilon^{-1} D_h^\varepsilon)$. Both metric marginals in that limit are the same measurable function of its field coordinate, so the joint limit is supported on the diagonal. The same

reasoning applies to the countably many global surface restrictions. Choose a deterministic further subsequence ε_n on which these convergences hold almost surely.

For rational endpoints in an open U, define the local formula

$$F_U(g;p,q) = \inf_{V \Subset U} \limsup_{n\to\infty} a_{\varepsilon_n}^{-1} D_g^{\varepsilon_n}(p,q;V), \tag{13.9}$$

using the countable domains in (12.7). For all sufficiently large n, each term depends only on $g|_U$, since the convolution support is smaller than the clearance from V to ∂U. On a good base realization and for every continuous f, the local version of weighted recovery and intrinsic reconstruction give

$$F_U(h+f) = (e^{\psi f} \cdot E_h)(\cdot,\cdot;U). \tag{13.10}$$

Only compact path pieces enter this argument, so f need not be globally bounded. It is a deterministic assertion on the same event for all f.

Use the rational envelope to define F_U at all endpoints and use Lemma 12.12 for a Borel convention on invalid inputs. The global assignment is the formula with $U = \mathbb{R}^d$. On every relevant input, (13.10) identifies its actual intrinsic restrictions with F_U. The definition depends on $h + f$, not on a representation of that input as a field plus a continuous function, so it is consistent even when two such representations exist. Lemma 12.14 gives its extended length structure, lower semicontinuity, and the composition rule. These prove locality and the simultaneous Weyl identity for dependent perturbations as well.

For a fixed translation z, the prelimit translation identity is exact. The translated and re-pinned field has the base law, so the same almost-sure subsequence construction applies to it. Uniqueness of its lower-semicontinuous limit gives the translated identity on the base field. The simultaneous Weyl formula extends it to all continuous perturbations of a good base realization. □

14 Weak metrics and deterministic comparison

The construction yields an auxiliary class with uniform crossing and surface estimates. We retain its deterministic scale constants until uniqueness identifies them. Definition 1.3 remains the class used for the final characterization.

Definition 14.1 (Regular weak assignment)**.** A regular weak assignment is a Borel assignment $g \mapsto D_g$ to Euclidean lower-semicontinuous extended metrics. On a base field plus a possibly dependent continuous function, it has the length, locality, and simultaneous continuous Weyl properties of Axioms I–III, together with translation covariance.

On the base pinned field there are positive Borel constants c_r satisfying (4.9), with $Q > 0$, and an exponent $\beta > 0$, such that for every fixed coordinate complex $S \Subset U$ the actual intrinsic restrictions in (13.1) are finite and have uniformly tight β-Hölder upper constants. Every fixed annulus crossing and its reciprocal are tight in units

$$s_r(z) = c_r e^{\psi h_r(z)}, \tag{14.1}$$

uniformly in deterministic $r > 0, z \in \mathbb{R}^d$.

A family is uniformly regular if it has one positive exponent, common constants in (4.9), and common quantile bounds for each fixed surface and annular template. There is no assertion of simultaneous samplewise regularity for every real translated surface.

Propositions 13.1 and 13.7 place every LFPP subsequential assignment in this class. An arbitrary original-V competitor is not assumed to belong to it.

Lemma 14.2 (Coercivity without a surface upper bound)**.** *Let D be a separating lower-semicontinuous length metric. Suppose the normalized reciprocals of concentric global crossings are tight, with constants satisfying (4.9) for $Q > 0$. Almost surely, every finite D-rectifiable path is Euclidean continuous and bounded, every D-bounded set is Euclidean bounded, every finite-distance pair admits a geodesic, and every finite-distance component is complete.*

Proof. The covariance formula gives $\operatorname{Var} h_R(0) = \log R + O(1)$ as $R \to \infty$. The raw scale lower bound and reciprocal tightness therefore imply

$$D(S_R, S_{2R}) \longrightarrow \infty \quad \text{in probability as } R \to \infty.$$

Choose deterministic $R_n \uparrow \infty$, with $R_{n+1} > 4R_n$, so that $\mathbb{P}[D(S_{R_n}, S_{2R_n}) < n] \le 2^{-n}$. Almost surely these crossing costs exceed n eventually.

Let $P : [0, T] \to \mathbb{R}^d$ be metric-continuous and rectifiable. It cannot meet both boundaries of any sufficiently large one of these annuli. Choose an omitted sphere from each such annulus. For any Euclidean compact K, the inverse image $P^{-1}(K)$ is closed: if $t_j \to t$ and $P(t_j) \in K$, a Euclidean subsequential limit $x \in K$ satisfies

$$D(x, P(t)) \le \liminf_j D(P(t_j), P(t)) = 0,$$

so $x = P(t)$. The closed radial bands between the omitted spheres consequently give a countable disjoint closed partition of $[0, T]$ by their inverse images.

Such a partition has only one nonempty member. To verify this directly, start with a closed interval meeting two members. If the next specified closed member misses it, retain the interval. Otherwise choose a complementary interval component adjacent to that member. This component cannot be contained in a single other closed member, because its closure meets the specified member. It contains a smaller closed interval meeting two members and avoiding the specified one. Iteration gives nested nonempty closed intervals avoiding the first n members. Their intersection contradicts the purported partition. Hence the image of P lies in one bounded band. The closed-graph argument and compactness of this range now give Euclidean continuity.

A finite near-minimizing path from a fixed center to a sufficiently distant point must cross a large annulus. Its crossing cost shows that each metric ball is Euclidean bounded. Near-minimizers for a finite pair thus lie in one common Euclidean compact set. Parametrize them at bounded constant speed; (12.1) gives equicontinuity and lower semicontinuity gives a minimizing limit. A D-Cauchy sequence is Euclidean bounded and, by the same inverse modulus, Euclidean Cauchy. Lower semicontinuity gives convergence in D to its Euclidean limit, which lies in the same finite-distance component. □

Lemma 14.3 (Consequences of regular weak estimates)**.** *A regular weak metric has the small separators and rational core of Lemma 12.7. Its actual intrinsic metrics are lower semicontinuous and have the reconstruction (12.8). For every fixed $m < \infty$, there is $B < \infty$ such that, outside probability $O(\varepsilon^m)$, every pair in a fixed macroscopic compact region at Euclidean separation at least εR has distance at least*

$$\varepsilon^B c_R e^{\psi h_{L_* R}(0)}, \tag{14.2}$$

where $L_* > 0$ is fixed sufficiently large. These probability statements are uniform in R and in uniformly regular families.

Proof. Lemma 14.2 supplies Euclidean-continuous geodesics. An annular separator diameter divided by its crossing is tight, so the event $H \le \Lambda X$ has probability as close to one as desired for

sufficiently large Λ. The crossing can now be trimmed, by Euclidean continuity, between its last inner-boundary contact and first outer-boundary contact. It is therefore determined by the field in an open annular buffer. The determining block includes the averaging sphere, by the convention after (1.4).

Lemma 4.4, on a summable sequence of shrinking radius blocks, gives good separating shells at every point. Thin the blocks so the enlarged shells are nested and disjoint. Lemma 12.7 applies. The endpoint-gluing part of Lemma 12.8, which needs no approximating sequence, gives actual intrinsic lower semicontinuity and rational reconstruction.

For (14.2), include a fixed positive crossing lower quantile in each good event. At candidate radii in $[\varepsilon^2 R, c\varepsilon R]$, the Gaussian increments $h_r(z) - h_{L_* R}(0)$ have variance $O(\log \varepsilon^{-1})$. The ratio bound (4.9) and a union bound over the polynomial-size candidate grid make all units at least the right side of (14.2), after increasing B. A good ball containing one endpoint in its inner half separates it from any point at the stated distance if c is small enough. Every finite path must cross this shell. The covering failure and the Gaussian unit failure can each be made $O(\varepsilon^m)$. □

Proposition 14.4 (Comparison of raw constants on many levels)**.** *Let E, D be regular weak assignments with the same Weyl parameter and raw constants c_r^E, c_r^D. There are deterministic $0 < m_0 < M_0 < \infty$ such that, for all sufficiently small ε, at least $9/10$ of the geometrically spaced radii in $[\varepsilon^2, \varepsilon]$ satisfy*

$$m_0 \le c_r^E / c_r^D \le M_0. \tag{14.3}$$

Any fixed fraction below one can replace $9/10$.

Proof. At each radius, impose a normalized E-separator upper bound and a normalized D-crossing lower bound. Their joint probability can be as close to one as required, uniformly in center and radius. The common factor $e^{\psi h_r(z)}$ cancels. If $c_r^E / c_r^D \le m_0$, the separator cost is at most a fixed constant times m_0 times the competitor crossing.

Suppose such radii occupy a fixed positive fraction of a small logarithmic block. Lemma 4.4 applies to that deterministic subset. The actual internal distance $D(0, 3e_1; B_4)$ has a finite deterministic upper quantile, by the surface bound on a fixed connecting complex. The fixed E-crossing of $B_2 \setminus \overline{B}_1$ has a positive deterministic lower quantile. Choose each quantile event to have probability at least $9/10$.

On their intersection with the covering event, route a near-minimizing D-path in B_4 by Lemma 12.4. For sufficiently small ε, its new endpoints are inside B_1 and outside B_2. The routed E-path must therefore pay the fixed positive crossing lower quantile. Its length is at most a fixed multiple of m_0 times the D upper quantile. Choose m_0 so small that these inequalities contradict each other. The intersection has positive probability once the covering failure is sufficiently small. Thus the low-ratio levels cannot have the prescribed positive density.

Reverse the roles of the metrics to rule out a prescribed positive density of levels with ratio at least M_0, for M_0 sufficiently large. Allocate the two excluded densities to total $1/10$. All radius subsets are deterministic and the cover estimate is uniform in them, so this proves (14.3) for every sufficiently small block. □

Theorem 14.5 (Deterministic bi-Lipschitz comparison)**.** *Two regular weak assignments E, D with the same (ψ, Q) are almost surely globally bi-Lipschitz with deterministic positive constants, simultaneously on all point pairs. Their raw constants are uniformly comparable at every radius. They may consequently be measured in a common family c_r of units, with adjusted uniform bounds.*

Proof. Use the majority levels in Proposition 14.4. At those levels, the separator-to-crossing comparison has uniformly high probability with a deterministic factor, in either direction. The

cover lemma and the router give the corresponding path comparison between approximating endpoints. Let the maximal radius tend to zero along a sequence with summable cover failures, and exhaust space by compact sets. Lower semicontinuity gives both global comparisons simultaneously for all pairs. Infinite distances on the right are harmless.

The comparisons hold for lengths and hence for all actual intrinsic metrics. Apply them to the same annular crossing after dividing by the common field factor $e^{\psi h_r(0)}$. Its lower medians are comparable. Uniform crossing and reciprocal tightness compare each raw constant with the corresponding median by deterministic factors. Thus the raw constants are comparable at every radius, not merely on the majority levels. Replacing one by the other preserves (4.9) and the uniform local estimates. □

Remark 14.6 (Optimal comparison constants). The event $E \le cD$ on all pairs is measurable. A violation with finite D is witnessed in a compact endpoint box by a rational $a > 0$ such that $D(x,y) \le a$ and $E(x,y) > ca$. For fixed metrics the section is a relatively open subset of a compact set, hence sigma-compact. Joint Borel evaluation and measurable projection apply. A countable union detects every violation. Consequently the deterministic essential upper and lower comparison constants are well defined. Their endpoint inequalities hold by countable intersection along approximating constants; no countable determination of an arbitrary competitor is being assumed.

15 Annular witnesses and occupation estimates

In this section the ordered pair is denoted by (A, B) and satisfies

$$B \le CA, \qquad 0 < C < \infty. \tag{15.1}$$

Both metrics have the coercive geometry of Lemma 14.2 and positive crossing lower quantiles in common units $s_r(z) = c_r e^{\psi h_r(z)}$. Only B needs surface upper bounds, except in the endpoint lemma explicitly stated for a regular weak A.

Set

$$K_{z,r} = \{r/4 \le |x - z|_\infty \le 4r\}, \qquad U_{z,r} = \{r/3 < |x - z|_\infty < 7r/2\}. \tag{15.2}$$

Choose quantiles $A_0 < \infty$ and $0 < a_H \le 1$ so that, outside a probability as small as required,

$$\operatorname{diam}_{B(\cdot,\cdot;U_{z,r})} S_{3r}(z) \le A_0 s_r(z), \qquad A(S_r(z), S_{2r}(z)) \ge a_H s_r(z). \tag{15.3}$$

Take $A_H \ge \max\{1, 2A_0/C, 2a_H\}$. The quantiles are fixed before choosing a ratio threshold.

Definition 15.1 (Annular witness)**.** For $0 < c < C$, let $H_r(c; z)$ be the event that an $A^{K_{z,r}}$-geodesic, from a point of $S_{r/2}(z)$ to a point of $S_{3r}(z)$, has length at most $A_H s_r(z)$ and has a middle subpath $P|_{[u,v]}$ satisfying

$$P(u) \in S_r(z), \quad P(v) \in S_{2r}(z), \quad P([u,v]) \subset \overline{B_{2r}(z)} \setminus B_r(z), \quad B^{K_{z,r}}(P(u), P(v)) > c \operatorname{len}_A(P|_{[u,v]}). \tag{15.4}$$

The parent minimizes between its own endpoints, not necessarily between the two entire surfaces. Write $H_r(c) = H_r(c; 0)$.

The event is measurable in the completed sigma field of a fixed open annular block containing $K_{z,r}$ and the averaging sphere. Lemma 12.3 supplies locality of the constrained metric and path coordinates, without assuming open-domain intrinsic continuity for an arbitrary competitor. An existential path assertion is analytic in the compact path coordinates; its complement is also measurable in the completion. Translation covariance makes the probability independent of z.

Lemma 15.2 (A local saving in the absence of a witness)**.** *Suppose* (15.3) *holds and* $H_r(c;z)$ *fails. Let* P *be an* A*-geodesic constrained to a larger compact domain containing* $K_{z,r}$*, with both endpoints outside* $\overline{B_{3r}(z)}$*, and suppose it visits* $B_{r/2}(z)$*. Let* I *be its full first-to-last* $S_{3r}(z)$ *time span in* A*-arclength. Its endpoints admit a* B*-connection in the larger domain of length at most*

$$(C-\gamma(c))|I|, \qquad \gamma(c)=\min\left\{\frac{C}{2},\frac{(C-c)a_H}{A_H}\right\}>0. \tag{15.5}$$

Proof. If $|I|>A_H s_r(z)$, use the boundary connector in (15.3). Its cost is at most $A_0 s_r(z)\le C|I|/2$.

Otherwise choose an inner-ball visit, the first subsequent $S_{3r}(z)$ contact, and the last preceding $S_{r/2}(z)$ contact. The intervening path lies between these two surfaces and hence in $K_{z,r}$. It is minimizing there, since a shorter constrained connection would shorten the original geodesic. Its length is at most $A_H s_r(z)$. The last $S_r(z)$ contact before the first subsequent $S_{2r}(z)$ contact gives a middle crossing of length $\ell\ge a_H s_r(z)$. Failure of the witness event gives a $B^{K_{z,r}}$ connection for it of cost at most $c\ell$. Retain the other portions of I, using (15.1) for their lengths. The total cost is at most

$$C(|I|-\ell)+c\ell\le C|I|-(C-c)a_H s_r(z)\le\left(C-\frac{(C-c)a_H}{A_H}\right)|I|.$$

Approximate intrinsic infima and let the total excess decrease to zero. No exterior portion of the old path is claimed to be determined by the local field. □

Fix $L_*>10\sqrt{d}$ and put

$$S_R=c_R e^{\psi h_{L_*R}(0)}. \tag{15.6}$$

For $0<\tau<1/10$, let $G_R(\tau,c)$ be the event that an $A^{K_{0,R}}$-geodesic $P:[0,L]\to K_{0,R}$ satisfies

$$\begin{aligned} \tau S_R\le L\le\tau^{-1}S_R, \qquad & |P(0)-P(L)|_2\ge\tau R,\\ \operatorname{dist}_2(P([0,L]),\partial K_{0,R})\ge\tau R, \qquad & \operatorname{dist}_2(P([L/4,3L/4]),\{P(0),P(L)\})\ge\tau R,\\ B^{K_{0,R}}(P(0),P(L))>cL. & \end{aligned} \tag{15.7}$$

The central half is fixed in the time parameter. This definition does not ask for a Euclidean displacement of order τR after only τL metric time.

Lemma 15.3 (Density of witness levels)**.** *There is* $p_*>0$*, independent of* $c\in[C/2,C)$*, with the following property. Put*

$$c^\sharp=C-\gamma(c)/8. \tag{15.8}$$

For every $\tau>0$ *there is* $\varepsilon_0>0$ *such that, if* $\mathbb{P}[G_R(\tau,c^\sharp)]\ge\tau$*, then, for* $\varepsilon<\varepsilon_0$*, at least three quarters of the levels in a sufficiently separated geometric lattice in* $[\varepsilon^2R,\varepsilon R]$ *satisfy*

$$\mathbb{P}[H_r(c)]\ge p_*. \tag{15.9}$$

The bounds are uniform in R *and in families with common regularity quantiles. With the choice* $A_H\ge 2a_H$*, one has* $c<c^\sharp<C$*.*

Proof. Suppose at least one quarter of the levels violate (15.9). At those levels require (15.3) and failure of $H_r(c;z)$. First choose the regularity failure probabilities and then p_* so small that the common probability of these good events meets Lemma 4.4 for a density of one quarter. This choice is independent of the positive number $C-c$.

With probability tending to one, the good inner balls cover a fixed compact macroscopic region. On (15.7), for ε small relative to τ, every good ball whose inner half meets the central half of P has

$K_{z,r} \subset K_{0,R}$ and has both endpoints of P outside its outer ball. The full first-to-last outer-boundary intervals have interiors covering $[L/4, 3L/4]$, and each admits the saving in Lemma 15.2.

There are finitely many candidate balls. An inclusion-minimal subcover by their intervals has multiplicity at most two: after removing containments, three intervals with a common interior point would make the middle interval contained in the union of the other two. The interval family can be colored with two colors having disjoint interiors. Its total interval length is at least $L/2$, so one color carries at least $L/4$. Replace that color by the local B-connections and retain the other original portions. The result stays in $K_{0,R}$ and has length at most

$$CL - \gamma(c)L/4 < c^\sharp L,$$

contradicting the last condition in (15.7). Make the covering failure smaller than τ to contradict the assumed witness probability. Finally, $a_H/A_H \le 1/2$ gives $\gamma(c) \le (C-c)/2$, proving the stated order of thresholds. □

Lemma 15.4 (Optimality supplies a macroscopic witness)**.** *If C is the optimal deterministic upper constant in (15.1), then for every $c < C$ there are a radius R on any fixed geometric lattice and $\tau > 0$ with*

$$\mathbb{P}[G_R(\tau, c)] \ge \tau. \tag{15.10}$$

Consequently a density of at least three quarters of sufficiently long small-scale blocks satisfies (15.9) for every fixed threshold $c < C$.

Proof. Optimality gives positive probability of a finite A-distance pair with $B(x,y) > cA(x,y)$. Take an A-geodesic between them. Its compact image fits with positive clearance in some $K_{z,R}$ with rational center and geometric radius: choose R sufficiently large and place the center at a distance comparable to R from the image. The path remains minimizing in that compact annulus, and its constrained B endpoint distance is at least its global B distance. Its central half has positive Euclidean separation from its endpoints, by injectivity and compactness.

Its length divided by the positive finite unit $c_R e^{\psi h_{L_* R}(z)}$ is positive and finite. Thus it satisfies all inequalities in (15.7) for some positive rational τ. A countable union over centers, radii, and tolerances gives fixed data of positive probability. Translation covariance removes the center; decrease τ so that the probability is at least τ. Apply Lemma 15.3 using the larger threshold $c^\sharp < C$. A fixed multiplicative shift of a geometric radius block changes only finitely many level indices, so the resulting density statement is on a common geometric lattice. □

15.1 Occupation contraction

Lemma 15.5 (A defective contraction)**.** *Let $P : [0, L] \to K$ be an A^K-geodesic and suppose*

$$B^K(P(s), P(t)) \ge C(t-s) - \delta S, \qquad s < t. \tag{15.11}$$

Consider J nested shells around a point visited by P, with pairwise disjoint enlarged annuli contained in K and with outer balls avoiding both endpoints. Number them from largest to smallest. Let X_j be their A crossings and suppose the middle-separator B diameters are at most $\Lambda C X_j$. The full visit span to the innermost middle ball is at most

$$\Lambda \left(\frac{\Lambda}{\Lambda+1}\right)^{J-1} L + (\Lambda+1)\delta S/C. \tag{15.12}$$

For an ordinary A-geodesic the same statement holds with $B = A$, $C = 1$, and $\delta = 0$.

Proof. Let I_j be the full first-to-last time span on the middle separator. Its endpoints have a B^K connection of cost at most $\Lambda C X_j$. Thus $|I_j| \le \Lambda X_j + \eta$, where $\eta = \delta S/C$. Before a visit to the innermost ball, the path crosses all smaller shells on disjoint time intervals inside I_j. Hence

$$\sum_{\ell>j} X_\ell \le \Lambda X_j + \eta.$$

Writing $T_j = \sum_{\ell\ge j} X_\ell$ and $q_0 = \Lambda/(\Lambda+1)$ gives $T_{j+1} \le q_0 T_j + (1-q_0)\eta$. Since $T_1 \le L$, iteration yields $T_J \le q_0^{J-1} L + \eta$. The final span is at most $\Lambda X_J + \eta \le \Lambda T_J + \eta$. □

Corollary 15.6 (Common polynomial visit bounds)**.** *Fix compact macroscopic buffers, a central region separated from the allowed endpoint surfaces, and $L \le MS_R$. For any prescribed $m > 0$, there are $\theta > 0$ and $K_0 < \infty$ such that, with probability $1 - O(\varepsilon^m)$, every path satisfying (15.11) has full visit span to every candidate ball of radius at most εR in the central region at most*

$$\varepsilon^\theta L + K_0 \delta S_R/C. \tag{15.13}$$

The event is common to all such paths, and its constants are uniform in R. For ordinary geodesics of a regular weak metric take $\delta = 0$.

Proof. Use enclosing shells with radii between $\varepsilon^{1/2}R$ and $\varepsilon^{1/4}R$. Their separator-to-crossing inequalities have arbitrarily high probability after increasing Λ. The Gaussian annular estimate and a finest-grid union bound give at least a fixed multiple of $\log \varepsilon^{-1}$ successful separated levels about every candidate center, outside probability $O(\varepsilon^m)$. The candidate ball lies in their innermost middle ball for small ε. The shells stay inside the macroscopic buffer and avoid the endpoint surfaces. Apply (15.12); its geometric factor is a positive power of ε. Decreasing the exponent absorbs fixed constants. The union bound is over the environment, not over random paths. □

Lemma 15.7 (Escape from deterministic endpoint surfaces)**.** *Suppose A also has regular weak surface estimates. For any $a, M > 0$ and any failure probability $\eta > 0$, there is $\tau > 0$ such that, outside that failure probability, every $A^{K_{0,R}}$-geodesic from S_R to S_{2R} with length in $[aS_R, MS_R]$ has its central half at Euclidean distance at least τR from both endpoints. The assertion is uniform in R.*

Proof. Choose uniform upper quantiles for the intrinsic A-Hölder constants on S_R, S_{2R} in fixed buffers inside $K_{0,R}$, and bound $s_R(0)/S_R$ and its reciprocal. The logarithm of this last ratio is a Gaussian increment of uniformly bounded variance.

For a small $t > 0$, the simultaneous cover at radii in $[t^2R, tR]$ gives about each endpoint x a good cube with a definite inner buffer about x. Its separator diameter is at most Λ times its crossing. On the same deterministic macroscopic surface, choose y at Euclidean distance between fixed multiples of the local radius from x, outside the local outer cube. A bounded chain of faces supplies such a point even at a corner. The macroscopic Hölder bound gives an A-path from x to y in the fixed buffer with cost at most $CHt^\beta s_R(0)$.

This path crosses the good shell. Its prefix gives access to the separator, and its entire length bounds the crossing. Every separator point consequently has constrained distance from x at most $C(1+\Lambda)Ht^\beta s_R(0)$. The geodesic's prefix to its last separator contact is no longer than this, by constrained minimality. Choose t so small that this time is below $aS_R/4$. After that contact the geodesic stays outside the local inner cube, which contains a Euclidean ball of radius at least a fixed multiple of t^2R about x. Reverse time for the other endpoint. Allocate the quantile and covering failures to total less than η. □

Lemma 15.8 (From annular to buffered witnesses)**.** *For two regular weak metrics with deterministic bi-Lipschitz comparison, there is $\tau > 0$ such that*

$$\mathbb{P}[H_r(c)] \geq p_* \quad \Longrightarrow \quad \mathbb{P}[G_r(\tau, c)] \geq \tau, \tag{15.14}$$

uniformly in r and $c < C$. A smaller threshold on the right is also allowed.

Proof. Use the middle subpath in Definition 15.1. It is minimizing in $K_{0,r}$, lies in the closed middle annulus, and has endpoints separated by at least r. Its normalized length is bounded above by the parent bound and below by a crossing lower quantile, outside arbitrarily small probability. Its image has a fixed positive clearance from $\partial K_{0,r}$. Lemma 15.7 supplies the central-half separation uniformly over all possible witnesses. Intersect these common events with $H_r(c)$, allocating total failure below $p_*/2$, and decrease τ below every resulting bound and below $p_*/2$. □

Corollary 15.9 (Uniform witnesses under exact covariance)**.** *If A, B both have exact scale covariance and C is optimal in (15.1), then, using common power units,*

$$\mathbb{P}[H_r(c)] \geq p_* \qquad (r > 0,\ C/2 \leq c < C). \tag{15.15}$$

Only B needs a surface upper estimate.

Proof. For fixed c, exact covariance makes $\mathbb{P}[H_r(c)]$ independent of r. If it were below p_*, every level would be bad in Lemma 15.3. Lemma 15.4 supplies a positive-probability buffered witness at $c^\sharp < C$, a contradiction. This argument does not use Lemma 15.7 or an A surface bound. □

16 Marked trees and a one-sided shortcut theorem

The deterministic argument in this section assumes $E \leq CD$. Both metrics have Euclidean-continuous finite paths and the compact inverse modulus. Only E has the small separators and continuous intrinsic surface restrictions. Every tree edge and every inserted shortcut is measured in E, never in D.

All canonical image and access bounds in this section are intrinsic in one fixed open working buffer. More precisely, when that buffer is U_c, write $\mathcal{E} = E(\cdot, \cdot; U_c)$. Lengths of paths contained in U_c are the same for $\mathcal{E}$ and E: intrinsic distances are no smaller than global ones, while the subpaths themselves give the reverse inequality for their partition lengths. In particular, a D-path contained in U_c has $\mathcal{E}$-length at most C times its D-length. Global closeness is not substituted for intrinsic closeness.

16.1 Avoidance with control of the canonical image

Lemma 16.1 (Compact subsets of a finite forest)**.** *If F is a finite topological forest and $A \subset F$ is compact, then $\check{H}^j(A; \mathbb{Z}) = 0$ for $j \geq 1$. If F is embedded in $\mathbb{R}^d$, $d \geq 3$, and S is a cubical boundary, then $S \setminus A$ is path connected.*

Proof. Give the forest any positive edge-length metric. For every $\eta > 0$, the open η-neighborhood of A in F has finitely many components, each a subtree and hence contractible. Distinct components contain points of A at distance at least 2η, so finiteness follows from compactness. These neighborhoods are cofinal among the neighborhoods of A in the finite polyhedron F. The neighborhood description of Čech cohomology gives its vanishing in positive degree. This reasoning includes disconnected and non-locally-connected compact subsets.

The set $S \cap A$ is a compact subset of a finite forest. Alexander duality for arbitrary compact subsets of a sphere uses Čech cohomology, rather than singular cohomology without a local-contractibility hypothesis [6, Section 3.3, pp. 255–257]. It gives

$$\widetilde{H}_0(S \setminus A; \mathbb{Z}) \cong \check{H}^{d-2}(S \cap A; \mathbb{Z}) = 0. \tag{16.1}$$

The empty intersection is harmless. The complement is open in the manifold S, so it is locally path connected; (16.1) makes it path connected. The use of a forest, rather than a graph with cycles, matters in dimension three. □

Lemma 16.2 (Avoidance with intrinsic image control)**.** *Let $O \subset U_c$ be open, let $F \Subset O$ be an embedded finite E-rectifiable forest, and let $A \subset F$ be compact. If γ is a finite E-rectifiable path in O whose endpoints avoid A, then, for every $\eta > 0$, there is a finite E-rectifiable path with the same endpoints, avoiding A, contained in O, and with image in the $\mathcal{E}$-distance-η neighborhood of the image of γ.*

Proof. For each point of the compact set $A \cap \gamma([0,1])$, choose a small rational separator from Lemma 12.7. Its closure is in O, avoids the endpoints, and each boundary point has intrinsic access of cost below $\eta/10$ to its anchor on γ. Choose finitely many whose interiors cover the intersection, and remove cubes contained in others.

For one boundary S, Lemma 16.1 makes $S \setminus A$ path connected. Two overlapping nonnested cubes have a boundary intersection outside A. Otherwise, on the first boundary, its relatively open portions inside and outside the second cube would separate $S \setminus A$. Both portions are nonempty and neither can be contained in a forest: an open patch of a manifold of dimension at least two contains an embedded circle, whereas a forest does not. This contradicts (16.1). Only overlapping interiors are used; a tangency of closures does not connect an open cube union.

The intersection graph routes an entrance contact to an exit contact of each component of the finite open cube union along finitely many boundaries avoiding A. Those entrance and exit contacts avoid A, because every point of A on the original path is in the open union. Process the first entry and last exit of a component, and continue chronologically. Each component is used at most once; visits to other components may be swallowed. This gives finitely many Euclidean-continuous boundary detours. Every new boundary point is intrinsically close to its anchor on the original image.

The boundary detours are not yet assumed rectifiable. Their images lie in a finite union of rational boundaries on which $\mathcal{E}$ is continuous. Each detour is therefore $\mathcal{E}$-continuous and compact. It has positive $\mathcal{E}$-distance from A and from the boundary of a fixed open buffer compactly contained in O, by compact separation. Partition it into finitely many $\mathcal{E}$-small steps and replace the steps by almost-minimizing paths in U_c, with costs below these barriers and below $\eta/10$. The replacements avoid A, remain in O, and stay intrinsically close to the detour. Their concatenation with the retained original pieces proves the claim. □

16.2 Embedding protected marks

A protected mark is a simple E-rectifiable arc with a designated closed middle subarc $R = [u_R, v_R]$ and nonempty open attachment stubs on its two sides. It carries a number $a_R > 0$ such that

$$\operatorname{len}_E(R) < c_0 a_R, \qquad D(u_R, v_R) > a_R, \qquad c_0 < C. \tag{16.2}$$

The second distance may be infinite. Marks and their stubs will be small in intrinsic canonical image; their total lengths need only be finite.

Lemma 16.3 (An embedded marked spanning tree)**.** *Put*

$$V = A_{3/2,5/2}(0), \qquad U_c = A_{1/2,4}(0). \tag{16.3}$$

Suppose a finite connected coordinate mesh in V has vertices forming an intrinsic canonical net accessible from ∂V at cost at most η_0. Each edge of an abstract spanning tree of this mesh has a protected mark in its own disjoint box. The mark and stubs have $\mathcal{E}$-image diameter at most a, and each incident vertex has an E-path to its designated stub in V whose image is within $\mathcal{E}$-distance a of that vertex. Neighboring vertices have $\mathcal{E}$-distance at most a.

For a sufficiently small relative to a prescribed $\delta_{\mathrm{im}} > 0$, these data yield an embedded finite E-rectifiable tree $T \Subset V$ containing all protected middles, with

$$\sup_{x \in \partial V} E(x, T; U_c) \le \eta_0 + O(a), \tag{16.4}$$

$$\mathcal{E}(x, y) \le \delta_{\mathrm{im}} \quad \text{if the simple tree arc from } x \text{ to } y \text{ contains no full protected middle.} \tag{16.5}$$

Its total intrinsic edge length L_T is finite; no prior upper bound on L_T is required.

Proof. Start with the disjoint forest consisting of all marked arcs and mesh vertices. Attach the incidences of the abstract spanning tree one at a time. Permit contacts with the old forest only in a small relative neighborhood of the intended source vertex cluster and an open subinterval of the designated target stub. Choose their closures inside those sets and disjoint from each other. The remaining forbidden set is a compact subset of the current finite forest.

Apply Lemma 16.2 to the prescribed connection, with intrinsic image tolerance at most a. Its contacts with the old forest are now in the two permitted neighborhoods. Take its last source-cluster contact before its first target-stub contact. The intervening path meets the old forest only at its endpoints. Minimize length inside this path's compact rectifiable trace, using Lemma 12.2, to remove loops without changing its image bounds. Add the resulting simple arc.

Each attachment joins different components. Contracting the initial marked arcs and vertex clusters gives a subdivision of the chosen abstract spanning tree; an incidence in this abstract tree cannot close a cycle. The final object is thus an embedded finite tree. Each attachment meets the assigned open stub beyond its protected middle, and contact with all other protected pieces was forbidden. No protected middle is bypassed.

Every new incidence starts from its original mesh vertex before the last-contact trimming, and its image stays within a fixed multiple of a of that vertex. The image bound for a vertex cluster therefore does not accumulate over earlier attachments. After deleting the protected open middles, the remaining components are vertex clusters with their incident stubs and partial marks. A simple tree arc containing no full mark either stays in one mark or meets at most one such cluster. Its endpoints are at intrinsic distance at most a fixed multiple of a. Decrease a to get (16.5). The original net vertices remain in T, giving (16.4). Finitely many finite-length arcs give $L_T < \infty$. □

The mesh need not lie on a rectifiable hypersurface. Intrinsic Hölder bounds on a fixed coordinate complex in V give short ambient connections between nearby mesh vertices. Net points on ∂V are moved slightly into V along finitely many normal coordinate plaques, which give access in U_c. The surface modulus covers face edges and corners. Section 17 makes these choices with fixed high probability and a deterministic finite number of trials.

16.3 Boundary-access neighborhoods and cut neighborhoods

Give T its intrinsic edge-length metric d_T. The topology of this finite metric tree is its Euclidean embedded topology. Since $\mathcal{E} \le d_T$ and $\mathcal{E}$ separates points, compactness gives

$$\omega_T(t) := \sup\{d_T(x,y) : x,y \in T,\ \mathcal{E}(x,y) \le t\} \longrightarrow 0 \qquad (t \downarrow 0). \tag{16.6}$$

This inverse continuity is on one fixed compact tree, not on an open volume.

Lemma 16.4 (Controlled boundaries and compulsory cut neighborhoods)**.** *Let $\lambda_n \downarrow 0$ and choose $\eta_n = o(\lambda_n)$. There are open neighborhoods $N_n \Subset V$ shrinking Euclideanly to T, such that every point of ∂N_n has intrinsic E-access inside U_c to an anchor on T of cost at most η_n.*

For each protected cut $c \in \{u_R, v_R\}$ there is a rational cubical cut neighborhood about c, included in N_n, with boundary access to c at most η_n, diameter tending to zero, and closure disjoint from the other cut neighborhoods. Outside that cut neighborhood, the two sides of the tree cut lie in disjoint neighborhood regions. A path changing sides within N_n must enter the cut neighborhood. Boundary anchors outside the cut neighborhood can be chosen on the assigned side.

Proof. Use the small-separator lemma at each of the finitely many cuts. Choose rational cubes with positive inner buffers, costs below η_n, radii below $1/n$, and pairwise disjoint closures. Make the cut neighborhood about any other cut lie on the assigned side of the present cut. This is possible because distinct cuts have positive separation from the opposite compact branch.

The two compact branches outside a smaller inner cube about a cut are disjoint. Choose disjoint neighborhoods of those branch pieces, with disjoint closures outside the larger cut neighborhood. Do this for each cut before choosing the remaining covering cubes. Cover the rest of T by rational good cubes of diameter at most $1/n$, each with an anchor on T and boundary access below η_n. Make the part of each cube outside every cut neighborhood stay in the assigned branch neighborhood for that cut. An anchor strictly inside a cut neighborhood needs no additional cube. An anchor on a cut neighborhood boundary belongs to exactly one of the compact branch pieces and has an open neighborhood on that assigned side outside the cut neighborhood. The finitely many cut constraints are compatible after shrinking the cube.

Compactness gives a finite cover. Let N_n be the union of the cut neighborhoods and the covering cubes. Its boundary is contained in their boundaries, giving the one-cube access bound. Outside a cut neighborhood, the constituent cube and its anchor are on the assigned branch, or are another cut neighborhood already chosen on that branch. The disjoint branch regions force every internal change of sides to pass through the cut neighborhood. Choose all access costs below the positive exit barriers of the fixed working buffer. All paths then remain in U_c. The maximal cube diameter tends to zero, so the union shrinks to T. □

16.4 The one-sided all-path dichotomy

Fix $c_0 < c_1 < C$, $a_0 \in (0,1)$, and $\epsilon_0 > 0$. Choose a smooth baseline weight $w_0 \in [a_0, 1]$ such that $-\log w_0$ has compact support in V and $w_0 \le a_0$ on the fixed gain slab

$$W = A_{15/8,17/8}(0).$$

The compact support is important: it gives positive clearance from the collector boundary. Put

$$K_c = \{1 \le |x|_\infty \le 7/2\}, \qquad V^+ = A_{5/4,11/4}(0). \tag{16.7}$$

Assume the D crossings of the untouched boundary bands between $S_1, S_{9/8}$ and between $S_3, S_{7/2}$ cost at least $a > 0$. These boundary bands are disjoint from V, and the outer one lies outside V^+. Choose parameters with strict reserves so that

$$\delta_0 < a/4, \qquad 2C\delta_0/a_0 + \delta_{\mathrm{im}} < \epsilon_0/2, \qquad 3\eta_0 < c_1\delta_0/2. \tag{16.8}$$

Absorb the $O(a)$ mesh-access error in (16.4) into η_0 by making the mesh tolerance smaller; this mesh tolerance is unrelated to the boundary band constant a.

Choose smooth weights with

$$\lambda_n \le w_n \le w_0 \le 1, \qquad w_n = \lambda_n \text{ on } T, \qquad w_n = w_0 \text{ outside } N_n. \tag{16.9}$$

A smooth cutoff equal to one near T and compactly supported in N_n gives such a weight by interpolation. Its logarithmic discount is compactly supported in V.

Theorem 16.5 (Shortcut or small contact)**.** *For every sufficiently large n there is $b_n > 0$ such that every finite D-rectifiable path P with endpoints on ∂V has at least one of the following properties:*

(a) a subinterval, with endpoints in V^+, has w_n-weighted D-length greater than b_n and admits a w_n-weighted E-shortcut in U_c of cost strictly less than c_1 times that interval length;

(b) its endpoints have original intrinsic distance $E(P(0), P(1); U_c) < \epsilon_0$.

Only the tree's E-rectifiability is assumed.

Proof. 1: long paths and exterior portions. Boundary access followed by a tree route has weighted canonical cost at most

$$2\eta_0 + \lambda_n L_T < 3\eta_0 < c_1\delta_0/2$$

for all large n. Every path of weighted D-length at least δ_0 therefore has a whole-interval shortcut. A path leaving K_c crosses an untouched boundary band and has cost at least $a > \delta_0$. It remains to treat short paths confined to K_c.

Outside N_n such a path has original D-length at most δ_0/a_0. If it never enters N_n, length comparison in the working buffer gives (b). Otherwise let x_n, y_n be its first and last ∂N_n contacts, and let Q_n be the intervening path. The two exterior portions have total original canonical length at most $2C\delta_0/a_0$. Choose boundary anchors $\alpha_n, \beta_n \in T$ of cost at most η_n. A weighted canonical connection for Q_n costs at most $2\eta_n + \lambda_n L_T$. Fix $T_0 > 2L_T/c_1$. If its weighted D-length exceeds $T_0\lambda_n$, this is a strict shortcut for large n.

2: the limiting path lies in the tree. Suppose failing paths exist for infinitely many n in the remaining case. Parametrize Q_n at weighted D-speed λ_n on $[0, T_0]$, extending constantly after its endpoint. Since $w_n \ge \lambda_n$, these paths are 1-Lipschitz for the original D. Their images lie in the fixed compact K_c. The inverse modulus and Arzelà–Ascoli give a Euclidean-uniform limit Q, which is D-Lipschitz by lower semicontinuity.

The total original D-length outside N_n is at most $T_0\lambda_n/a_0$. If $Q(t) \notin T$, then $Q_n(t)$ eventually lies outside N_n. Its exterior excursion has a boundary endpoint at D-distance at most this quantity. A subsequential limit of those boundary endpoints belongs to T, because N_n shrinks to T. Lower semicontinuity makes its D-distance from $Q(t)$ zero, a contradiction. Thus $Q \subset T$. The boundary anchors converge to its endpoints, both Euclideanly and in d_T.

If the simple tree arc between the limiting endpoints contains no full protected mark, (16.5) gives an original intrinsic canonical connection of cost at most δ_{im}. Join the actual anchors to the

limiting endpoints by short tree pieces, add the boundary access and the two exterior portions, and obtain

$$E(P_n(0), P_n(1); U_c) \le 2C\delta_0/a_0 + \delta_{\mathrm{im}} + o(1) < \epsilon_0.$$

The middle connection here need not follow a possibly long tree arc. This is (b).

3: ordered access to a protected mark. Otherwise the limiting tree arc contains a full protected middle $R = [u_R, v_R]$. First suppose a cut c lies strictly between the limiting endpoints. Choose fixed short pre- and post-stubs on its two sides, using disjoint stubs in tree order for the two cuts of R. Any continuous tree path between the limiting endpoints has chronological passages through these stubs. Choose a last pre-stub contact before a first post-stub contact.

The corresponding prelimit points are Euclidean-close to those stubs. A point outside N_n can be moved to the appropriate endpoint of its exterior excursion at original D-cost $O(\lambda_n)$. Its intrinsic canonical cost is $O(\lambda_n)$ because the excursion stays in $K_c \subset U_c$. The inverse compact modulus preserves Euclidean proximity to the stub, and its boundary anchor converges in the intrinsic tree topology. These adjustments preserve the order: an exterior excursion of length tending to zero cannot contain points in two disjoint fixed stub neighborhoods, whose compact closures have a positive D separation.

Between the negative and positive branch contacts, the path either hits the cut neighborhood boundary or changes branches outside the cut neighborhood. A cut neighborhood hit has access at most η_n to c. In the second case, the last negative-region contact before the first positive-region contact bounds an exterior excursion. For fixed n the two branch regions on the compact path portion avoiding the closed cut neighborhood have positive separation, so this single excursion exists. Its boundary anchors k_n^-, k_n^+ are on opposite branches and satisfy

$$\mathcal{E}(k_n^-, k_n^+) \le 2\eta_n + CT_0\lambda_n/a_0.$$

Their tree arc passes through c. The weighted canonical access from either actual excursion endpoint to c is therefore bounded by

$$\eta_n + \lambda_n \omega_T(2\eta_n + CT_0\lambda_n/a_0) = o(\lambda_n). \tag{16.10}$$

If a cut is a limiting endpoint of Q, use the corresponding actual endpoint and its anchor instead; its access is $\eta_n + \lambda_n o(1) = o(\lambda_n)$. The construction at the other cut remains chronologically ordered.

We have obtained ordered actual points z_n, z_n' with weighted canonical access $o(\lambda_n)$ to u_R, v_R. Their unweighted access tends to zero too: the tree part has length bounded by $\omega_T(O(\lambda_n)) + o(1)$. Thus the points converge Euclideanly to the cuts. By (16.2) and lower semicontinuity, $D(z_n, z_n') > a_R$ for large n. The intervening weighted D-length is at least $\lambda_n a_R$, whereas the route through the protected middle costs

$$o(\lambda_n) + \lambda_n \operatorname{len}_E(R) + o(\lambda_n) < c_1\lambda_n a_R.$$

The endpoints are in V^+ and the shortcut is in U_c, giving (a), a contradiction.

Finally take b_n smaller than

$$\tfrac{1}{2}\min\{\delta_0, T_0\lambda_n, \lambda_n \min_R a_R\}.$$

All produced inequalities have strict reserves. Failure for arbitrarily large n would give exactly the counterexample sequence just excluded. □

16.5 The local formulation used by the counting argument

Definition 16.6 (A strict working discount)**.** A nonnegative $f \in C_c^\infty(V)$, with $w = e^{-\psi f}$, is a strict working discount with constants $(a, A, g, b, \epsilon_0, c_1, \delta_0)$, where $0 < b < \delta_0/2$, if the following hold in unit scale, with spare strict margins:

(i) The two untouched boundary band crossings cost at least a, the gain-slab crossing costs at least a_W, and $w \le a_0 < 1$ on that slab, where $g < (1 - a_0)a_W$.

(ii) The original intrinsic E diameter of ∂V in U_c is at most A, and every two points of ∂V have a weighted canonical connection in U_c of cost less than $c_1\delta_0/2$, with $\delta_0 < a/4$.

(iii) Every constrained $(w \cdot D)^{K_c}$-geodesic between points of ∂V, of length below δ_0, has either a shortcut as in Theorem 16.5(a) with threshold b, or original intrinsic endpoint E-distance less than ϵ_0.

The inequalities are multiplied by the local unit after a translation and dilation. The event is determined in a fixed open annular block containing $\overline{U_c}$ and the averaging sphere.

The universal geodesic assertion is measurable in the completed local sigma field. Compact constrained metrics and path fibers are Borel. On a short geodesic confined to K_c, strict shortcut tests can be restricted to rational times, by length comparison along that path. Existential failure is analytic and its universal complement is coanalytic. Both are universally measurable. The original all-path theorem, rather than only its geodesic restriction, will be used to approximate a random discount by deterministic smooth functions.

Lemma 16.7 (Stability under other discounts)**.** *Place finitely many working discounts in pairwise disjoint closed working balls. Let P be a geodesic for the simultaneously discounted competitor. For any two collector-boundary contacts at one site, either their path interval has an active shortcut with that site's constants, measured in the actual multiply discounted metric, or the contacts have original intrinsic canonical distance at most the site's ϵ_0 bound. The shortcut image lies in that site's working buffer.*

Proof. An interval leaving the site's K_c crosses an untouched boundary band. Other discounts vanish there, so its actual length is at least $a > \delta_0$. The cheap boundary connection in Definition 16.6(ii) is then a whole-interval shortcut. The same argument covers an interval staying in K_c but of length at least δ_0. In both cases its length is strictly above b, by $b < \delta_0/2$.

A shorter interval stays in K_c, where the other bumps vanish. It is a constrained geodesic for the single local discount, since a shorter constrained replacement would shorten the global discounted geodesic. Definition 16.6(iii) applies. All active lengths are measured where they are unchanged, or by the preceding boundary-band alternative. The inactive bound is in the original canonical metric. □

In particular, if no active interval occurs at a site, its entire collector contact set has small original intrinsic E diameter. This is stronger than a bound for one first–last contact pair, and remains valid after earlier chronological deletions change that pair.

17 Uniform local discounts

The weak uniqueness argument must work before exact dilation covariance is known. A positive-probability configuration at one radius is not enough: the counting argument needs a fixed finite

collection of discounts at a positive density of radii. We obtain this uniformity by enhanced compactness and configurations on fixed deterministic surfaces. Throughout the two-sided part of this section,

$$c_* D \le E \le C D, \qquad 0 < c_* < C < \infty, \tag{17.1}$$

and the pair has uniform regular weak estimates, with a common family of raw constants c_r. The one-sided strong case is treated separately at the end.

17.1 Enhanced compactness with the normalization data retained

Lemma 17.1 (Closure of uniformly regular weak pairs)**.** *Let (E^j, D^j) be regular weak pairs with uniform constants in the relative-power estimates, uniform surface Hölder quantiles, and uniform crossing and reciprocal-crossing quantiles. Normalize their common raw unit-scale constant to one, and suppose* (17.1) *holds with fixed constants. There is a subsequence converging in an enhanced coupling to a regular weak pair (E, D) with the same comparisons. The limiting pair is jointly conditionally local, field measurable, and has simultaneous continuous Weyl assignments.*

The enhanced convergence includes the global restrictions to every fixed rational coordinate complex and the buffered intrinsic arrays. Any fixed finite or countable set of field averages and raw scale ratios can be retained. At those retained radii the limiting raw constants can be taken to be the actual limits of the prescribed constants, rather than replaced by comparable medians. Compact-domain paths of bounded length may also be retained.

Proof. 1: arrays and a global lower-semicontinuous limit. Extract the restrictions to all rational finite coordinate complexes, their intrinsic restrictions in rational open buffers, distances between rational cube boundaries, and the field with its averaging processes. Surface Hölder tightness gives tightness of the functional coordinates. A fixed finite surface connection bounds each boundary distance from above. A finite family of annuli separating two disjoint boundaries gives reciprocal tightness. The relative-power estimates allow diagonal extraction of the deterministic raw constants at the stated countable radii. All coordinates live in a countable product of Polish spaces.

In a Skorokhod representation, define the candidate distance between distinct x, y as the increasing limit of the extracted set distances between shrinking nested rational cube boundaries about x, y. Equivalently, it is the supremum of the extracted boundary distances over disjoint rational cube neighborhoods containing the two points. This is a jointly measurable lower-semicontinuous function, positive off the diagonal by the limiting annular barriers. A path between nearby endpoint pairs crosses each fixed pair of surrounding boundaries; hence the lower-limit inequality holds. Slowly shrinking recovery boundaries give recovery pairs. This proves epigraph convergence, without yet using the triangle inequality for the candidate.

2: the triangle inequality uses the arrays before intrinsic identification. Good-separator estimates for the prelimit pairs follow from Lemmas 4.2 and 4.4. Pass their closed inequalities to each finite collection of extracted coordinates. A finest-grid union bound and Borel–Cantelli give, at every point, arbitrarily small nested shells with disjoint closures and extracted separator diameter H_i at most Λ times the extracted crossing X_i.

Suppose the candidate distances from x to z and from z to y are finite. Recovery paths cross any fixed finite collection of these shells about z, on disjoint time intervals. Thus $\sum_i X_i < \infty$, and $H_i \to 0$. In the prelimit, truncate the two recovery paths at the middle separator and join their contacts using the extracted internal coordinate for that separator. The resulting cost is at most the sum of the two recovery costs plus $H_i + o(1)$. First pass to the limit, then let $i \to \infty$. This proves the triangle inequality. In particular, this step does not assume that an extracted internal array is already the actual intrinsic metric of the candidate.

3: fixed-point consistency and length structure. Fix a rational point p and a rational plaque through it. A plaque Hölder bound connects p to a point of a small surrounding rational cube boundary at cost tending to zero in probability, uniformly in j. The whole boundary has tight diameter in its own unit, and that unit tends to zero in probability by Lemma 4.5. Consequently, for every $a > 0$,

$$\lim_{r\downarrow 0} \sup_j \mathbb{P}\left[\sup_{x\in S_r(p)} E^j(p,x) > a\right] = 0, \tag{17.2}$$

and likewise for D^j, along rational radii. Choose a sufficiently fast sequence of such radii and retain its arrays. The corresponding limiting failure probabilities are summable. The prelimit triangle inequalities and the vanishing limiting access costs identify every extracted rational-point value with the boundary-defined global limit.

On a fixed rational coordinate complex the lower-limit inequality bounds the global limit above by its continuous extracted restriction. Rational-point consistency, the triangle inequality and the common Hölder modulus give the reverse inequality at every pair. Thus the extracted global restriction is the actual restriction of the limiting metric.

At large radii the raw lower-power bound, the Gaussian average estimate and reciprocal crossing tightness make crossing costs tend to infinity in probability, uniformly in j. Choose a sparse sequence with summable lower-tail failures. Its limiting barriers confine bounded-cost recovery paths to a Euclidean compact set. Positive rational annular barriers in that set give equicontinuity. Compact path limits therefore supply a geodesic for every finite pair. The deterministic part of Lemma 14.2, applied to the large barriers just obtained, gives completeness and the remaining coercive properties; it does not wait for a choice of raw constants at every real radius. The same compactness retains any additional bounded-length path coordinates.

For a fixed complex $S \Subset U_0 \Subset U$, Lemma 12.6 gives

$$E(\cdot,\cdot;U)|_{S^2} \le L^{U_0}|_{S^2}.$$

This proves the actual intrinsic surface upper bounds. Enlarge the separator buffers and thin the shells if necessary, then apply Lemma 12.7. Lemma 12.8 now reconstructs all actual intrinsic metrics. This order avoids using local access before the internal arrays have been related to genuine paths.

4: raw constants and local Gaussian structure. The retained positive raw constants obey the same relative-power bounds. At other radii choose the median of the normalized limiting crossing. Uniform crossing and reciprocal quantiles compare these medians with any extracted raw constant by fixed factors. The resulting Borel family, modified on the retained countable set, satisfies the relative-power estimates with enlarged uniform constants. To check a marginal estimate at an additional fixed radius and center, extract its tight surface arrays as well. Lemma 12.9 identifies the resulting global restrictions even when their coordinates are nonrational. The already constructed limiting global law is unchanged, and the same median comparison gives the required estimate. No simultaneous extraction over all real radii or centers is asserted.

Before taking limits each metric is a local measurable function of the field. Fix an open conditioning region and subtract the smooth Gaussian conditional mean on exterior compact buffers, as in Proposition 13.3. The joint interior block and the residual-field exterior block are independent. Retain those Gaussian coordinates in the extraction. Local weighted recovery identifies the shifted limiting metrics, so independence of the joint blocks gives conditional locality for the limiting pair. Lemmas 13.4 and 13.5, followed by Proposition 13.6, remove residual randomness, one metric coordinate at a time.

For a single Borel assignment, use the intrinsic coordinates of the present sequence in place of the LFPP expressions in (13.9). On arbitrary distributional inputs, define each open-domain prelimit coordinate as the countable infimum of compactly constrained metrics over rational compact polycubical subsets of that domain. Lemma 12.2 makes this a Borel formula. On every relevant field-plus-continuous-function input it is the actual intrinsic coordinate, because finite paths are Euclidean continuous and have compact images.

Measurability of the limit gives an almost-sure further subsequence on the original field space. Intersect the simultaneous Weyl events for the countably many prelimit assignments. Exact prelimit Weyl scaling and local weighted recovery then give the limiting formula simultaneously for every continuous perturbation on a good base-field event. This establishes a Borel assignment and its Weyl rule without asserting pointwise locality of a prelimit version on every distributional input.

To check locality, now fix any possibly dependent random continuous f and any deterministic open U. For this input law, every prelimit intrinsic coordinate in a buffer $V \Subset U$ is measurable from $(h+f)|_U$ by the prelimit locality axiom. Their countable limsup and exhaustion infimum are measurable from the same restriction. The simultaneous reconstruction identifies that formula with the actual limiting intrinsic metric. Thus locality holds for each required input law, exactly with the quantifiers of Axiom II. No common exceptional set over all laws of f is needed.

Translation covariance passes in the joint law and becomes the pathwise identity for each fixed translation after field measurability. Finally, the two comparisons pass to the lower-semicontinuous limits by using a recovery sequence for the metric on the right of each inequality. □

17.2 Low-ratio configurations on fixed surfaces

Choose

$$1/C < b_0 < b_1 < b_2 < 1/c_*. \tag{17.3}$$

A low-ratio witness is the event $H_r^{\mathrm{low}}(b_2)$ of Definition 15.1 for the ordered pair $(A,B) = (E,D)$.

Lemma 17.2 (Finitely many surface configurations)**.** *For every $p_0 > 0$ there is a finite collection of deterministic rational surface configurations and positive constants $a_-, a_+, A_{\mathrm{acc}}, \tau$ with the following property. If*

$$\mathbb{P}[H_r^{\mathrm{low}}(b_2)] \geq p_0,$$

then with probability at least $p_0/2$ one of the configurations, in coordinates centered at the witness and divided by r, has two disjoint small rational cube boundaries $S^-, S^+ \subset B_4$ and a number $a \in [a_-, a_+]$ from a finite list such that:

(i) Their mutual D distance, and their D distance to S_6, are at least $(1+\tau)as_r$.

(ii) An E-rectifiable path in $\overline{B}_5$ joins S^- to S^+ with length at most $(1-\tau)as_r/b_0$.

(iii) An E-rectifiable path in $\overline{B}_6$ connects the center to that path with length at most $A_{\mathrm{acc}}s_r$.

The lower bounds can be tested in B_7 using the exit bounds in (i), so the configuration is local. Its finite geometry, constants and probability are uniform in r. With a fixed relaxation of the reserved margins, these configurations persist in every enhanced limit of Lemma 17.1 that retains their radii and units.

Proof. On the low-ratio event take its middle E-geodesic, of length $L \leq A_H s_r$, for which $D^{K_{0,r}}(u,v) > b_2 L$. Intersect with uniform regularity events of total failure probability below $p_0/8$. They give a lower bound $L \geq a_H s_r$ and a positive lower bound $a_{\mathrm{exit}} s_r$ for the D distance

from the closed middle annulus to $\partial K_{0,r}$ and to S_{6r}. The latter are uniform compact-separation quantiles from Lemma 14.3.

Choose a fixed integer N so large that $A_H/(Nc_*) < a_{\mathrm{exit}}/10$ and $b_1 A_H/N < a_{\mathrm{exit}}/10$, and partition the middle geodesic into N equal E-length pieces. These choices are made before choosing the trimming fraction and the surface-access error below. The inequality $D \le E/c_*$ bounds the constrained D distance of each piece by its E length divided by c_*. This is below the exit barrier. Global and constrained D distances of its endpoints therefore agree. The constrained triangle inequality forces one piece, of length $\ell = L/N$, to satisfy $D(x,y) > b_2\ell$.

Trim equal small fractions from its two ends, retaining nonempty attachment stubs. The loss in the lower D bound is at most the total trimmed E length divided by c_*. Choose the fraction so that the core still has endpoint D distance greater than $b_1\ell$, while its E length is at most ℓ. The length ℓ/s_r lies in the fixed positive interval $[a_H/N, A_H/N]$.

Apply Corollary 15.6 to the diagonal ordered pair (E,E), with $C = 1$ and defect zero, along the original parent. The core and stub cuts are a fixed positive metric time apart and stay in the middle annulus, away from the parent's endpoint surfaces. Outside an additional probability below $p_0/8$, their Euclidean separations are bounded below uniformly. Good rational separators around the two core cuts can then be selected from a fixed finite radius block and spatial grid. Choose the block sufficiently small. Their intrinsic E diameters and their access from the cuts are at most ηs_r, with a further failure probability below $p_0/8$, by the same occupation estimate and good-separator inequality. Their access paths stay in a fixed buffer inside B_{5r}, and the two cubes are disjoint.

Every point of a selected boundary is within E distance ηs_r of its cut, and hence within D distance $\eta s_r/c_*$. The full-boundary mutual D distance is at least the core distance minus $2\eta s_r/c_*$; the exit bounds lose no more than the same access error. The core and its two access paths give an E connection of cost at most $\ell + 2\eta s_r$. Since $b_0 < b_1$, choose η sufficiently small and then a finite grid of values of a so that the lower bounds exceed $(1+\tau)as_r$ and the upper bound is below $(1-\tau)as_r/b_0$, for one $\tau > 0$. The choice of N ensures that the grid can be chosen below the exit quantile.

There are only finitely many selected boundary pairs. Connect the center to each possible initial boundary through a fixed coordinate complex in B_6. Uniform intrinsic surface bounds give a common access quantile A_{acc} with failure probability below $p_0/8$. Trimming a connecting path at its last initial-boundary contact and first final-boundary contact, and minimizing in its compact trace, makes it simple when required. The four allocated error probabilities leave probability at least $p_0/2$.

For persistence, the lower bounds in (i) are distances between fixed full surfaces. They pass under uniform convergence of the retained global restrictions. The bounded-length paths in (ii)–(iii) have compact images in fixed closed cubes, so their limits have no greater E length. The retained raw ratios and averages make the units converge. Pass the non-strict inequalities with the factors $1 \pm \tau$ as closed events, then relax those factors. In particular, no lower bound at moving endpoints is passed merely by lower semicontinuity. A path below the tested threshold cannot leave the local buffer because of the exit bounds; this proves locality of the test as well. □

Remark 17.3. The implication $D^j(x_j, y_j) > a \Rightarrow D(x,y) > a$ is false for general lower-semicontinuous convergence. The configuration avoids it by first replacing moving endpoints by entire fixed boundaries, using $D \le E/c_*$. This two-sided step belongs only to the weak construction. It is not used for the final original-V competitor.

17.3 Independent fine trials and marked configurations

Lemma 17.4 (Fine-field amplification)**.** *Suppose a finite-menu local configuration has probability at least p_0 at a microscopic radius, uniformly in translations, and has fixed strict reserves. Given finitely many macroscopic edge boxes and an error probability $\eta > 0$, one can choose a fixed number of trials in each box, a coarse/fine separation factor L, and a sufficiently small microscopic ratio $\rho > 0$, so that every box contains a relaxed configuration with probability at least $1 - \eta$. Its microscopic unit and access costs may simultaneously be required to be as small a fraction of the macroscopic unit as prescribed.*

For weak metrics the assertion applies at those macroscopic radii R where the configuration probability at ρR has the stated lower bound. All choices are uniform over such R.

Proof. Split the compact-kernel white noise at scale $L\rho R$. On a trial region of diameter $O(\rho R)$ subtract one value of the smooth coarse field. Lemma 4.5 makes its oscillation smaller than any fixed tolerance with probability tending to one as $L \to \infty$, uniformly in radius and center. Removing or restoring such an oscillation changes every normalized local length and distance by factors $e^{\pm C\psi t}$. Include the averaging sphere in the trial region and choose t below the configuration's reserve. A full-field success then implies a relaxed fine-field success except on probability below $p_0/2$.

The fine field below $L\rho R$ restricted to a trial is determined by a spatial footprint of radius $O(L\rho R)$. The fine field is the original LCGF plus a possibly dependent smooth function and constant, so locality and the simultaneous Weyl rule apply to it. Trials with disjoint footprints are therefore independent, including their local metric events. Each has success probability at least $p_0/2$.

Choose a finite trial count N such that $(1-p_0/2)^N$ is smaller than the allocated error probability for each edge box. Next choose L so that the sum of the coarse-oscillation failures over these finitely many trials is below the next allocated error probability. Finally choose ρ small enough that the footprints fit disjointly into the boxes. The centers may lie on the corresponding coordinate mesh edges. Restoring the coarse fields spends the second half of the reserve. Lemma 4.5 and a finite union bound make all microscopic units, and hence the configuration access costs in those units, smaller than the prescribed macroscopic tolerance. No independence between this last event and the trial successes is needed. The ratio ρ may be chosen to be a power of the fixed geometric scale spacing. □

Lemma 17.5 (High-probability marked configurations)**.** *Fix c_0 with $1/b_0 < c_0 < C$, and then fix $c_1 \in (c_0, C)$. For each $p < 1$ there is a fixed geometric ratio $\rho > 0$ such that, at every radius R with*

$$\mathbb{P}[H^{\mathrm{low}}_{\rho R}(b_2)] \geq p_0, \tag{17.4}$$

there is, with probability greater than p, a marked tree and a strict working discount from Section 16. The boundary band, boundary, gain, inactive and ratio constants can be fixed independently of R. The discount and its positive active threshold may initially depend on the configuration.

Proof. Choose high-probability quantiles for the two untouched D boundary bands, the D gain-slab crossing, and the intrinsic E diameter of ∂V in U_c. This fixes a, A and an insertion gain $g > 0$. Choose $\epsilon_0 < Cg/16$, then $\delta_0, \eta_0, \delta_{\mathrm{im}}$ with strict reserves in (16.8). A fixed Gaussian quantile converts between s_R and the macro unit in (15.6).

The mesh and number of marks are deterministic. A uniform intrinsic Hölder quantile on ∂V and finitely many normal coordinate plaques supplies a finite face-interior boundary net and inward

displacements with access below $\eta_0/4$. Two parallel interior cubical boundaries and a coordinate bridge give a connected complex $S_0 \Subset V$ containing the displaced net, after a smaller adjustment if necessary. A Hölder quantile on S_0 in V permits a fixed fine subdivision with neighboring intrinsic E distances below the tolerance in Lemma 16.3. Choose an abstract spanning tree of this mesh and disjoint small edge boxes contained in V.

Apply Lemmas 17.2 and 17.4, requiring the microscopic units and access costs to be below that mesh tolerance. Trial centers lie on S_0, so its intrinsic connections give access from the neighboring vertices; no rotational covariance of the metrics is used. Make each configuration path simple in its compact trace and trim short ends to leave stubs. The strict lower bound between its original endpoint sets persists at the core cuts, by lower semicontinuity for that fixed metric (or by the two-sided comparison). The strict inequality $1/b_0 < c_0$ gives (16.2). Its entire intrinsic image is small, because its length and its access to the trial center are small in the macroscopic unit.

Lemma 16.3 embeds the protected spanning tree while preserving those local image bounds. Theorem 16.5 supplies a sufficiently deep smooth discount and a positive active threshold. Construct with an inactive bound $\epsilon_0/2$ and a shortcut ratio strictly below c_1 to preserve reserves. Its baseline gives the prescribed insertion gain and leaves the boundary bands untouched. All failures are among the finitely budgeted quantile, trial, oscillation and small-unit events. No upper bound on the total tree length was required before choosing the deep discount. □

17.4 Persistence and one finite family for all admissible radii

Lemma 17.6 (Persistence of a strict working discount)**.** *Suppose (E^j, D^j) converges enhancedly to (E, D), with $E^j \le C D^j$. Fix a smooth positive weight bounded away from zero. If it is a strict working discount for the limiting pair with positive threshold b, then for all sufficiently large j it is a working discount with any fixed relaxation of the strict bounds, active threshold $b/2$, and a slightly smaller short-path cutoff. The constants used in the transfer may depend on this fixed weight.*

Proof. The boundary band and gain lower bounds concern fixed surfaces and pass by their enhanced convergence. The original and weighted canonical boundary upper bounds persist by buffered upper recovery. More explicitly, for each limiting endpoint pair choose a connecting path inside U_c with a reserved excess. Its compact image is in a smaller buffer. The retained intrinsic arrays in that buffer converge uniformly to continuous arrays, so their moduli attach nearby endpoints at uniformly small cost. This uses actual enhanced array convergence, not an assertion that a merely tight sequence of Hölder constants is bounded almost surely. Finitely many endpoint neighborhoods cover $(\partial V)^2$, so finitely many such paths prove the required uniform upper bound in the prelimit. This uses upper recovery only, not convergence of a single-domain internal array to an intrinsic metric.

If the short-geodesic assertion failed, take a sequence of failing geodesics with endpoints on the fixed collector boundary, parametrized on $[0, 1]$ at constant speed. Apply Lemma 12.11 with $(A_j, B_j) = (D^j, E^j)$, $K_0 = K = K_c$, and $U = U_c$. Join the two components of ∂V by a fixed rational coordinate plaque in $\overline{V}$ to obtain a connected rational coordinate complex $S \subset K_c$, retained by Lemma 17.1. Every path from S leaving K_c crosses an untouched boundary band. The length bound $\delta_0 < a/4$ is below this exit barrier. The finite-path continuity and Euclidean boundedness hypotheses follow from Lemma 14.2 and the closure construction. The lemma therefore gives a limiting weighted D-geodesic with convergence of its length. Use a prelimit short-path cutoff strictly below the limiting cutoff.

If the limiting length is zero, the two limiting endpoints coincide; uniform convergence of the

buffered intrinsic canonical arrays makes the prelimit endpoint pair inactive. Otherwise apply the limiting strict dichotomy. In its inactive case, buffered upper recovery of the endpoint connection, with surface attachment of the moving endpoints, gives the prelimit inactive bound. In its active case, Lemma 12.11 transfers a strictly improving shortcut and its positive-length interval to the prelimit paths. The ratio and threshold survive the fixed relaxation. Both alternatives contradict failure. Only $E^j \le CD^j$ is used for the added path lengths. □

Theorem 17.7 (A uniform finite deterministic discount family)**.** *Under* (17.1) *and* (17.3)*, for each $p < 1$ there are a fixed geometric ratio $\rho > 0$, finitely many nonnegative functions $f_1, \dots, f_J \in C_c^\infty(V)$, and positive constants*

$$a, A, g, b, \epsilon_0, \delta_0, \qquad c_1 < C, \qquad \epsilon_0 < Cg/8,$$

such that at every radius satisfying (17.4) *their local working events have union of probability at least p. The family and constants are independent of the center and radius. The events are determined by a fixed annular field block modulo constants, and Lemma 16.7 applies on them.*

Proof. Choose the macro constants, mesh, trial count, coarse separation and microscopic ratio as in Lemma 17.5, with success probability $p' > p$. Make these choices using a conservative trial probability, for example $p_0/8$, and a quarter of the configuration reserve. Thus the losses from finite-menu extraction, taking a limit, and removing and restoring the coarse field are already included in the fixed choices.

For each admissible radius R, rescale the assignments by

$$E_g^{(R)}(x,y) = c_R^{-1} E_{g(\cdot/R)}(Rx, Ry), \qquad D_g^{(R)}(x,y) = c_R^{-1} D_{g(\cdot/R)}(Rx, Ry). \tag{17.5}$$

Their raw constants are c_{Rr}/c_R, and they form a uniformly regular weak family. Lemma 17.1 gives enhanced compactness. The configuration at the fixed normalized radius ρ passes to each limit by Lemma 17.2, retaining its unit and relaxing its reserve. Every limiting pair is local and field measurable. Fine-field amplification consequently applies to it with the same finite trial geometry. It has marked configurations with probability greater than p', and hence random strict working discounts with positive thresholds on that event.

Enumerate a countable dense family of nonnegative smooth compactly supported bumps in V, together with positive rational active thresholds. The all-path conclusion of Theorem 16.5 is important here. A uniform relative perturbation $e^{-t} \le w'/w \le e^t$ changes every path length by factors $e^{\pm t}$, every shortcut ratio by at most e^{2t}, and the positive length threshold by at most e^{-t}. It therefore preserves the relaxed conclusion even for a geodesic of the new weight. The inactive conclusion is in the original intrinsic E metric and is unchanged. The compactly supported baseline and its reserved gain can also be approximated. Thus at least one enumerated deterministic discount works strictly on each marked configuration. The countable union of these completed-local measurable events has probability greater than p'.

Suppose no finite family with a common positive threshold worked uniformly. For each integer j, choose an admissible radius at which the first j discounts, with threshold $1/j$ and the final relaxed constants, have success probability at most p. Extract an enhanced limit of these rescaled pairs. For the limit, a finite initial subfamily and some positive threshold b_0 have strict success probability greater than p. Lemma 17.6 implies that, on each of these finitely many strict events, the relaxed prelimit event with threshold $b_0/2$ occurs eventually. Fatou's lemma makes the lower limit of their union probabilities greater than p. For large j the chosen finite family is among the first j and $1/j < b_0/2$, a contradiction.

Translation covariance gives the same probability at every deterministic center. All averages used for the local units are inside the determining annular block. Each distance, threshold and unit acquires the same factor under a constant Weyl shift, so the events are invariant under constants. The boundary-band argument in Lemma 16.7 proves their use under other discounts. □

17.5 The exact-scaling one-sided version

Proposition 17.8 (Templates for an original-V competitor)**.** *Suppose E and D have exact scale and translation covariance with the same $Q > 0$, E has the canonical regularity estimates, D has the coercive crossing geometry, and $E \le CD$. If equality fails with positive probability, there is a finite-template theorem of the form of Theorem 17.7 at every radius, with a ratio $c_1 < C$. No lower comparison $c_* D \le E$ is required.*

Proof. A failure gives an E-finite pair with $E < cD$ for some deterministic $c < C$, on an event of positive probability; $D = \infty$ is allowed. The canonical rational core approximates its endpoints in E. Lower semicontinuity of the fixed competitor preserves a strict lower D threshold. A countable choice therefore gives fixed rational endpoints, a positive threshold and a cheap E path with positive probability. Confine that path to a deterministic compact box. Enlarge the box until the relevant D exit barriers exceed the threshold on a positive-probability subevent, using coercivity. Lower semicontinuity then supplies fixed small rational endpoint neighborhoods with the same lower bound. Their boundaries give a local configuration: the local lower bound and the exit bound together imply the global lower bound. A finite canonical connection to the center, restricted to a further upper-quantile event, gives local access.

Make the cheap arc simple in its compact trace and trim short ends. Lower semicontinuity preserves the strict competitor lower bound at the core cuts; no competitor length of the arc is needed. Normalize the absolute threshold by the positive local unit, restrict its ratio to a fixed compact interval, and choose a sufficiently fine rational threshold while preserving the strict ratio inequality. After one fixed initial rescaling of its box, the configuration has fixed local geometry and probability. Exact covariance transports it to every dilation and translation.

Apply Lemma 17.4 at sufficiently small microscopic scale. Macro regularity uses only canonical surface estimates and positive finite competitor crossing quantiles. The marked-tree construction and Theorem 16.5 are one-sided. At unit scale, approximate their random strict discounts by a countable deterministic family as above, then truncate this union to a finite family of success probability as close to one as prescribed and with a common positive threshold. Exact covariance transports the same family and constants to every radius. This argument uses the one-sided hypotheses throughout. □

18 Restoration and weighted counting

Assume $E \le CD$, canonical intrinsic surface Hölder estimates, positive competitor crossing quantiles in the common units, and a finite family of working discounts. The family may come from Theorem 17.7 on a positive fraction of the candidate radius levels, or from Proposition 17.8 on every level. Its union probability is chosen high enough for the simultaneous covers. This section does not use a lower comparison of E with D.

18.1 Near-maximal endpoint problems

Work in $K = K_{0,R}$ and put $S = c_R e^{\psi h_{L_* R}(0)}$, with $L_* > 10\sqrt{d}$. Choose common macro quantiles for the surface Hölder constants, positive crossings, and $s_R(0)/S$ and its reciprocal. Let M be a resulting upper bound for the normalized parent lengths. For a defect $\delta > 0$, cover S_R and S_{2R} by connected compact coordinate-face patches of diameter at most a fixed multiple of $\delta^{1/\beta} R$. On the macro event their intrinsic E^K diameters are at most $\delta S/8$. The number of ordered patch pairs is at most

$$C_0 \delta^{-2(d-1)/\beta}. \tag{18.1}$$

If $H_R(c)$ occurs with $c = C - \delta/(4M)$, its middle endpoints belong to one of these pairs (K_1, K_2), and

$$0 < L := D^K(K_1, K_2) \le MS, \qquad F := E^K(K_1, K_2) \ge CL - \delta S. \tag{18.2}$$

Indeed, if ℓ is the middle length, then $L \le \ell \le MS$ and the two patch errors give $F \ge c\ell - \delta S/4 \ge CL - \delta S/2$. No one fixed patch pair is required to have a probability bounded below independently of δ.

Choose the D^K-minimizer consistently by Lemma 12.2, and parametrize it by its original D length on $[0, L]$. These lengths agree with its constrained arclength: the ambient length is at most the constrained length and at least the constrained endpoint minimum. The endpoint triangle inequality and $E \le CD$ imply, for all $s < t$,

$$C(t - s) - \delta S \le E^K(P(s), P(t)) \le C(t - s). \tag{18.3}$$

18.2 A bin carrying a positive mass of metric units

Use geometrically spaced radii in $[\varepsilon^2 R, \varepsilon R]$. At each radius take a lattice of mesh κr, and a fixed finite coloring such that the closed balls $\overline{B}_{4r}(z)$ of one color are pairwise disjoint. Restrict centers to

$$9R/8 \le |z|_\infty \le 13R/8.$$

For small ε their working balls lie inside K and away from both endpoint surfaces. A label includes its center and its designated template; a bin will also specify the radius and color. The local unit at a label is $s_i = c_r e^{\psi h_r(z)}$.

Lemma 18.1 (Common bounds for the units and visit spans)**.** *For any prescribed polynomial failure exponent there is $B < \infty$ such that, outside that failure probability,*

$$\varepsilon^B S \le s_i \le \varepsilon^{-B} S \tag{18.4}$$

for every candidate. Set

$$\delta = \varepsilon^{B+1}. \tag{18.5}$$

There is $\theta_0 > 0$ such that every good site visited in its inner ball by a path satisfying (18.3) has $s_i/S \le \varepsilon^{\theta_0}$, on a further common event with polynomially small failure probability. The events are simultaneous over all the endpoint problems and their near-maximal paths.

Proof. There are only polynomially many candidate centers and logarithmically many radius levels. The increments $h_r(z) - h_{L_* R}(0)$ have variance $O(\log \varepsilon^{-1})$. The relative-power bounds for c_r/c_R and a Gaussian union bound give (18.4), with B large enough.

Corollary 15.6, applied to the larger working ball, bounds its full visit span by $\varepsilon^{\theta} MS + K_0 \delta S/C$. A path visiting the inner ball crosses an untouched outer boundary band during that span, at cost at least as_i. Hence

$$as_i \le \varepsilon^{\theta} MS + K_0 \delta S/C.$$

Choose $0 < \theta_0 < \min\{\theta, B+1\}$ and then take ε small to absorb the fixed constants. This is a consequence of one common separator environment, not a union bound over paths. □

Lemma 18.2 (The unit-mass bin)**.** *On the common macro and micro events and* (18.2)*, some deterministic bin contains N good labels visited by P in their inner balls, with units in $[qS, 2qS]$, such that*

$$\varepsilon^B/2 \le q \le \varepsilon^{\theta_0}, \qquad Nq \ge v_\varepsilon := c(\log \varepsilon^{-1})^{-2}. \tag{18.6}$$

The bin records one radius, one color, one template and one dyadic unit interval. There are $O((\log \varepsilon^{-1})^2)$ bins. Each deterministic bin universe has at most $C\varepsilon^{-2d}$ labels and pairwise disjoint working balls.

Proof. The path has a central segment from its last $S_{5R/4}$ contact before its first subsequent $S_{3R/2}$ contact. A positive canonical crossing quantile and $E \le CD$ make this segment's D time at least $\ell_0 S$ on the macro event, for some $\ell_0 > 0$.

The successful inner balls cover this segment. For each selected good site use the full first-to-last collector-boundary interval I_i. It contains the inner-ball visits: the endpoint surfaces are outside the site's working ball, while the collector surrounds its inner ball. The original canonical boundary connection and (18.3) give the essential upper bound

$$|I_i| \le (A/C)s_i + \delta S/C \le (A+1)s_i/C, \tag{18.7}$$

since $s_i \ge \varepsilon^B S$. The finite intervals cover the central segment, so

$$\ell_0 S \le \sum_i |I_i| \le (A+1)C^{-1} \sum_i s_i.$$

No bounded-overlap assertion is needed for this inequality. Assign a site with several successful templates to the first one, and partition by the stated bins. Some bin carries at least a fixed multiple of $S/(\log \varepsilon^{-1})^2$ in total unit mass. Its mass is at most $2NqS$, proving the lower bound in (18.6). The upper unit bound follows from Lemma 18.1. Lattice cardinality at radius at least $\varepsilon^2 R$ gives the deterministic universe bound. □

18.3 Chronological restoration and surviving discounts

The next operation is deterministic. Let U_i be pairwise disjoint open working regions, V_i their collectors, and $f_i \ge 0$ continuous bumps with

$$\operatorname{supp} f_i \Subset V_i \Subset U_i.$$

The endpoints of the path problems are outside all working regions. Absorb ψ into the bumps when convenient.

Lemma 18.3 (Chronological boundary restoration)**.** *Let $w = \exp(-\psi \sum_i f_i)$ and let P be a finite $(w \cdot D)$-path. Suppose every two points of its collector contact set $P([0,1]) \cap \partial V_i$ admit an original*

E connection inside U_i of length at most b_i, up to arbitrary excess. Then its endpoints admit an original E connection of length at most

$$C \operatorname{len}_{w\cdot D}(P) + \sum_i b_i. \tag{18.8}$$

Each collector is used at most once. No competitor length of an inserted connection is used.

Proof. Start at the initial time. If the remaining original path meets no bump support, retain it and stop. Otherwise choose the earliest remaining support contact, at site i. There is a last preceding contact with ∂V_i after the current time, because the support is compactly contained in V_i and the current point is outside V_i. There is a last subsequent exit from V_i, since the final endpoint is outside its working region. Replace the portion between these two contacts by the stated E connection, and resume at the original last-exit time.

The retained prefix meets no support. There is no later original visit to a collector already processed. Visits to other sites may be swallowed, but any remaining visits are processed later. Inserted connections stay in disjoint working regions, so they meet no other collector or support. All subsequent deletion endpoints are therefore still original path contacts. The procedure stops after at most the number of sites.

On retained original pieces every bump vanishes. Their D lengths equal their discounted lengths, and their E lengths are at most C times these lengths. Summing them and the connection costs gives (18.8), after sending the finitely many approximation errors to zero. The hypothesis on the entire contact set is what makes the new contact pair legitimate after earlier deletions. □

Fix a bin from Lemma 18.2 and choose any k of its N labels, forming I. Along the original selected path, a confined gain-slab crossing at each selected site has its weight decreased by the prescribed amount. The crossings lie in disjoint working balls. Thus

$$(w_I \cdot D)^K(K_1, K_2) \le L - gkqS. \tag{18.9}$$

Choose the discounted minimizing path consistently. Restore each discount whose positive bump support is not visited by this path. The path retains its length while the metric increases, so the stable selector keeps the same geometric path. At termination an output $Z \subset I$, of size m, remains, and its selected path visits every positive support in Z.

Call a surviving site active if the path has an interval with the shortcut and threshold in Lemma 16.7. If it is inactive, that lemma bounds its entire original intrinsic canonical contact set by $\epsilon_0 s_i$. Every contact set, active or not, has the general bound As_i.

Proposition 18.4 (Surviving and active fractions)**.** *Enlarge A so that $A \ge Cg$. If $\epsilon_0 \le Cg/8$ and $\delta/q \le Cg/2$, every k-subset has a surviving output satisfying*

$$m \ge \kappa_1 k, \qquad a_Z \ge \kappa_2 m, \qquad \kappa_1 = \frac{Cg}{4A}, \quad \kappa_2 = \frac{Cg}{8A}, \tag{18.10}$$

where a_Z is its number of active labels.

Proof. The output path still has discounted length at most $L - gkqS$. Apply Lemma 18.3, with cost As_i at active sites and $\epsilon_0 s_i$ at inactive ones, and compare with (18.2). Since $s_i \le 2qS$,

$$CL - \delta S \le C(L - gkqS) + 2qS\{Aa_Z + \epsilon_0(m - a_Z)\}.$$

Cancel CL and divide by qS:

$$Cgk - \delta/q \le 2Aa_Z + 2\epsilon_0(m - a_Z). \tag{18.11}$$

The left side is at least $Cgk/2$, and the right side is at most $2Am$, proving the first bound. Since $m \le k$, the same inequality yields

$$2Aa_Z \ge Cgk/2 - 2\epsilon_0 k \ge Cgk/4.$$

This proves the second bound. The argument never estimates competitor length on the canonical tree or on a canonical shortcut. □

For a deterministic set Z in the full bin universe, let A_Z be the following necessary output event: (18.2) holds; every designated local working event holds and its unit lies in $[qS, 2qS]$; the consistently selected jointly discounted path visits every positive support in Z; and at least $\lceil \kappa_2 |Z| \rceil$ labels are active. These events are measurable in the completed field sigma field. Write

$$A_m(h) = \sum_{|Z|=m} \mathbf{1}_{A_Z}(h).$$

The common environment is not included in this event definition: it will only be used to produce many outputs.

Lemma 18.5 (A lower count weighted by surviving size)**.** *For fixed k and sufficiently small ε, on the near-maximality and common good event some bin satisfies*

$$\sum_{m=\lceil \kappa_1 k \rceil}^{k} q^m A_m(h) \ge \left(\frac{v_\varepsilon}{2k}\right)^k. \tag{18.12}$$

Proof. Every k-subset of the actual visited universe has an output $Z \subset I$ of size at least $\kappa_1 k$. An output of size m has at most $\binom{N-m}{k-m}$ input preimages. Therefore

$$\binom{N}{k} \le \sum_m A_m \binom{N-m}{k-m}, \qquad 1 \le \sum_m A_m \frac{(k)_m}{(N)_m}.$$

For $N \ge 2k$ the last ratio is at most $(2k/N)^m$. Since $Nq \ge v_\varepsilon$ and $v_\varepsilon \le 1$ for small ε,

$$1 \le (2k)^k v_\varepsilon^{-k} \sum_m q^m A_m.$$

Here $N \ge 2k$ eventually because $N \ge v_\varepsilon \varepsilon^{-\theta_0} \to \infty$. This proves the claim. The weight is q^m, not q^k. □

18.4 All shifted outputs use the same path

For each deterministic Z put $f_Z = \sum_{i \in Z} f_i$ and

$$B_m(h) = \sum_{|Z|=m} \mathbf{1}_{A_Z}(h + f_Z).$$

Each bump vanishes on its own averaging sphere, on every other site's averaging sphere in the bin, and on the macro sphere of radius $L_* R$. Thus all s_i and S are unchanged by the unpinned

shift. Re-pinning multiplies every metric and unit by the same Weyl factor and changes neither the events nor the selected geometric paths.

Discounting Z after the shift $h \mapsto h + f_Z$ gives exactly D_h. Hence all shifted outputs use the same selected D_h^K-geodesic P. Write $L_0 = \mathrm{len}_{D_h}(P) = D_h^K(K_1, K_2)$ for its length in the unshifted field. If any shifted output occurs, monotonicity and its length bound give

$$L_0 \le MS. \tag{18.13}$$

Every label in a shifted output is visited by this path and requires an untouched outer boundary-band crossing of cost at least aqS. Choose one confined crossing per visited label. Their working balls are disjoint, so the crossing intervals are disjoint. The entire visited-label universe consequently has at most

$$M/(aq) \tag{18.14}$$

elements.

Lemma 18.6 (Restoration after canonical shortcuts)**.** *Fix a shifted output Z_0 of size at most k. If n distinct labels outside Z_0 are active in at least one shifted output, then*

$$\nu n \le 2Ak + \delta/q, \qquad \nu = \frac{(C - c_1)\min\{a, b\}}{3} > 0. \tag{18.15}$$

Proof. Each such label supplies an E_h shortcut along the common path P, with ratio at most $c_1 < C$, for an original D_h interval of length at least bqS. Its image is in its own working region. Indeed, start with the local working event in the raised field of an output containing that label, and then apply all the output's discounts. Inside the working region the other bumps vanish and its own discount returns the field to h. For a tested interval that leaves the region, Lemma 16.7 uses an untouched boundary band and measures the length in the actual common metric D_h, not in a partially restored metric.

The left endpoints of intervals at distinct sites are separated in D_h time by at least aqS. A path starting in one site's V^+ and reaching another working ball crosses the first site's outer boundary band, which was placed outside V^+. Thus the initial portions of length $\min\{a, b\}qS$ of these intervals are disjoint, and their union has length at least $n \min\{a, b\}qS$.

Select a longest interval, discard all intervals meeting it, and repeat. Each discarded interval has no greater length than the one selected when it was discarded and lies in its triple enlargement about its center. The selected disjoint intervals therefore have total length at least one third of the union.

Insert their E_h shortcuts into P. This is an E_h-rectifiable concatenation; it is not assumed D_h-rectifiable. Restore the field on Z_0 by the chronological boundary procedure applied to this already modified path, using the original canonical boundary connections in the target field $h + f_{Z_0}$. Their total target cost is at most $2AkqS$ by the working events of Z_0.

A restoration boundary cannot meet any inserted shortcut image, since their working balls are disjoint. Every deletion endpoint is on an unchanged original portion of P. Each inserted shortcut is retained whole or swallowed whole. Let J_r, J_s be the retained and swallowed intervals, and let d_0 be deleted original time outside them. The target canonical length is at most

$$\begin{aligned} C\left(L_0 - d_0 - \sum_{I \in J_r \cup J_s} |I|\right) + c_1 \sum_{I \in J_r} |I| + 2AkqS \\ \le CL_0 - (C - c_1) \sum_{I \in J_r \cup J_s} |I| + 2AkqS. \end{aligned}$$

On retained original pieces the target bumps vanish, so their lengths are bounded using the original D_h time. Swallowed shortcuts only improve the bound. Put $L_{Z_0} = D^K_{h+f_{Z_0}}(K_1, K_2)$. The reference output gives target endpoint distance at least $CL_{Z_0} - \delta S$, and monotonicity gives $L_{Z_0} \geq L_0$. Comparing this lower bound with the displayed upper bound and the interval sum proves (18.15). □

Corollary 18.7 (Few shifted outputs)**.** *For $m \leq k$, almost surely,*

$$B_m(h) \leq C_k q^{-(m-\lceil \kappa_2 m \rceil)}, \tag{18.16}$$

provided δ/q is bounded. The constants are uniform in the bin and endpoint patches.

Proof. There is nothing to prove if no shifted output occurs. Otherwise (18.15), including the at most k labels of Z_0, bounds the active-label universe by $C_1 k$. Equation (18.14) bounds the whole visited universe by C_2/q. Every size-m output has at least $u = \lceil \kappa_2 m \rceil$ active labels. Its number is at most

$$\sum_{j=u}^{m} \binom{\lfloor C_1 k \rfloor}{j} \binom{\lfloor C_2/q \rfloor}{m-j} \leq C_k q^{-(m-u)}.$$

Here $m \leq k$ and $0 < q \leq 1$; all remaining combinatorial factors are absorbed into C_k. □

18.5 Gaussian change of measure and endpoint entropy

Theorem 18.8 (Counting at a fixed endpoint problem)**.** *Let G_ε be the common good environment above. Choose $\alpha < 1$ sufficiently close to one that*

$$\gamma = \alpha\kappa_2\theta_0 - 2d(1-\alpha) > 0. \tag{18.17}$$

For each fixed positive integer k,

$$\mathbb{P}[(18.2) \text{ and } G_\varepsilon] \leq C_{\alpha,k}(\log \varepsilon^{-1})^{2k+2}\varepsilon^{\gamma\kappa_1 k}. \tag{18.18}$$

The estimate is uniform over the deterministic endpoint patches and over macro radii having a fixed positive density of admissible template levels.

Proof. For one bin and size m, there are at most $C_k\varepsilon^{-2dm}$ deterministic candidate subsets. Lemma 4.1 applies separately to A_Z and its deterministic shift f_Z. Its squared Cameron–Martin norm is at most a fixed multiple of k^2, uniformly in position and scale. Finite-sum Hölder and (18.16) give

$$\begin{aligned} q^m \mathbb{E}A_m &\leq K_{\alpha,k} q^m \sum_{|Z|=m} \mathbb{P}[A_Z(h+f_Z)]^\alpha \\ &\leq C_{\alpha,k}\varepsilon^{-2dm(1-\alpha)} q^{m-\alpha(m-\lceil\kappa_2 m\rceil)} \leq C_{\alpha,k}\varepsilon^{\gamma m}. \end{aligned} \tag{18.19}$$

The exponent of q is $(1-\alpha)m + \alpha\lceil\kappa_2 m\rceil \geq \alpha\kappa_2 m > 0$, so $q \leq \varepsilon^{\theta_0}$ gives the last inequality. The factor $\varepsilon^{-2dm(1-\alpha)}$ is the deterministic candidate entropy and cannot be omitted.

Sum over $m \geq \lceil\kappa_1 k\rceil$ and the $O((\log\varepsilon^{-1})^2)$ bins. On the event on the left of (18.18), one bin has (18.12). Integrating that pointwise lower bound proves the estimate. □

With (18.5), the patch entropy is at most $C\varepsilon^{-P_*}$, where

$$P_* = 2(B+1)(d-1)/\beta. \tag{18.20}$$

After fixing the geometric constants and α, choose a fixed integer k with

$$\gamma\kappa_1 k > P_* + 1. \tag{18.21}$$

The union of all near-maximal patch events on G_ε then has probability tending to zero. The complement of G_ε is added once, not multiplied by the patch count. Its macro quantile failures can be made arbitrarily small, and its micro cover, occupation and unit failures tend to zero.

The order of choices is noncircular. Fix a micro failure exponent, for example one, before choosing the local success probability and the constants B, θ_0. Then choose α, then k, and only then send ε to zero. The change-of-measure cost $\exp(O_\alpha(k^2))$ is a finite constant in this limit.

Corollary 18.9 (Suppression at an admissible macro radius)**.** *For every $\eta > 0$, one can choose the macro quantiles and local success probability, and then $\varepsilon_0 > 0$, such that the following holds. If a fixed positive fraction of the levels in $[\varepsilon^2 R, \varepsilon R]$ admit the finite-template conclusion, then for $\varepsilon < \varepsilon_0$ and*

$$c_\varepsilon = C - \varepsilon^{B+1}/(4M) \tag{18.22}$$

one has $\mathbb{P}[H_R(c_\varepsilon)] < \eta$. All choices are uniform in R.

Proof. On the common macro event, each witness gives a near-maximal patch problem by (18.2). Apply Theorem 18.8, the patch union bound and (18.21). Choose the macro failure below $\eta/2$, and then ε small enough that the micro failures and the counting bound total less than $\eta/2$. □

19 Uniqueness of weak metrics

Theorem 19.1 (Uniqueness in the regular weak class)**.** *Let $d \geq 3$ and $Q > 0$. Two regular weak assignments with the same Weyl parameter and scale exponent are deterministic positive multiples of one another, almost surely simultaneously on all point pairs. The equality persists simultaneously under every continuous Weyl perturbation of a good base field. Exact dilation covariance is not an assumption.*

Proof. Theorem 14.5 gives positive finite optimal deterministic constants

$$\begin{aligned} c_* &:= \sup\{c > 0 : cD \leq E \text{ almost surely on all pairs}\}, \\ C &:= \inf\{c > 0 : E \leq cD \text{ almost surely on all pairs}\}. \end{aligned} \tag{19.1}$$

The endpoint inequalities hold at c_*, C, by countable intersections along approximating constants. Their events are measurable as explained in Remark 14.6. Suppose $c_* < C$.

Use both ordered comparisons. The upper pair $(A, B) = (D, E)$ has optimal constant C and witness probability threshold p_{up} from Lemma 15.3. The reverse pair $(A, B) = (E, D)$ has optimal constant $C_{\mathrm{rev}} = 1/c_*$. Choose

$$1/C < b_0 < b_1 < b_2 < C_{\mathrm{rev}}, \qquad b_2 \geq C_{\mathrm{rev}}/2,$$

and put

$$b^\sharp = C_{\mathrm{rev}} - \gamma_{\mathrm{rev}}(b_2)/8.$$

The choice of witness constants makes $b_2 < b^\sharp < C_{\mathrm{rev}}$.

By Lemmas 15.4, 15.3 and 15.8, there are a fixed $\tau > 0$ and a macro radius R_0 on the geometric lattice such that, in every sufficiently small block $[t^2R_0, tR_0]$, at least three quarters of the levels R satisfy

$$\mathbb{P}[G_R^{\mathrm{low}}(\tau, b^\sharp)] \geq \tau. \tag{19.2}$$

Here the superscript denotes the reverse ordered pair. The conversion from H to G is uniform in the radius because both weak metrics have surface estimates.

For every radius satisfying (19.2), Lemma 15.3 at threshold b_2 gives, uniformly in that radius, a density of at least three quarters of small levels $s \in [t^2R, tR]$ with

$$\mathbb{P}[H_s^{\mathrm{low}}(b_2)] \geq p_{\mathrm{low}} > 0.$$

Theorem 17.7 supplies a fixed microscopic ratio ρ, a fixed finite discount family and common constants. A candidate level r is admissible when the last bound holds at $s = \rho r$. We chose ρ to be a power of the geometric spacing. This shifts the radius indices by a fixed number, so in $[\varepsilon^2 R, \varepsilon R]$ it loses only $O(|\log \rho|)$ levels. For sufficiently small ε, uniformly over the radii in (19.2), at least one half of the candidate levels are admissible.

Apply Corollary 18.9 to the upper ordered pair with $\eta = p_{\mathrm{up}}/2$. Fix the macro quantiles and the local success probability first, and then choose one sufficiently small ε. At the fixed threshold

$$c_\varepsilon = C - \varepsilon^{B+1}/(4M) < C,$$

every radius in (19.2) satisfies

$$\mathbb{P}[H_R^{\mathrm{up}}(c_\varepsilon)] < p_{\mathrm{up}}/2. \tag{19.3}$$

All constants, including this ε, are fixed before the next outer-scale limit.

On the other hand, optimality of C and the witness-density lemma imply that at least three quarters of the levels in all sufficiently long outer blocks $[t^2R_1, tR_1]$ satisfy

$$\mathbb{P}[H_R^{\mathrm{up}}(c_\varepsilon)] \geq p_{\mathrm{up}}, \tag{19.4}$$

for some other fixed macro radius R_1 on the same lattice. As t tends to zero through powers of that lattice, the index intervals for R_0 and R_1 differ by a fixed shift. The two density-three-quarter sets therefore intersect: their intersection has at least half the common block length minus a fixed error. Equations (19.3) and (19.4) contradict each other at any intersection level.

Thus $c_* = C$. The two global comparisons give $E = CD$ on every finite pair and on their common infinite-distance pairs. Both constants in (19.1) were deterministic, so the multiplier is deterministic. On the intersection of the good base-field events, simultaneous Weyl scaling transfers this equality to every continuous perturbation. □

The proof order matters. Construction, comparison and the finite-family argument retained raw constants and did not assume exact scale covariance. Enhanced compactness preserved fixed-surface configurations; the density argument handled varying local laws. Exact dilation covariance is derived next, rather than inserted into weak uniqueness.

20 Exact scale covariance and convergence

Fix a regular weak subsequential assignment E from Section 13. The weak uniqueness theorem now applies without any unproved exact scaling assumption.

20.1 Dilation covariance and the original finiteness condition

Proposition 20.1 (Exact power scaling)**.** *For each fixed deterministic $r > 0$ and $z \in \mathbb{R}^d$, almost surely simultaneously for all endpoints,*

$$E_h(rx + z, ry + z) = E_{h(r\cdot+z)+Q\log r}(x, y). \tag{20.1}$$

The same assertion holds for a base field plus a possibly dependent random continuous function.

Proof. For fixed $r > 0$ define

$$E_g^{[r]}(x, y) = E_{g(\cdot/r)}(rx, ry).$$

This is a regular weak assignment. Length, locality, Weyl scaling and translation covariance follow from the change of variables. Its raw constants at scale s can be taken to be c_{rs}. Indeed, for pinned g, the field $g(\cdot/r)$ differs from a pinned base field by $g_{1/r}(0)$, and this constant cancels in the normalized estimates. Its relative-power bounds and fixed-template quantiles are therefore uniform in s.

Theorem 19.1 gives $E^{[r]} = k(r)E$ with $k(r) > 0$ deterministic. This identity extends to continuous perturbations and hence to rescaled base fields, which are pinned base fields plus random constants. Composing coordinate changes gives $k(rs) = k(r)k(s)$. The function k is Borel, for instance as the ratio of the positive finite lower medians of $E_h^{[r]}(0, e_1)$ and $E_h(0, e_1)$; the field transformations and evaluation are jointly Borel. A positive Borel multiplicative function is $k(r) = r^\alpha$ for some $\alpha \in \mathbb{R}$.

Let $T_r h = h(r\cdot) - h_r(0)$. Then

$$E_h(rx, ry) = r^\alpha e^{\psi h_r(0)} E_{T_r h}(x, y). \tag{20.2}$$

To identify α, distinguish the actual crossing median

$$\mathfrak{c}(r) := \mathrm{Med}\big[e^{-\psi h_r(0)} E_h(S_r, S_{2r})\big]$$

from any raw constants chosen by a previous extraction. The field $T_r h$ has the base pinned law, so (20.2) gives exactly $\mathfrak{c}(r) = r^\alpha \mathfrak{c}(1)$. Uniform crossing and reciprocal tightness make $\mathfrak{c}(r)$ comparable to c_r by constants independent of r. Apply (4.9) as $r \downarrow 0$, with arbitrarily small exponent loss. It follows that $\alpha = \psi Q$. Translation covariance proves (20.1). The simultaneous Weyl rule, with the transformed perturbation $f(r\cdot+z)$, gives the statement for dependent continuous perturbations. No simultaneous assertion over all real transformations is needed by Axiom IV. □

Proposition 20.2 (The original continuum-finiteness axiom)**.** *The canonical assignment satisfies Axiom V of Definition 1.3, on every base field plus a possibly dependent random continuous function.*

Proof. On one countable full-probability event, each rational coordinate plaque has finite continuous actual intrinsic metric in each rational buffer containing it. Every two points of Σ in a connected open U can be joined by a finite canonical path in U: connect each locally on its plaque to a rational point and connect the rational points by an axis-parallel polygonal chain with compact neighborhood in U. Finitely many rational plaques cover that chain and supply the finite intrinsic connections.

A connected nonsingleton compact set has a coordinate projection that is a nontrivial interval. This interval contains a rational number. The set therefore meets a rational coordinate hyperplane, and hence Σ. Choose a point in each of $K_1 \cap \Sigma$ and $K_2 \cap \Sigma$. Their finite internal connection

gives $E(K_1, K_2; U) < \infty$. This is pathwise existence, so no measurable selection of these two points is needed. A continuous perturbation is bounded on the connecting path's compact image and preserves finiteness. The same countable base event consequently works for all such perturbations. □

Theorem 20.3 (Normalized convergence)**.** *Under (1.1), the metrics $a_\varepsilon^{-1} D_h^\varepsilon$ converge in probability in the topology (1.8), jointly with h, to a field-measurable extended metric E_h. Their restrictions converge uniformly in probability on each fixed deterministic finite connected coordinate-hypersurface complex. Moreover,*

$$a_\varepsilon^{-1} D_h^\varepsilon(0, e_1; B_2) \longrightarrow E_h(0, e_1; B_2) \tag{20.3}$$

in probability. The limiting variable is positive and finite, is atomless, and has unique median one. The same deterministic normalizers give global convergence after adding a possibly dependent random bounded continuous function to h.

20.2 The internal point-distance normalizer

Lemma 20.4 (Stochastic continuity of a Borel field functional)**.** *Suppose $T_t h$ has the same law as h and $T_t h \to h$ in probability in a Polish distribution space as $t \to 0$. For any finite Borel field functional Y, one has $Y(T_t h) \to Y(h)$ in probability.*

Proof. Lusin's theorem gives a compact set of arbitrarily large field probability on which Y is continuous, and hence uniformly continuous. Both field arguments lie in that set except on twice its complement probability, because their marginal laws agree. Their convergence then proves the assertion. Truncate Y first if necessary. □

Lemma 20.5 (Continuity of the normalizing domain)**.** *For $R > 1$ put $X_R = E_h(0, e_1; B_R)$. Almost surely,*

$$\lim_{R \uparrow 2} X_R = X_2 = \lim_{R \downarrow 2} X_R. \tag{20.4}$$

Proof. The variables are finite by a fixed coordinate-complex connection. Exact scaling gives, for r near one,

$$X_{2r} = r^{\psi Q} e^{\psi h_r(0)} E_{T_r h}(0, e_1/r; B_2).$$

The transformed field has the pinned base law and tends to h in the local distribution topology. Continuity of sphere averages gives $h_r(0) \to 0$. Lemma 20.4 applies to the finite Borel variable $Y(h) = E_h(0, e_1; B_2)$. The moving point e_1/r stays on a fixed deterministic plaque compactly contained in B_2. Its intrinsic Hölder constant under $T_r h$ has the same tight law as under h, so $E_{T_r h}(e_1, e_1/r; B_2) \to 0$ in probability. The triangle inequality yields $X_{2r} \to X_2$ in probability from both sides. Since $R \mapsto X_R$ is nonincreasing, its almost-sure one-sided limits exist; convergence in probability identifies them with X_2. □

Proposition 20.6 (Convergence of the normalizing functional)**.** *Along every enhanced subsequential coupling with limiting assignment E,*

$$a_\varepsilon^{-1} D_h^\varepsilon(0, e_1; B_2) \longrightarrow E_h(0, e_1; B_2). \tag{20.5}$$

Proof. For each small fixed rational $\eta > 0$, compact path limits and buffered recovery give

$$\begin{aligned} E_h(0, e_1; B_{2+\eta}) &\le \liminf_\varepsilon a_\varepsilon^{-1} D_h^\varepsilon(0, e_1; B_2) \\ &\le \limsup_\varepsilon a_\varepsilon^{-1} D_h^\varepsilon(0, e_1; B_2) \le E_h(0, e_1; B_{2-\eta}). \end{aligned}$$

For the first inequality, bounded-cost prelimit paths have images in $\overline{B}_2 \Subset B_{2+\eta}$ and positive limiting annular barriers give equicontinuity. For the last, a finite limiting path in $B_{2-\eta}$ has positive boundary clearance and is recovered by finitely many short links with endpoints on retained surfaces. Both fixed endpoints are nonsingular. Let $\eta \downarrow 0$ and use (20.4). □

Lemma 20.7 (Atomlessness and a unique internal-distance median)**.** *The variable $X = E_h(0, e_1; B_2)$ is positive and finite, has no atoms, and has a unique median.*

Proof. Choose a nonnegative $g \in C_c^\infty(B_1^{\mathrm{Eucl}})$ which is at least $c > 0$ on a closed Euclidean annulus separating 0 from e_1. Its unit-sphere average is zero, so tg is a Cameron–Martin shift of the pinned field. On a good base field put $X_t = X(h + tg)$. For $t > s$, each connecting path has a portion crossing this annulus, and the weighted length identity gives

$$X_t \geq X_s + (e^{\psi c(t-s)} - 1)Y_s, \tag{20.6}$$

where $Y_s > 0$ is the annulus crossing in the metric of $h + sg$. The same path's existence makes that crossing finite. Thus X_t is strictly increasing. The two-sided weight bound using $\|g\|_\infty$ makes it continuous, and (20.6) makes it tend to infinity as $t \to \infty$.

For each fixed value a, a good field has at most one time with $X_t = a$. Fubini gives $\int_{-T}^{T} \mathbb{P}[X(h + tg) = a]dt = 0$. If $\mathbb{P}[X = a] > 0$, equivalence of the shifted Gaussian laws would make every integrand positive, a contradiction. Hence X is atomless.

If $a < b$ and $\mathbb{P}[X \leq a] > 0$, then on that event the continuous function $t \mapsto X_t$, for $t \geq 0$, passes through (a, b) during a nonempty open interval of times. Fubini and Gaussian equivalence imply $\mathbb{P}[a < X < b] > 0$. The distribution function therefore has no flat interval at a value strictly between zero and one. Its median is unique. □

Proposition 20.8 (Identification of the scalar and the full limit)**.** *Every subsequential assignment has* $\operatorname{Med} E_h(0, e_1; B_2) = 1$. *All such assignments agree almost surely on the same field. The normalized LFPP metrics converge in probability, jointly with the field, to this assignment.*

Proof. Each prelimit normalizing variable has lower median one. Its probabilities of belonging to $[0, 1]$ and $[1, \infty)$ are both at least one half. Proposition 20.6 and the closed-set part of Portmanteau make one a median of the limit; Lemma 20.7 makes it the unique median.

Any two subsequential assignments are regular weak and hence deterministic multiples by Theorem 19.1. Their intrinsic point-domain medians are both one, so the multiplier is one. Tightness therefore gives the unique joint limiting law (h, E_h) for every cutoff sequence.

To obtain convergence in probability on the original field space, adjoin the fixed variable E_h to any allegedly nonconvergent subsequence. Every further joint limit of $(h, a_\varepsilon^{-1} D_h^\varepsilon, E_h)$ has both metric marginals equal to the same Borel function of its field coordinate. It is supported on the diagonal. Applying Portmanteau to a positive-distance closed set contradicts the alleged failure of convergence in probability. □

Proof of Theorem 20.3. Proposition 20.8 gives the global and normalizing-functional assertions. Retain the restrictions to any fixed deterministic coordinate complex in every enhanced extraction. Lemma 12.9 identifies its unique possible uniform limit with E_h restricted to that complex, including nonrational coordinates. Tightness and the field-measurable identification give uniform convergence in probability.

For a possibly dependent bounded continuous f, take any cutoff sequence and a further subsequence with almost-sure enhanced convergence for the base field. The mollifications of f converge locally uniformly and are bounded by its realized global supremum. Lemma 12.10 applies

pathwise, simultaneously for all such f, and yields the limit $e^{\psi f} \cdot E_h = E_{h+f}$. The subsequence criterion gives convergence in probability, without independence of f and h. □

Corollary 20.9 (Regular variation and compact-mollifier universality)**.** *For every fixed $\lambda > 0$,*

$$\frac{a_{\lambda\varepsilon}}{a_\varepsilon} \longrightarrow \lambda^{1-\psi Q}. \tag{20.7}$$

The convergence is locally uniform in $\lambda \in (0, \infty)$. Each fixed nonnegative smooth radial compactly supported probability mollifier, normalized by its own B_2 internal point-distance median, converges to the same assignment E_h.

Proof. For fixed r, the deterministic factor $b_\varepsilon(r) = r a_{\varepsilon/r}/a_\varepsilon$ in (13.3) is bounded above and below. On any subsequence where it converges, full convergence on the original and rescaled field, together with exact covariance, identifies its limit as $r^{\psi Q}$ by evaluation at a fixed positive finite point pair. Thus this is the limit on every subsequence, proving (20.7). The normalizer is positive and Borel in the cutoff. Apply the uniform convergence theorem for measurable slowly varying functions to $a_\varepsilon/\varepsilon^{1-\psi Q}$, after replacing ε by $1/x$ [8, Theorem 1.2.1]. This gives the stated local uniformity. No convergence of $a_\varepsilon/\varepsilon^{1-\psi Q}$ to a constant is asserted.

The construction in Sections 5–11 applies separately to each stated compact mollifier, with the same exponent. Its limits are regular weak assignments. Weak uniqueness compares them, and their unique point-domain medians fix the multiplier to one. Apply the full-convergence argument to each fixed mollifier. □

Corollary 20.10 (Orthogonal covariance)**.** *For each fixed $O \in O(d)$, almost surely simultaneously for all endpoints,*

$$E_h(Ox, Oy) = E_{h \circ O}(x, y).$$

Proof. The radial mollifier and Euclidean path-length element give this identity exactly for the unnormalized global prelimit metrics. Dividing both sides by the same deterministic a_ε preserves it. The field $h \circ O$ has the base law because the field and sphere pin are orthogonally invariant. Full convergence on both fields proves the result, including reflections. Rotation of the cubical normalizing domain is not used in this argument. □

21 Uniqueness under the five axioms

The reference assignment E now satisfies the original strong axioms: length, locality and Weyl scaling come from Section 13, exact covariance from Proposition 20.1, and continuum finiteness from Proposition 20.2. Let D be an arbitrary second assignment satisfying Definition 1.3. No surface upper bound, deterministic-point finiteness or membership in the regular weak class is assumed for D.

Lemma 21.1 (Crossing geometry from I–V)**.** *Every deterministic cubical annulus has positive finite D crossing almost surely. In units $r^{\psi Q} e^{\psi h_r(z)}$, the law of a fixed-template crossing is independent of r, z. Finite D paths are Euclidean continuous and bounded, D-bounded sets are Euclidean bounded, finite pairs admit geodesics, and finite-distance components are complete.*

Proof. Disjoint compact boundary sets have positive global D distance by lower semicontinuity and separation. Axiom V makes their internal infimum finite in a deterministic connected open buffer, and hence makes their global distance finite. Axioms III–IV and the scale-invariant field

law modulo constants identify the normalized global crossing law. Its reciprocal is tight. The exact power constants satisfy (4.9), so Lemma 14.2 applies without any surface upper estimate.

Only after this step do we localize crossings. Euclidean continuity permits truncation of a path between its last inner-boundary contact and its first subsequent outer-boundary contact. Thus the global crossing infimum equals the confined crossing infimum. It is a local random variable by Axiom II and Lemma 12.3. This ordering avoids using Euclidean continuity to prove the barriers from which that continuity was obtained. □

Lemma 21.2 (One deterministic comparison)**.** *There is a deterministic $M < \infty$ such that, almost surely simultaneously on all point pairs,*

$$E_h(x, y) \le M D_h(x, y). \tag{21.1}$$

Proof. A canonical intrinsic separator diameter has a tight normalized upper law. A competitor crossing has a tight normalized reciprocal law by Lemma 21.1. They use the same power unit $r^{\psi Q} e^{\psi h_r(z)}$. Therefore, by increasing a deterministic M, their separator-to-crossing inequality has probability as close to one as prescribed, uniformly in center and radius.

The event is local modulo constants. Lemmas 4.2 and 4.4 give simultaneous covers. The router, Lemma 12.4, applies to every finite D path. Use a summable sequence of cover failures and a countable compact exhaustion, then let the endpoint approximations converge. Lower semicontinuity of E gives (21.1) on all finite D pairs. The inequality is vacuous when D is infinite. □

Define the optimal deterministic constant

$$C = \inf\{c > 0 : E \le cD \text{ almost surely on all pairs}\}. \tag{21.2}$$

It is finite and the inequality holds at C by countable intersection. It is positive: Axiom V supplies a finite connection between two disjoint compact continua, whereas E has positive distance between those compact sets. Were $C = 0$, the comparisons along a decreasing sequence of constants would give zero E distance for the distinct endpoints of such a finite D connection.

Theorem 21.3 (One-sided strong rigidity)**.** *Almost surely simultaneously for all $x, y \in \mathbb{R}^d$,*

$$E_h(x, y) = C D_h(x, y). \tag{21.3}$$

Proof. Suppose equality fails with positive probability. Since $E \le CD$, the discrepancy includes an E-finite pair with $E < cD$ for some deterministic $c < C$; the case $D = \infty$ is included. Proposition 17.8 then gives a fixed finite family of smooth discounts at every radius, with local success probability arbitrarily close to one. The shortcut ratio $c_1 < C$, threshold, untouched boundary bands, gain, and original canonical contact bounds are exactly the inputs to Section 18. Its construction used no lower comparison of D by E.

For the ordered pair $(A, B) = (D, E)$, Corollary 15.9 gives a fixed $p_* > 0$ such that

$$\mathbb{P}[H_R(c)] \ge p_* \qquad \text{for every } R > 0 \text{ and every } C/2 \le c < C.$$

Exact covariance is now available for E as a proved conclusion, and is an original axiom for D. Only E needs surface upper bounds in that corollary.

Choose the common macro failure probability below $p_*/4$ and the local template probability high enough for the covers. All candidate levels are admissible. Corollary 18.9, with a single sufficiently small fixed ε, gives

$$\mathbb{P}\left[H_R\left(C - \frac{\varepsilon^{B+1}}{4M}\right)\right] < p_*/2,$$

contradicting the preceding lower bound. Hence the strict discrepancy event is impossible.

Every E-finite pair has $E = CD$, and in particular cannot have infinite D distance. If $E(x, y) = \infty$, the initial comparison forces $D(x, y) = \infty$. This proves (21.3) on all pairs. The discrepancy localization in Proposition 17.8, using the canonical rational core and competitor lower semicontinuity, makes this an all-pairs statement on one measurable full-probability event, not merely a statement for each fixed deterministic pair. □

Proof of Theorem 1.4. The canonical assignment supplies existence. Apply Theorem 21.3 separately to two arbitrary strong assignments $D, \widetilde{D}$. For deterministic $C_D, C_{\widetilde{D}} > 0$,

$$E = C_D D = C_{\widetilde{D}}\widetilde{D}, \qquad \widetilde{D} = (C_D/C_{\widetilde{D}})D.$$

Intersect the good base-field events of these assignments. Simultaneous Weyl scaling transfers the equality to every continuous perturbation on that same event, including any dependent random continuous perturbation. Section 20 identifies the canonical point-domain median normalization. □

Remark 21.4. Surface regularity plays different roles in the two uniqueness arguments. For regular weak assignments, it provides compactness of low-ratio configurations for both metrics. For an arbitrary strong competitor, exact scale covariance supplies configurations at every scale, while only the reference metric supplies the tree paths and boundary connections. Thus the characterization uses precisely the five axioms in Definition 1.3.

22 Closed-cube crossing normalization

Recall the closed-cube crossing functional Z_{C_0} from (1.16). We prove convergence of this functional by first recovering paths in a smaller cube and then joining their endpoints to the original faces. The joining estimates must keep the paths inside the closed cube.

22.1 Chaining from one side of the boundary

Lemma 22.1 (Constrained boundary and parallel-face estimates)**.** *Let C' range over translates of axis-parallel closed cubes in a fixed bounded region, with side lengths in a fixed compact subinterval of $(0, \infty)$. Let S be a union of a bounded number of coordinate faces or parallel coordinate plaques in C'. For each exponent β in (11.7), there are $r_0 > 0$ and uniformly tight random constants $H_{\varepsilon,C',S}$ such that*

$$a_\varepsilon^{-1} D_h^\varepsilon(x, y; C') \le H_{\varepsilon,C',S}|x - y|_2^\beta, \qquad x, y \in S, \quad |x - y|_2 \le r_0. \tag{22.1}$$

The paths on the left are constrained to C', including its boundary. A finite boundary chain also gives tight constrained boundary diameters. These estimates hold for the actual limiting compact intrinsic metric $E^{C'}$, with uniform marginal laws over the stated geometries.

Proof. We adapt the fine-field chaining of Lemma 5.1 to paths constrained to the cube. Start with the stationary field at dyadic cutoff 2^{-n} and its fixed high reference quantile b_n. Write $r_\ell = 2^{-\ell}$ and $L_\ell = (1+\ell)^2$. The good and consecutive-level nice site processes in that lemma have, outside probability

$$C \sum_{\ell \geq M} 2^{d\ell} e^{-c\ell^2},$$

bad star-components of diameter at most a small fixed multiple of L_ℓ, simultaneously for $M \leq \ell < n$. The estimates are uniform over the bounded geometric family.

For $x \in C'$, clip each coordinate inward to a boundary band of width

$$t_\ell = K L_\ell r_\ell$$

from the corresponding face, and denote the resulting point by $T_\ell x$. Choose M so t_ℓ is less than a quarter of every allowed side length, and choose K larger than the fixed routing and edge-domain constants. The clipping map is 1-Lipschitz.

There is a good vertex within $CL_\ell r_\ell$ of $T_\ell x$, and a common nice vertex for adjacent levels at the same controlled distance. Join the selected vertices in a neighborhood of the straight segment between their clipped anchors, using good-component routing. The segment is at distance comparable to $KL_\ell r_\ell$ from the cube boundary, including for adjacent levels because $t_{\ell+1}$ is comparable to t_ℓ. The routing neighborhood has width at most $CL_\ell r_\ell$. For K sufficiently large, every route and every internal edge domain lies in C'. Loop erasure bounds the number of edges by a fixed polynomial in ℓ.

At the finest level, connect x to its selected vertex by a straight segment. Convexity keeps this segment in C', and its length is at most $CL_n r_n$. At each coarser level the construction stays within $CL_\ell r_\ell$ of x. If $|x-y| \leq r_m$, the clipped anchors at level m are at distance at most r_m, so the same routing joins their vertices. Consequently, for a fixed exponent J,

$$\begin{aligned} D_n(x,y;C') \leq C \sum_{\ell=m}^{n-1} (1+\ell)^J 2^{-\ell} b_{n-\ell} \exp\{\psi \sup_{T_\ell(S)} H_\ell\} \\ + C(1+n)^J 2^{-n} \exp\{\psi \sup_{T_n(S)} X_n\}. \end{aligned} \tag{22.2}$$

Here $T_\ell(S)$ is an ambient tube of radius $C(1+\ell)^2 2^{-\ell}$, used only to bound the coarse field. Every actual path remains in the closed cube.

The tube needs at most $C(1+\ell)^J 2^{(d-1)\ell}$ cutoff cells, uniformly even when parallel faces approach. Gaussian concentration therefore gives, for each $a > \sqrt{2(d-1)}$,

$$\mathbb{P}[\sup_{T_\ell(S)} H_\ell > a\ell \log 2] \leq C e^{-c\ell},$$

and the same bound for X_ℓ. Divide (22.2) by b_n and use the normalization bound (5.3). The positive separation between Q and $\sqrt{2(d-1)}$ gives the same summable series and every exponent in (11.7). Below the cutoff use the straight segment from x to y, still inside the convex cube, and the terminal density estimate.

For separated parts of a boundary, a fixed finite chain gives a diameter bound. For a disconnected family of parallel plaques, adjoin a bounded number of coordinate segments inside C' connecting chosen points of the components. Their total Euclidean length and cutoff covering numbers are uniformly bounded by the same surface-dimensional estimate. The connected enlargement controls

a reference value for all component pairs as well as the local modulus. Real cutoffs and the sphere pin transfer by the bounded-domain comparisons in Sections 2–11.

Finally extract the constrained arrays on a countable dense subset of the faces, jointly with the identified global metric. Their modulus gives continuous limiting arrays. Bounded-cost paths have images in the compact C' and positive limiting global barriers give equicontinuity. Their limits are E-rectifiable paths in C' with no greater length. Hence the actual $E^{C'}$ is bounded above by the extracted arrays. Lower semicontinuity of the compact intrinsic metric and its triangle inequality extend the bounds from the dense subset to all endpoints. This argument does not identify every constrained array with $E^{C'}$; its upper bounds suffice here. □

Lemma 22.2 (Convergence of the constrained crossing)**.** *The variable $X_{C_0} = Z_{C_0}(E_h)$ is positive and finite, and*

$$a_\varepsilon^{-1} Z_{C_0}(D_h^\varepsilon) \longrightarrow X_{C_0} \tag{22.3}$$

in probability, jointly with the global metric and field.

Proof. Positivity follows from the global distance between the disjoint compact faces. Lemma 22.1 gives a finite constrained boundary connection. Lemma 12.2 gives a Borel crossing functional and an attained minimum.

For $0 < t < 1/4$, let $C_t = [t, 1-t]^d$ and let X_t be its own opposite-face crossing constrained to C_t. Put $r_t = 1 - 2t$ and $z_t = t(1, \dots, 1)$. Exact covariance yields

$$X_t = r_t^{\psi Q} e^{\psi h_{r_t}(z_t)} X_{C_0}(T_{z_t, r_t} h).$$

The transformed field has the base law and tends to h in the distribution topology, and its scalar average tends to zero. Lemma 20.4 gives

$$X_t \longrightarrow X_{C_0} \quad \text{in probability.} \tag{22.4}$$

This is stochastic continuity under homotheties, not a monotonicity assertion for crossings with moving faces.

In an enhanced subsequential coupling, bounded-cost prelimit crossings in C_0 have compact path limits, so

$$X_{C_0} \le \liminf_\varepsilon a_\varepsilon^{-1} Z_{C_0}(D_h^\varepsilon).$$

For the upper bound fix a small rational t. A limiting crossing of C_t has positive clearance from ∂C_0. Its endpoints lie on fixed coordinate faces. Buffered recovery gives prelimit paths in C_0 between the same endpoints with upper limiting cost at most X_t. Project each endpoint normally to the corresponding original face of C_0. Lemma 22.1, applied to the union of the inner and original faces, bounds the two additional normalized costs by $2H_{\varepsilon,t} t^\beta$, with $H_{\varepsilon,t}$ tight uniformly in both variables.

Given a positive error tolerance, first choose t so small that (22.4) and the two boundary band costs exceed that tolerance only with small probability. Then use buffered recovery at this fixed t. This bounds the probability that the upper prelimit crossing exceeds X_{C_0} by more than three tolerances. Together with the lower bound it identifies the crossing limit along every enhanced subsequence. Since the limit is a Borel function of the field, the same diagonal joint-limit argument as in Proposition 20.8 gives convergence in probability on the original space. □

Lemma 22.3 (A unique closed-cube crossing median)**.** *The law of $\log X_{C_0}$ is a convolution of a nondegenerate Gaussian law and a probability law. In particular X_{C_0} has a strictly positive density on $(0, \infty)$ and a unique median $m_{C_0} \in (0, \infty)$.*

Proof. Choose a smooth compactly supported function equal to one near C_0. Adjust its sphere average by a bump near a negative-coordinate point of the unit sphere, away from C_0. The resulting nonzero pinned Cameron–Martin direction g equals one near C_0. Since all crossing paths are constrained to the cube,

$$X_{C_0}(h+tg) = e^{\psi t} X_{C_0}(h), \qquad t \in \mathbb{R}.$$

Decompose the Gaussian field as $h = h^{\perp} + Ng$, where N has variance $\|g\|_{\mathcal{H}}^{-2}$ and is independent of $h^{\perp}$. Explicitly, $N = I(g)/\|g\|_{\mathcal{H}}^2$ for the Gaussian linear functional corresponding to g.

The residual $h^{\perp} = h - Ng$ is a base field plus a dependent smooth perturbation. The simultaneous Weyl identity therefore applies on the good original field event and gives

$$\log X_{C_0}(h) = \psi N + \log X_{C_0}(h^{\perp}).$$

The second summand is a Borel function of $h^{\perp}$ and is independent of N. Both crossing variables are positive and finite. Gaussian convolution has a strictly positive density on the real line, so the original crossing has a continuous strictly increasing distribution function on its positive support and a unique median. No moment of its logarithm is required. □

Theorem 22.4 (Closed-cube median normalization)**.** *Let* $\mathfrak{a}_\varepsilon^\rho = \operatorname{Med} Z_{C_0}(D_h^\varepsilon)$. *Then*

$$\frac{\mathfrak{a}_\varepsilon^\rho}{a_\varepsilon} \longrightarrow m_{C_0}, \qquad (\mathfrak{a}_\varepsilon^\rho)^{-1} D_h^\varepsilon \longrightarrow m_{C_0}^{-1} E_h \tag{22.5}$$

in probability in the lower-semicontinuous topology. The limiting constrained unit-cube crossing has median one.

Proof. By Lemmas 22.2 and 22.3, the normalized crossing converges in law to a variable with continuous strictly increasing distribution function at its median. For each small $\eta > 0$, that distribution function is below one half at $m_{C_0} - \eta$ and above one half at $m_{C_0} + \eta$. Weak convergence places every prelimit lower median between those two numbers eventually. This proves the scalar convergence; combine it with Theorem 20.3 for the metric statement. □

23 Heat regularization

We now prove the heat-smoothing version of Theorem 1.5. The proof replaces the finite-range labels used for compact mollifiers by localized fine-field labels and controls the omitted heat tails. Put

$$p_t(x) = (2\pi t)^{-d/2} e^{-|x|_2^2/(2t)}, \qquad k_\varepsilon = p_{\varepsilon^2/2}, \qquad h_\varepsilon^{\mathrm{H}} = h * k_\varepsilon.$$

Write $D_h^{\mathrm{H},\varepsilon}$ for the exponential path metric and

$$a_\varepsilon^{\mathrm{H}} = \operatorname{Med} D_h^{\mathrm{H},\varepsilon}(0, e_1; B_2).$$

The convolution is defined for the pinned base field and for bounded continuous perturbations.

23.1 Spatial localization of the fine field

Use the compact-kernel white-noise representation (4.8). For $r > 0$ set

$$F_{r,\varepsilon}(x) = \int_0^r \int_{\mathbb{R}^d} (K_u * k_\varepsilon)(x-y) W(dy\, du),$$

and at a site z define

$$F^{\mathrm{loc}}_{r,\varepsilon,z}(x) = \int_0^r \int_{B_{Lr}(z)} (K_u * k_\varepsilon)(x-y) W(dy\, du).$$

Localized fields with disjoint spatial determining boxes are independent. They need not agree on overlaps: they define finite-range labels, which are compared with the single original field before any paths are concatenated.

Lemma 23.1 (Uniform fine-field localization error)**.** *Fix $C_0 < \infty$ and take $L > C_0 + 2R_K + 10$, where R_K is a support radius of K. On $B_{C_0 r}(z)$, the error $R_{r,\varepsilon,z} = F_{r,\varepsilon} - F^{\mathrm{loc}}_{r,\varepsilon,z}$ has a continuous version jointly in x and $0 \le \varepsilon \le r$, extended by zero at $\varepsilon = 0$. The variable*

$$Z_{r,z} := \sup_{0 \le \varepsilon \le r} \sup_{x \in B_{C_0 r}(z)} |R_{r,\varepsilon,z}(x)|$$

has a Gaussian upper tail uniformly in r, z. Its expected value and maximal standard-deviation scale are at most $C_1 e^{-c_1(L-C_0-R_K)^2}$. On a mesh of spacing $r = 2^{-m}$ in a fixed bounded region, for each fixed $\eta > 0$ and large m,

$$\mathbb{P}[\max_z Z_{r,z} > \eta m] \le C 2^{dm} e^{-c\eta^2 m^2}. \tag{23.1}$$

Proof. Since $\|K_u\|_1 = C u^{(d-1)/2}$ and its support radius is $R_K u$, the heat-kernel tail gives, for $x \in B_{C_0 r}(z)$ and $0 < \varepsilon \le r$,

$$\int_0^r \int_{\mathbb{R}^d \setminus B_{Lr}(z)} |(K_u * k_\varepsilon)(x-y)|^2 dy\, du \le C(r/\varepsilon)^d e^{-c(L-C_0-R_K)^2 (r/\varepsilon)^2}. \tag{23.2}$$

Indeed, outside that box the distance to the support of the translated K_u is at least a fixed multiple of $(L - C_0 - R_K)r$. Bound the convolution by its L^1 norm times the heat tail, square and integrate in y, then use $\int_0^r u^{d-1} du = r^d/d$.

Each spatial derivative, multiplied by the corresponding power of r, and each derivative in $a = \varepsilon/r$, satisfies the same estimate with an additional fixed power of a^{-1}. These powers are absorbed by a smaller Gaussian exponential constant. The bounds tend to zero faster than every power as $a \downarrow 0$. Gaussian Sobolev estimates on the fixed rescaled box times $[0,1]$ give a continuous version and the stated expected supremum. Gaussian concentration for the supremum gives its upper tail. A union bound over $O(2^{dm})$ sites proves (23.1). The supremum already includes every finer cutoff. □

Lemma 23.2 (Coarse heat errors and the exponent)**.** *At $r = 2^{-m}$ the process $H_m * k_\varepsilon - H_m$, for $0 \le \varepsilon \le r$, has uniformly bounded variances and rescaled derivative bounds on each r-cell. Its supremum on a fixed bounded region is $O_{\mathbb{P}}(\sqrt{m+1})$, uniformly over these cutoffs. The short-range heat field $X^{\mathrm{H}}_n = H_{0,\infty} * k_{2^{-n}}$ satisfies the same estimate for $X^{\mathrm{H}}_n - H_n$. Consequently,*

$$a^{\mathrm{H}}_\varepsilon = \varepsilon^{1 - \psi Q + o(1)}. \tag{23.3}$$

The short-range heat fields have nonnegative covariance and are positively associated.

Proof. For $u \geq \varepsilon$, smooth translation of K_u and the finite first moment of the heat kernel give

$$\|K_u * k_\varepsilon - K_u\|_2^2 \leq C\varepsilon^2 u^{-3}.$$

Integrating over $u \geq r$ bounds the coarse error variance by $C(\varepsilon/r)^2$. Rescaled spatial derivatives and derivatives in $\varepsilon/r \in [0,1]$ have uniform analogous bounds. Each rescaled cell thus has a bounded expected supremum and Gaussian upper tail; the maximum over $O(2^{dm})$ cells is $O_{\mathbb{P}}(\sqrt{m+1})$.

For $u < \varepsilon$, Young's inequality gives

$$\|K_u * k_\varepsilon\|_2^2 \leq \|K_u\|_1^2 \|k_\varepsilon\|_2^2 \leq Cu^{d-1}\varepsilon^{-d}.$$

Its integral below ε, together with the preceding estimate above ε, is bounded. Spatial increments at scale ε have bounded rescaled variance and entropy. Therefore $X_n^{\mathrm{H}} - H_n$ has compact-region supremum $O_{\mathbb{P}}(\sqrt{n+1})$. The corresponding multiplicative path-length error has logarithm $o_{\mathbb{P}}(n)$. The sharp-field point exponent (2.10) therefore gives the same exponent for heat smoothing.

By (2.2), the pinned field adds the large-scale correction $(g * k_\varepsilon) - C$. This correction has a uniform compact-region Gaussian supremum tail for $0 \leq \varepsilon \leq 1$. To see this directly, the kernel calculation gives

$$\operatorname{Var} g(y) \leq C \log(2 + |y|), \qquad \sup_y \mathbb{E}|\partial^\alpha g(y)|^2 < \infty \quad (|\alpha| \geq 1).$$

Write the convolution as a Gaussian-weighted average of $g(x - \varepsilon z)$ at unit scale. Minkowski's inequality and the logarithmic variance bound give a uniform second moment on compact (x, ε) sets. Derivatives in x and ε have the same bound; the latter insert Gaussian-integrable polynomials in z and positive-order derivatives of g. Gaussian Sobolev estimates and concentration give the supremum assertion, including $\varepsilon = 0$. The correction does not change the exponent. The same calculation on unit logarithmic cutoff intervals handles all real cutoffs.

Finally, $K_u * k_\varepsilon$ is nonnegative. The covariance of the stationary short-range field is an integral of products of these kernels. Gaussian positive association therefore applies [9]. This assertion concerns the original field and does not rely on independence of localized labels. □

23.2 Tightness estimates for heat smoothing

Proposition 23.3 (Heat tightness and boundary estimates)**.** *In the range* (1.1)*, statements (T1)–(T5) of Proposition 11.3 hold for* $D_h^{\mathrm{H},\varepsilon}$ *and* $a_\varepsilon^{\mathrm{H}}$*. Every surface exponent in* (11.7) *is allowed. The one-sided constrained boundary estimates of Lemma 22.1 hold as well.*

Proof. We verify each use of compact support in Sections 5–11. Work first with the stationary short-range heat field, and let b_t^{H} be one fixed sufficiently high quantile of its internal reference point distance. Lemma 23.2 gives $b_n^{\mathrm{H}} = 2^{-\chi n + o(n)}$, where $\chi = 1 - \psi Q$.

1: finite-range labels and surface chaining. At mesh $r = 2^{-m}$ and fine cutoff $\varepsilon \leq r$, call a site edge-good when all its localized fine-field coordinate edge distances are at most Mrb_{t-m}^{H}, where $t = -\log_2 \varepsilon$. The original fine field has the exact rescaled short-range heat law at cutoff $t - m$. Choose the reference quantile sufficiently high, and then M sufficiently large to absorb a fixed high quantile of $Z_{r,z}$. The label failure probability is as small as prescribed, uniformly in both cutoffs. The labels have a fixed finite dependence range in lattice units, since their determining noise lies in $B_{Lr}(z) \times (0, r)$. Consecutive-level nice labels have a fixed enlarged range. The finite-dependence path count and good-component routing in Lemma 5.1 therefore apply.

Return each chosen localized edge path to the original fine field before concatenation. Its cost increases by at most $e^{\psi Z_{r,z}}$. On the simultaneous event (23.1), this factor is at most $e^{\psi\eta m}$ eventually. The coarse heat convolution differs from the sharp coarse field by at most another ηm on all relevant cells, outside a summable failure probability, by Lemma 23.2. Thus the chain in the proof of Lemma 5.2 has only a fixed factor M and an additional factor $e^{2\psi\eta m}$ in its level-m summand. The terminal heat field has variance $n \log 2 + O(1)$ and bounded within-cell expected oscillation.

The tube covering number is still a polynomial times $2^{(d-1)m}$. Under the one-sided normalization ratio used at this stage, choose its exponent loss and η so that the effective surface-series parameter remains strictly between $\sqrt{2(d-1)}$ and Q. The same Gaussian summation gives tight Hölder bounds. It also gives a field-dependent separator upper series involving only the sharp coarse fields H_m: the heat and localization errors have been replaced by the deterministic factor $e^{2\psi\eta m}$ on a common eventual event. The independent-center estimate required by the zero-cluster argument is consequently preserved. Arbitrarily small losses yield every exponent in (11.7).

2: zero-cluster locality and spatial independence. The proof of Lemma 6.1 uses stationarity, radiality, a reference quantile, surface compactness and positive Gaussian association. These all hold for the heat field. Removing a fixed coarse white-noise band is exact and leaves a field independent of the removed noise. The original and coarse-deleted profiles satisfy the same multiplicative comparison, with the supremum of $H_a * k_\varepsilon$ on the compact cost region; that supremum is tight and converges locally uniformly as $\varepsilon \downarrow 0$.

For two fixed localization domains with disjoint closures at distance $\Delta > 0$, choose a so large that $R_K 2^{-a} < \Delta/10$. In each domain restrict the fine noise below 2^{-a} to a $\Delta/4$ enlargement of its compact nonzero-cost region. The two noise regions are disjoint. On those cost regions the omitted heat tails have every fixed derivative variance at most

$$C\varepsilon^{-N} \exp(-c\Delta^2/\varepsilon^2). \tag{23.4}$$

This is the calculation of (23.2) with fixed spatial separation. Their suprema tend to zero in probability. The normalized profile arrays are tight and differ by multiplicative factors tending to one, so their limits have the exact independence of the spatially truncated arrays. Cofinite domains cause no problem: the wired density vanishes outside a fixed outer cube, and only its compact nonzero-cost region enters.

Thus the limiting closed zero clusters are independent of each fixed coarse field, and local clusters in separated domains are independent. The first-exit inequality, source-clearance convention and closed-containment correction in Lemma 6.2 and the proof of Lemma 6.3 apply. The neighborhood argument in Proposition 6.4 gives $p \le p^2$ for the probability of a positive crossing. The positive-probability lower bound excludes $p = 0$. This proves reciprocal crossing tightness whenever the one-sided ratio needed for the preceding surface estimate holds.

3: record cutoffs and both normalizer comparisons. Choose a small ratio loss $\delta_1 > 0$, and use the successive records of $2^{(\chi+\delta_1)n} b_n^{\mathrm{H}}$. Along these records the required one-sided ratio is immediate, so Steps 1–2 give crossing noncollapse there.

At a coarse mesh, define crossing-good labels by the localized fine field with threshold $M^{-1} r b_k^{\mathrm{H}}$. Reciprocal tightness of the original fine crossings and a localization-error quantile make the failure probability arbitrarily small. Finite dependence bounds all bad components in the relevant region by diameter $O((1+m)^2)$ with exponentially small failure. Returning the lower crossing bounds to the actual field loses a factor $e^{\psi \max_z Z_{r,z}}$, whose logarithm is $O_{\mathbb{P}}(\sqrt{m})$. The coarse heat error has the same bound, while the sharp coarse oscillation over the routing neighborhood is $o_{\mathbb{P}}(m)$.

Lemma 7.1 and the argument of Lemma 7.2 therefore give, for each $\eta' > 0$,

$$b^{\mathrm{H}}_{m+k} \geq c_{\eta'} b^{\mathrm{H}}_k 2^{-(\chi+\eta')m} \tag{23.5}$$

when k is a record cutoff. The estimates are uniform in k, because both errors were controlled over all finer heat cutoffs. The finitely many bounded m follow from a single fixed separating fine annulus and a coarse-field quantile.

The record-envelope calculation in Proposition 7.3 now gives the one-sided ratio at every cutoff. Reapply Steps 1–2 at all cutoffs. For the opposite comparison, follow Lemma 7.4: route a sharp coarse near-minimizer through the infinite component of localized edge-good sites. Return every edge to the original fine field and the convolved coarse field to the sharp one. The total multiplicative loss is $e^{o_{\mathbb{P}}(m)}$, uniformly in the fine cutoff. The sharp point exponent gives

$$b^{\mathrm{H}}_{m+k} \leq C_{\eta'} b^{\mathrm{H}}_k 2^{-(\chi-\eta')m}. \tag{23.6}$$

This proves the two-sided ratios, point and annulus quantile bounds, and surface estimates at all cutoffs. No comparability with a compact-mollifier normalizer was assumed to obtain them.

4: quantitative lower tails. Apply the proof of Lemma 8.1, using crossing-good labels on a mesh 2^{-m} in a fixed annulus. Unless a bad star-component has macroscopic diameter, each crossing visits a good cell whose doubled shell is inside the annulus. Its cost is bounded below by the fine lower bound times the coarse minimum and the factor $e^{-\psi \max Z_{r,z}}$. Take $m = \lfloor 2 \log_2 \log T \rfloor$. The bad-component failure is at most $C 2^{dm} e^{-c2^m}$. The coarse minimum has failure at threshold of order $-\log T$ bounded by $C \exp[-c(\log T)^2/m]$. The localization and coarse-error suprema have the smaller bound $C 2^{dm} e^{-c(\log T)^2}$. For cutoffs below m, use the Euclidean shell width times the minimum smooth heat density. This proves the lower tail in (T3) and all negative moments of normalized crossings.

5: many good annuli. At shell scale $r = 2^{-m}$ put $u_m = r 2^{-\ell_0}$ and split off the fine noise with $u < u_m$. Restrict it spatially to the band $\{0.9r < |y - z|_\infty < 4.1r\}$, taking ℓ_0 large enough that $R_K u$ is much smaller than the band margins. At sufficiently separated scales these bands are disjoint. For $\varepsilon = 2^{-t} \leq u_m$, put $v = t - m - \ell_0$. Let H^{loc}_m be the internal diameter of the middle separator for the localized fine field, and let C^{loc}_m be its crossing constrained to the closed shell. The fixed-template surface and reciprocal crossing quantiles, together with the localization comparison, make the event

$$H^{\mathrm{loc}}_m \leq M_0 u_m b^{\mathrm{H}}_v, \qquad C^{\mathrm{loc}}_m \geq M_0^{-1} u_m b^{\mathrm{H}}_v$$

have probability as close to one as desired by increasing M_0. These events are independent across the separated bands. On the shell the omitted heat error has expected supremum and standard-deviation bounds of the form

$$C \exp[-c(r/\varepsilon)^2], \qquad \varepsilon \leq r 2^{-\ell_0},$$

after absorbing derivative powers. Over separated scales their standard deviations have a summable total, uniformly in the finest cutoff. Represent the sum of their absolute suprema as a Gaussian supremum. Its mean and maximal variance are bounded independently of the number of shells. Gaussian concentration therefore bounds the probability that a fixed positive fraction of k errors exceeds a sufficiently large allowance by e^{-Bk}, for any prescribed B.

For points x_r, x_s in the shells of radii $r \leq s$, let G_r, G_s be the corresponding coarse-field increments relative to z. With $A_{r,u}(y) = K_u(x_r - y) - K_u(z - y)$, the increment kernel satisfies

$\|A_{r,u} * k_\varepsilon\|_2 \le Cru^{-3/2}$. The common noise of the scale-r and scale-s coarse increments starts at $u \ge 2^{-\ell_0}s$. Consequently,

$$|\operatorname{Cov}(G_r, G_s)| \le Crs \int_{2^{-\ell_0}s}^{\infty} u^{-3}\, du \le C2^{2\ell_0} r/s\,.$$

The estimate holds uniformly for the spatial increments whose suprema define the coarse oscillations; centering by a spherical average follows by integrating the same increment bound. The expected suprema are bounded, and the sum of k oscillations is a Gaussian supremum with maximal variance $O(k)$ and mean $O(k)$. Gaussian concentration gives an arbitrarily large exponential rate for the event that more than $k/6$ coarse oscillations exceed a sufficiently large allowance R. The same holds for the localization errors. Increasing M_0 makes the binomial bound for more than $k/6$ failures of the localized fine event have the same rate.

If the fine event holds and both errors are at most R, comparison of path lengths before and after the two field corrections gives

$$H_{t,m}(z)/C_{t,m}(z) \le M_0^2 e^{4\psi R}.$$

Here $H_{t,m}$ and $C_{t,m}$ are the functionals from Section 9 for the heat metric. The common coarse additive value cancels, and the crossing can be trimmed to the closed shell. Choose each of the three failure-count rates to be at least $B + \log 3$. A union bound shows that at least $k/2$ shells satisfy the displayed ratio outside probability e^{-Bk}. This proves the estimate of Lemma 9.1. The finest-grid argument of Corollary 9.2 then gives (T5) simultaneously at every point.

6: real cutoffs, pinning and the metric limit. The real-cutoff argument in Lemma 8.2 uses only the established quantile estimates and bounded coarse suprema, so its rounding and fixed-family comparison apply. The uniform Gaussian pinning correction in Lemma 23.2 transfers the estimates to the prescribed field. The median and the fixed high reference quantile are comparable by upper and reciprocal tightness of the reference distance. The boundary-distance construction, separator gluing, fixed-point consistency and compact path limits then give (T1), by the proofs of Theorems 1.1 and 1.2 in Sections 10–11.

Finally, the inward-clipped routing in Lemma 22.1 uses the same localized finite-range labels. Its edge domains remain in the closed cube. The two heat errors are absorbed by the same arbitrarily small surface-series loss from Step 1. Thus the constrained boundary estimates have the unchanged admissible exponents. □

23.3 Locality despite the heat kernel's infinite support

Lemma 23.4 (Fixed-buffer truncation)**.** *Fix $\Delta > 0$ and a smooth cutoff χ_Δ equal to one on $B_{\Delta/2}$ and zero outside B_Δ. Set*

$$k_{\varepsilon,\Delta} = \frac{\chi_\Delta k_\varepsilon}{\int \chi_\Delta k_\varepsilon}.$$

On every fixed compact set, $h * k_\varepsilon - h * k_{\varepsilon,\Delta}$ tends to zero in probability uniformly with every fixed number of derivatives, faster than any power of ε. The same is true for the Gaussian residual field of Lemma 4.6.

Proof. The kernel difference has zero integral. Its L^1 norm, fixed polynomially weighted L^1 norms, L^∞ norm and the corresponding derivative norms are bounded by a polynomial in ε^{-1} times $e^{-c\Delta^2/\varepsilon^2}$. Bound its logarithmic covariance energy by splitting $|x-y| \le 1$ and its complement. On the first part use local integrability of $|\log|x-y||$ and the L^1–L^∞ bounds. On the second

use the polynomially weighted L^1 bounds. Each specified derivative error has variance at most $C\varepsilon^{-N}e^{-c\Delta^2/\varepsilon^2}$. Zero total mass makes the bound translation invariant. Gaussian Sobolev estimates on a compact neighborhood give the supremum assertion. Conditional Gaussian residual covariance is an orthogonal-projection contraction, so its variance for each test is no larger; the same estimates apply. □

Proposition 23.5 (Heat subsequential limits are regular weak)**.** *Every subsequential heat coupling from Proposition 23.3 admits a field-measurable regular weak assignment with parameters* (ψ, Q).

Proof. Proposition 23.3 and the exact heat change of variables give buffered intrinsic recovery by Lemma 12.8 and raw scale constants by Proposition 13.1. For conditional locality choose $W \Subset V$ and an exterior compact buffer $W_0 \Subset \mathbb{R}^d \setminus \overline{V}$. Replace heat smoothing on these regions by the kernels in Lemma 23.4, with support radius below their clearances. The interior coordinates are then functions of $h|_V$. After subtracting the smooth conditional mean with a compactly supported cutoff, the exterior coordinates are functions of the independent residual. The truncated joint blocks are exactly independent. The true heat coordinates differ by factors tending to one, because the field errors vanish uniformly and the normalized coordinates are tight. Their limits therefore have that same independence. Local weighted recovery restores the known conditional mean and proves conditional locality.

The supportwise comparison and cell variance argument of Section 13 remove residual metric randomness. For a Borel assignment under all continuous perturbations, use the fixed-buffer truncated heat kernels in (13.9). In each $V \Subset U$ choose their support within the clearance to ∂U. They approximate the true heat field on the base sample. For every continuous f, including unbounded f, their convolutions converge locally uniformly on the fixed buffer. The local recovery proof gives the simultaneous Weyl assignment without ever taking a global heat convolution of such an f. Intrinsic reconstruction and Lemma 12.14 give the metric structure and locality. Translation covariance follows on the base field from the true heat identity and then under perturbations by Weyl scaling. The weak estimates are precisely Proposition 23.3. □

Theorem 23.6 (Heat convergence)**.** *For* $d \geq 3$ *and* $Q_d(\psi) > \sqrt{2(d-1)}$,

$$(a_\varepsilon^{\mathrm{H}})^{-1} D_h^{\mathrm{H},\varepsilon} \longrightarrow E_h \tag{23.7}$$

in probability in the lower-semicontinuous topology, jointly with the field and uniformly on each fixed deterministic coordinate complex. The internal B_2 *point distance converges to its actual intrinsic value and has limiting median one. Global convergence holds after a possibly dependent bounded continuous perturbation.*

If $\mathfrak{a}_\varepsilon^{\mathrm{H}} = \operatorname{Med} Z_{C_0}(D_h^{\mathrm{H},\varepsilon})$, *then*

$$\frac{\mathfrak{a}_\varepsilon^{\mathrm{H}}}{a_\varepsilon^{\mathrm{H}}} \longrightarrow m_{C_0}, \qquad (\mathfrak{a}_\varepsilon^{\mathrm{H}})^{-1} D_h^{\mathrm{H},\varepsilon} \longrightarrow m_{C_0}^{-1} E_h. \tag{23.8}$$

Thus the heat-smoothed closed-cube convention gives the same original-I–V assignment with its indicated deterministic normalization.

Proof. Every heat limit is regular weak by Proposition 23.5. Theorem 19.1 compares it with the compact-mollifier assignment. Exact scaling, domain continuity and the unique point-median argument in Section 20 apply to each heat limit and fix the multiplier to one. Every subsequence therefore has the same field-measurable limit, giving convergence in probability by the diagonal

joint-limit argument. Surface convergence and bounded dependent perturbations follow from the same weighted recovery.

The heat boundary estimates in Proposition 23.3 permit the closed-cube recovery argument of Section 22. Its limit is the actual X_{C_0} of the already identified assignment. The unique crossing median gives both assertions in (23.8). All assertions concern each fixed regularization, not a simultaneous uncountable family of kernels. □

Proof of Theorem 1.5. Set $\mathcal{D}_h = m_{C_0}^{-1} E_h$. Theorems 22.4 and 23.6 prove the asserted convergence for the base field. Corollary 20.9 gives the same point-normalized assignment for each fixed compact mollifier; the scalar m_{C_0} is therefore common to all of them. Bounded dependent perturbations follow from the pathwise weighted recovery in Lemma 12.10, followed by the deterministic scalar convergence of the cube normalizers. The uniform convergence on each fixed coordinate complex is preserved by this scalar conversion. □

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